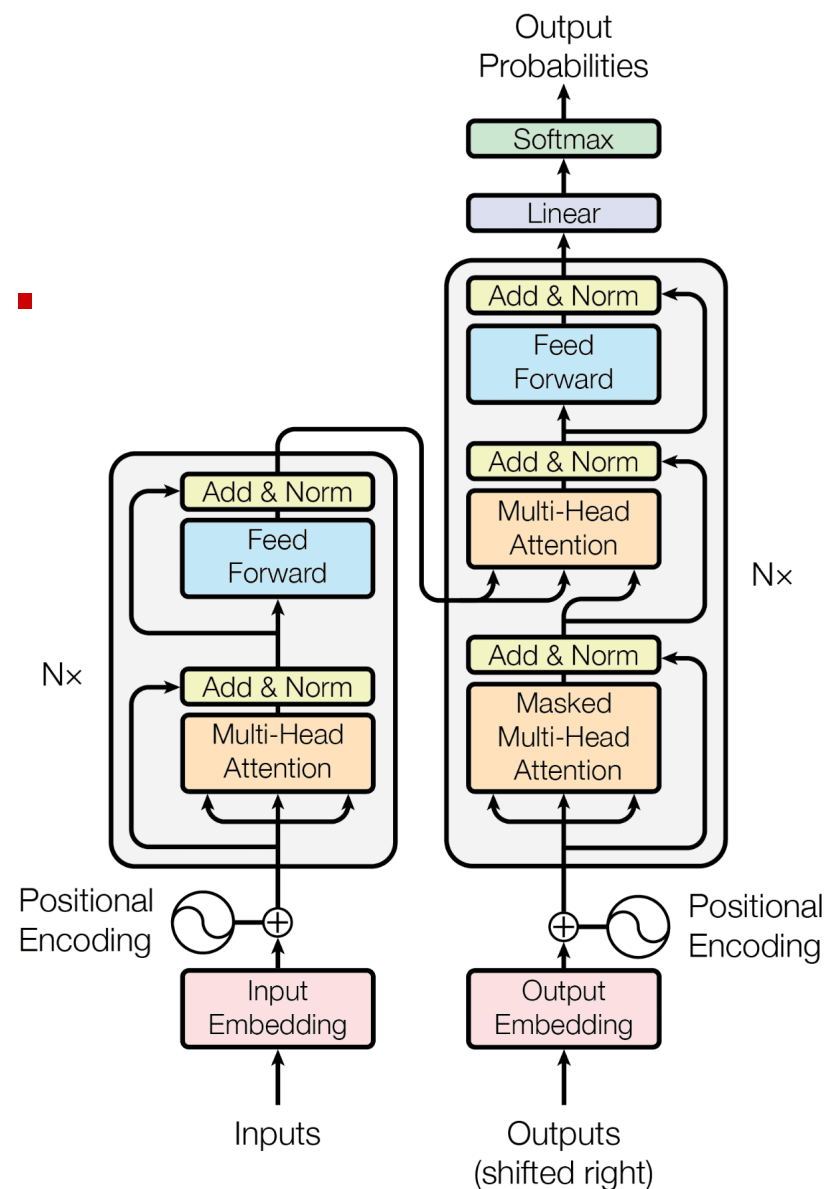
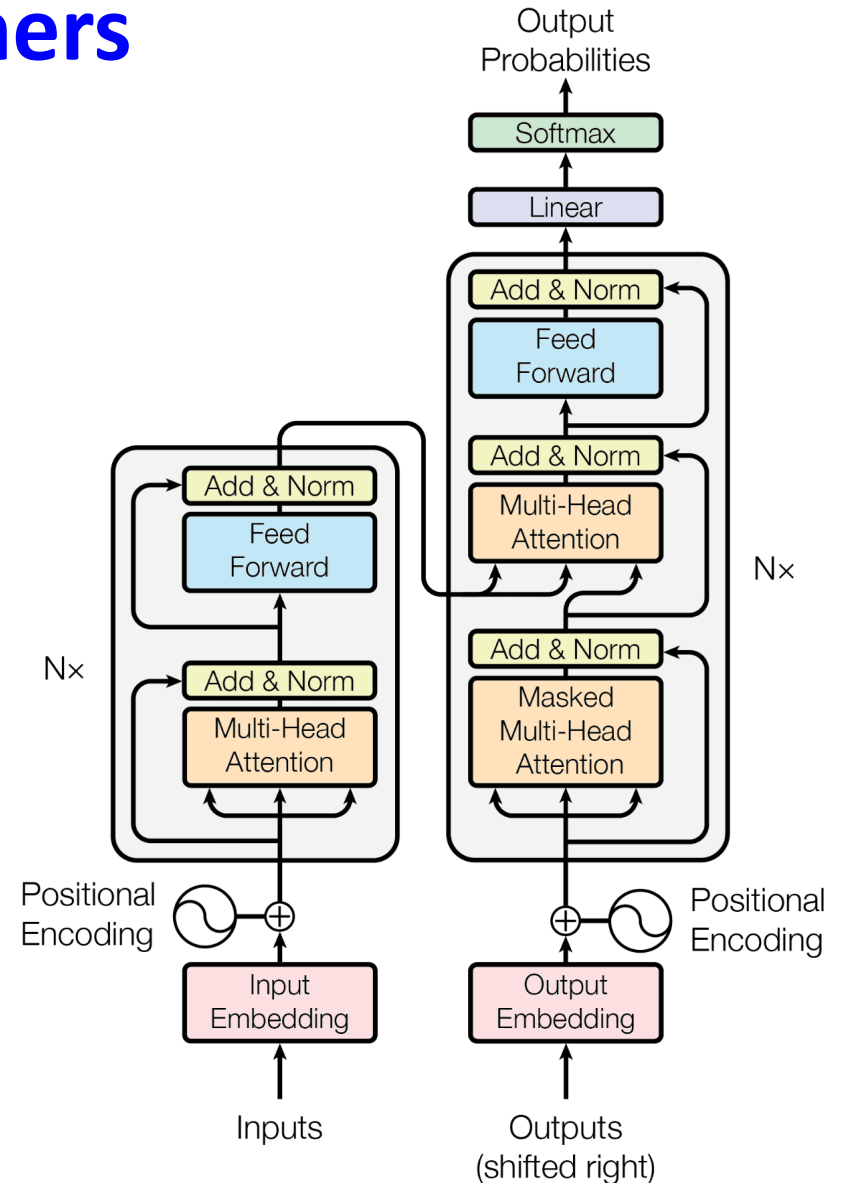


The most Comprehensive Lecture Notes on TRANSFORMERS....



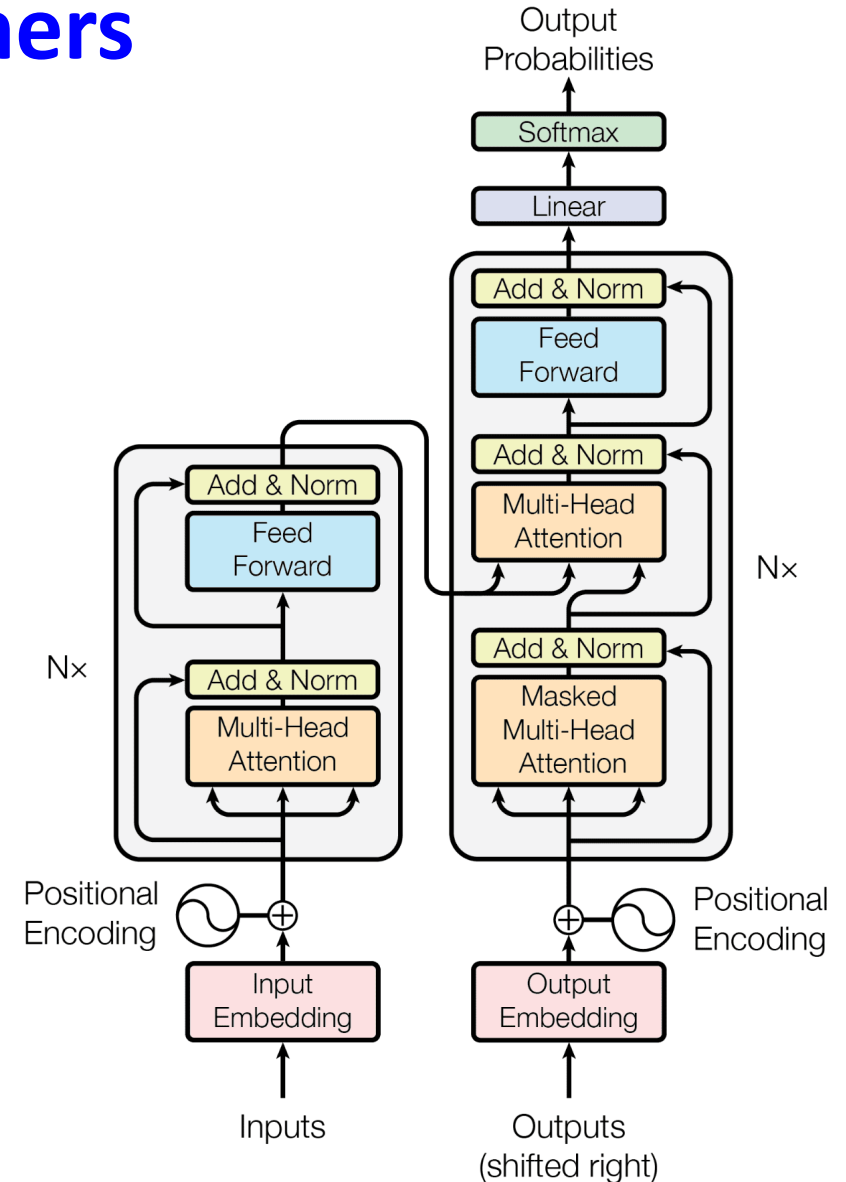
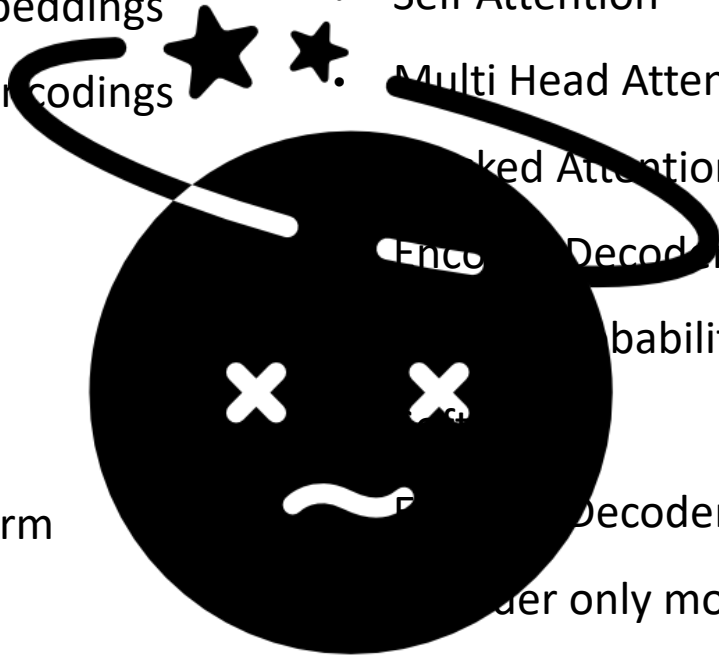
Transformers

- Tokenization
- Input Embeddings
- Position Encodings
- Residuals
- Query
- Key
- Value
- Add & Norm
- Encoder
- Decoder
- Attention
- Self Attention
- Multi Head Attention
- Masked Attention
- Encoder Decoder Attention
- Output Probabilities / Logits
- Softmax
- Encoder-Decoder models
- Decoder only models



Transformers

- Tokenization
- Input Embeddings
- Position Encodings
- Residuals
- Query
- Key
- Value
- Add & Norm
- Encoder
- Decoder
- Attention
- Self Attention
- Multi Head Attention
- Encoder Attention
- Decoder Attention
- Probabilities / Logits
- Encoder only models
- Decoder models

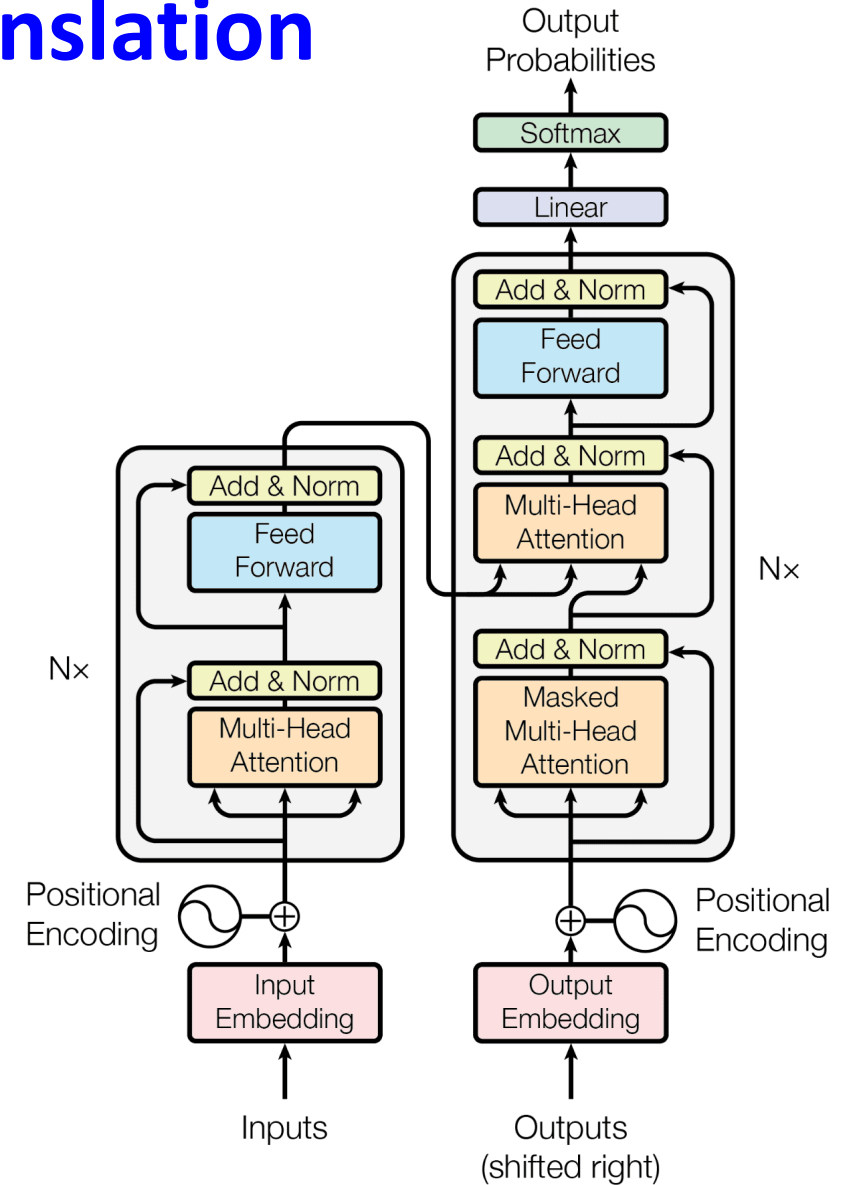


Machine Translation

Targets
Ich have einen apfel gegessen

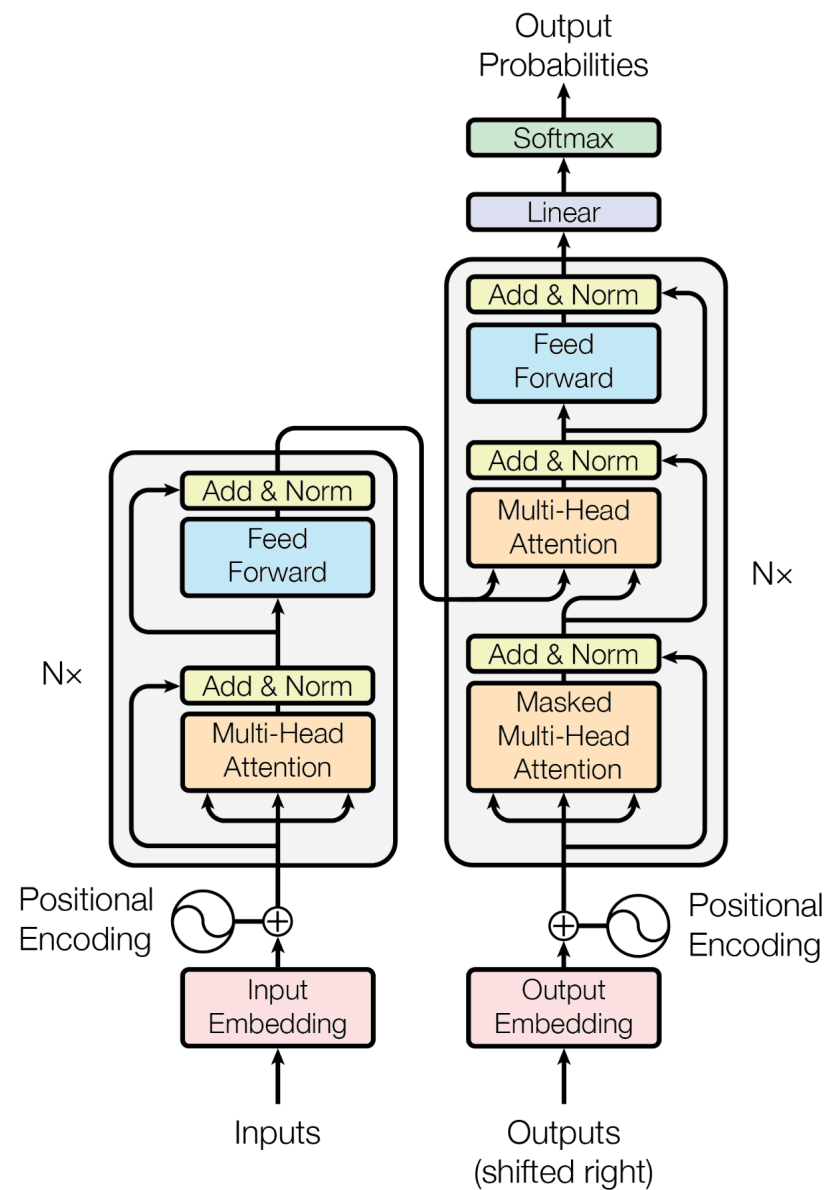
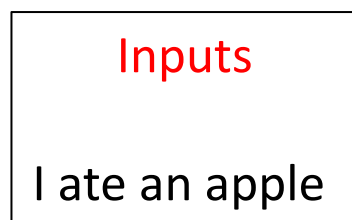


Inputs
I ate an apple

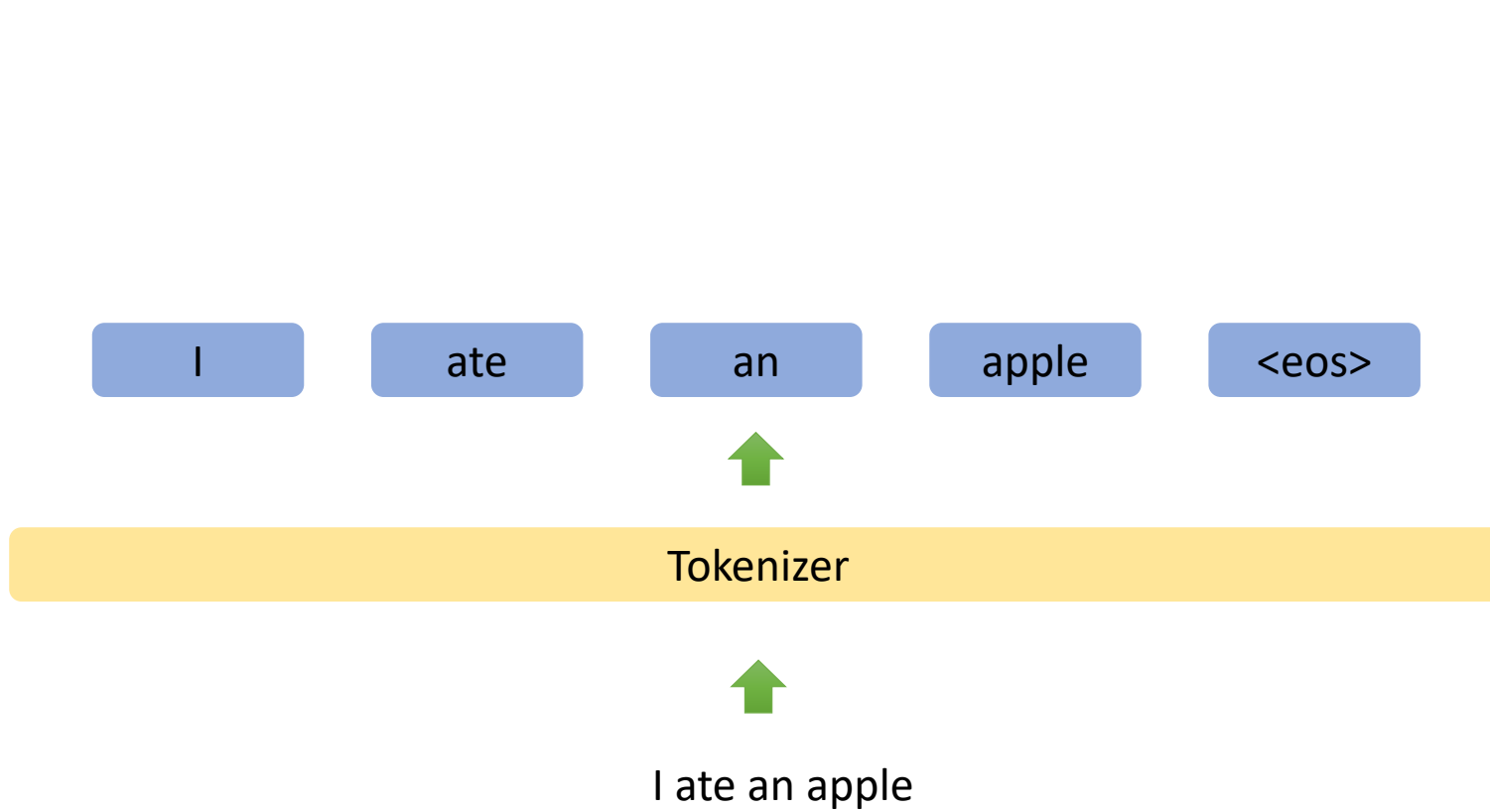


Inputs

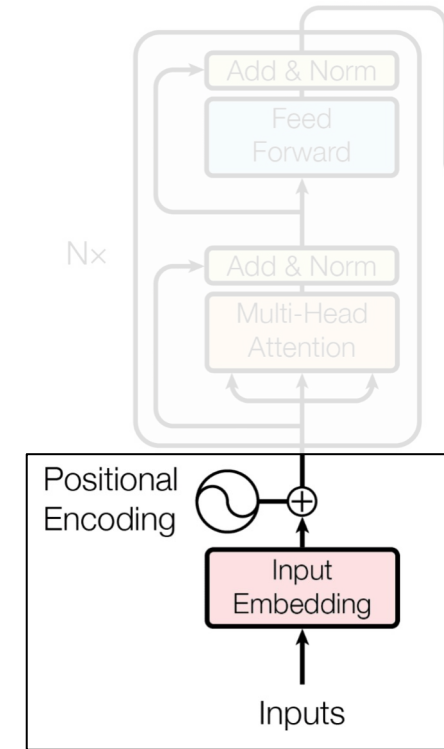
Processing Inputs



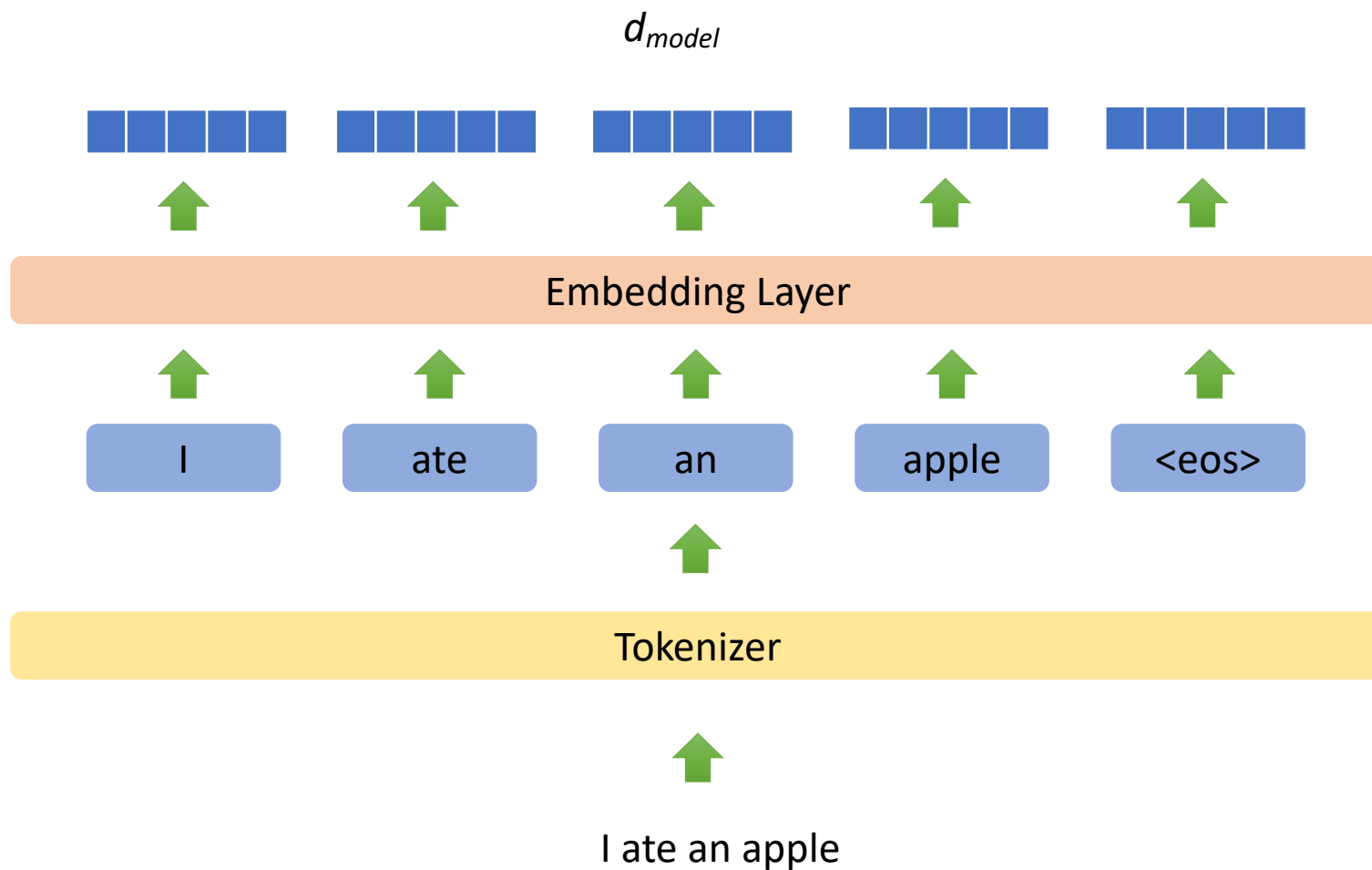
Inputs



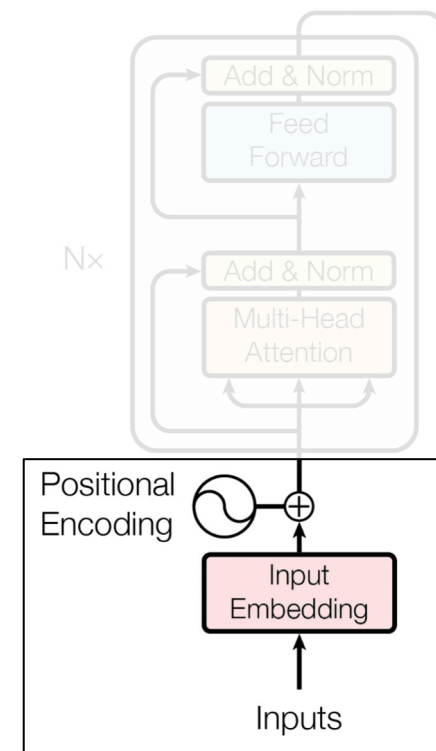
Generate Input Embeddings



Inputs

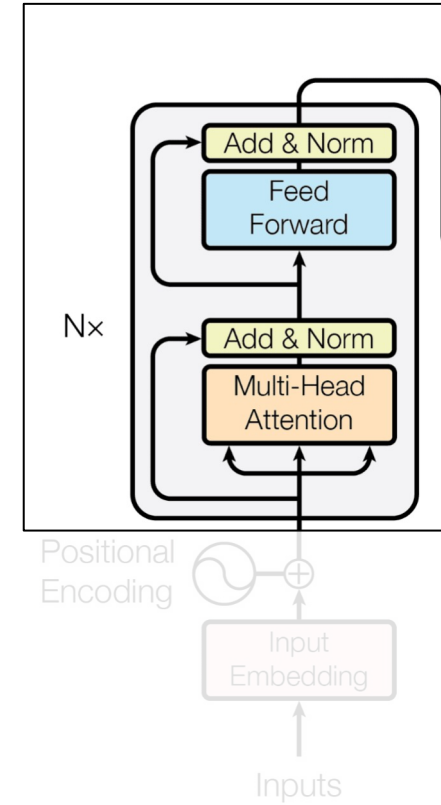
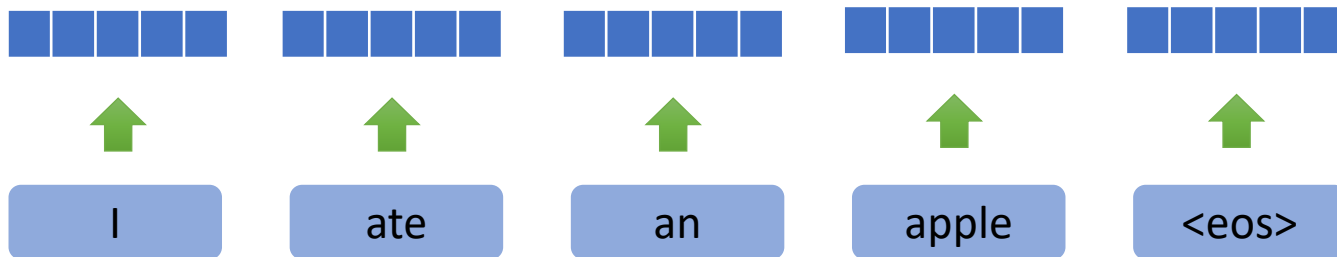


Generate Input Emembeddings

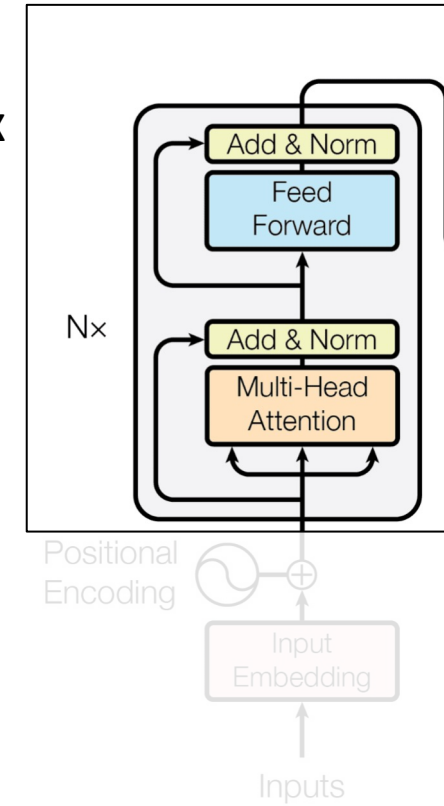
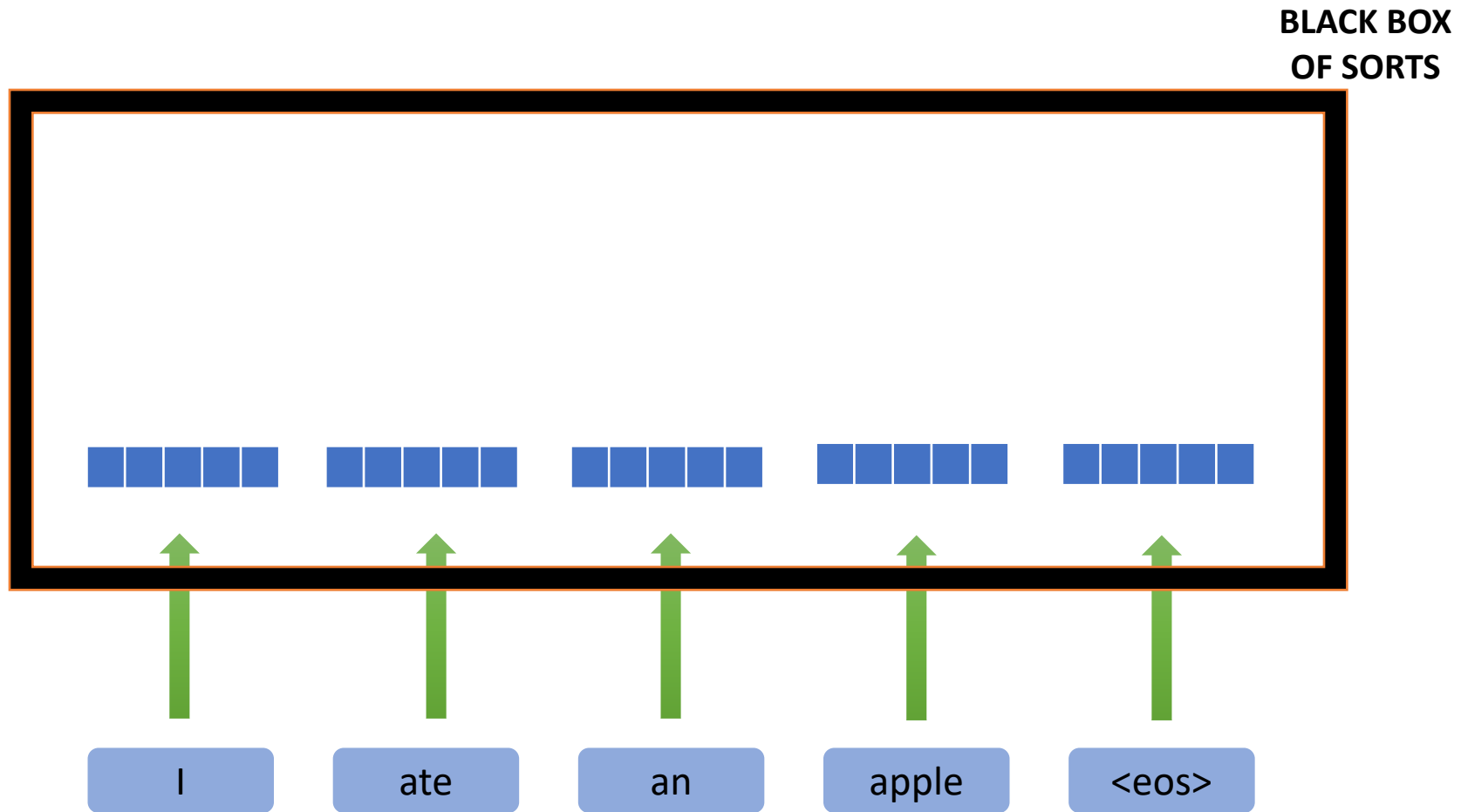


Encoder

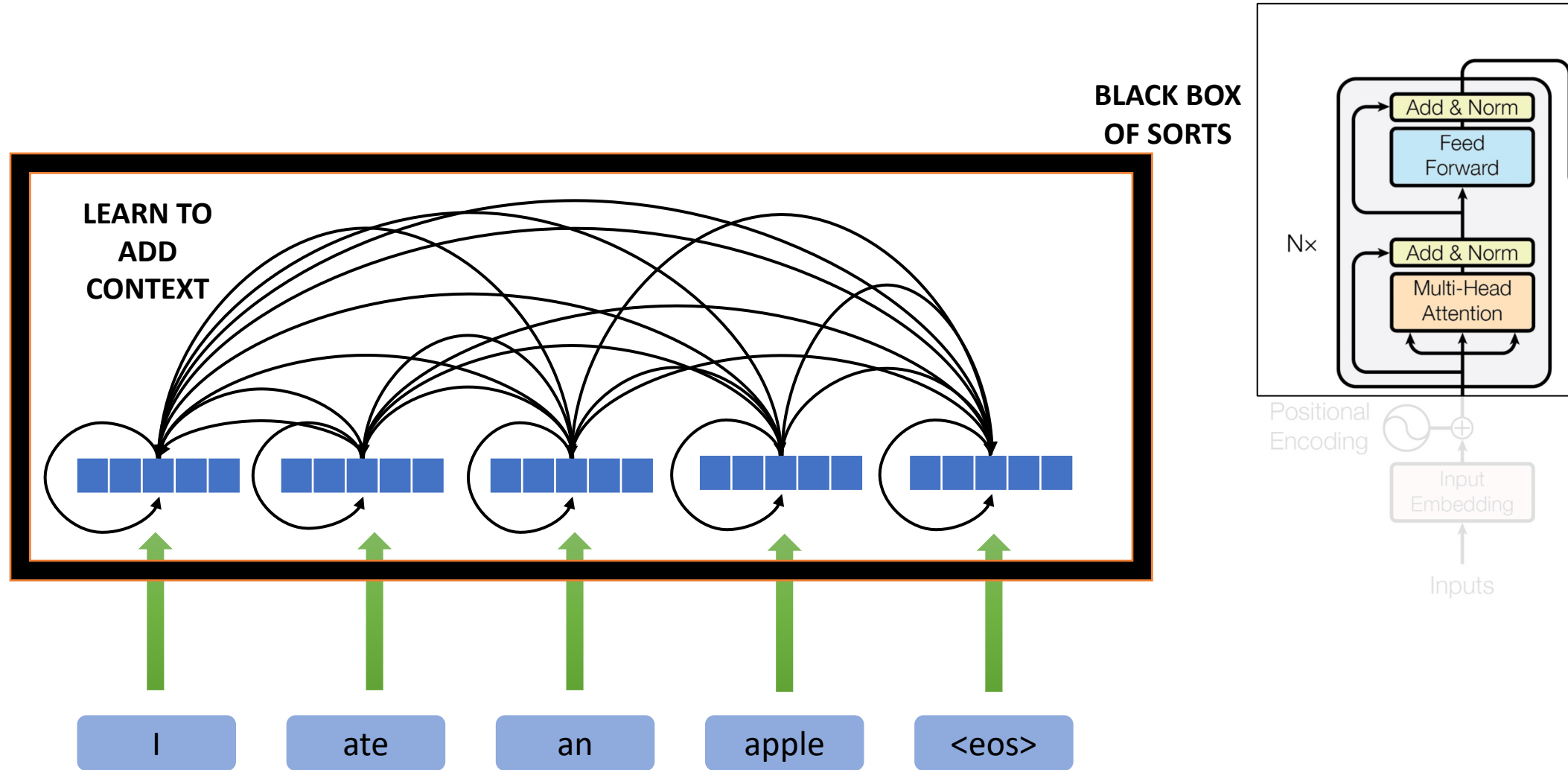
WHERE IS THE
CONTEXT ?



Encoder

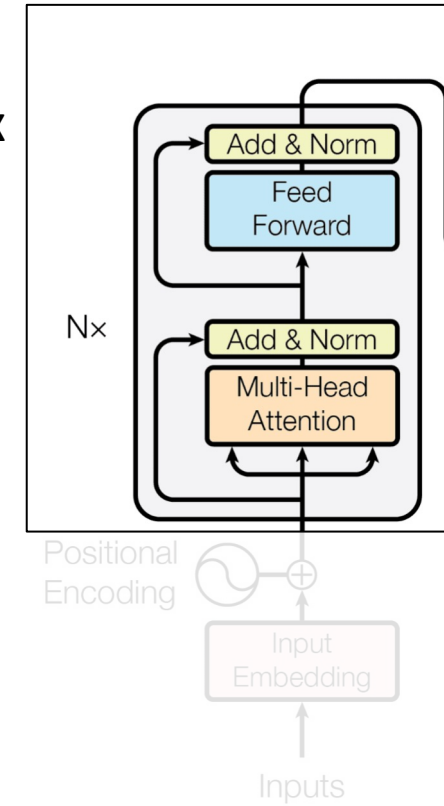
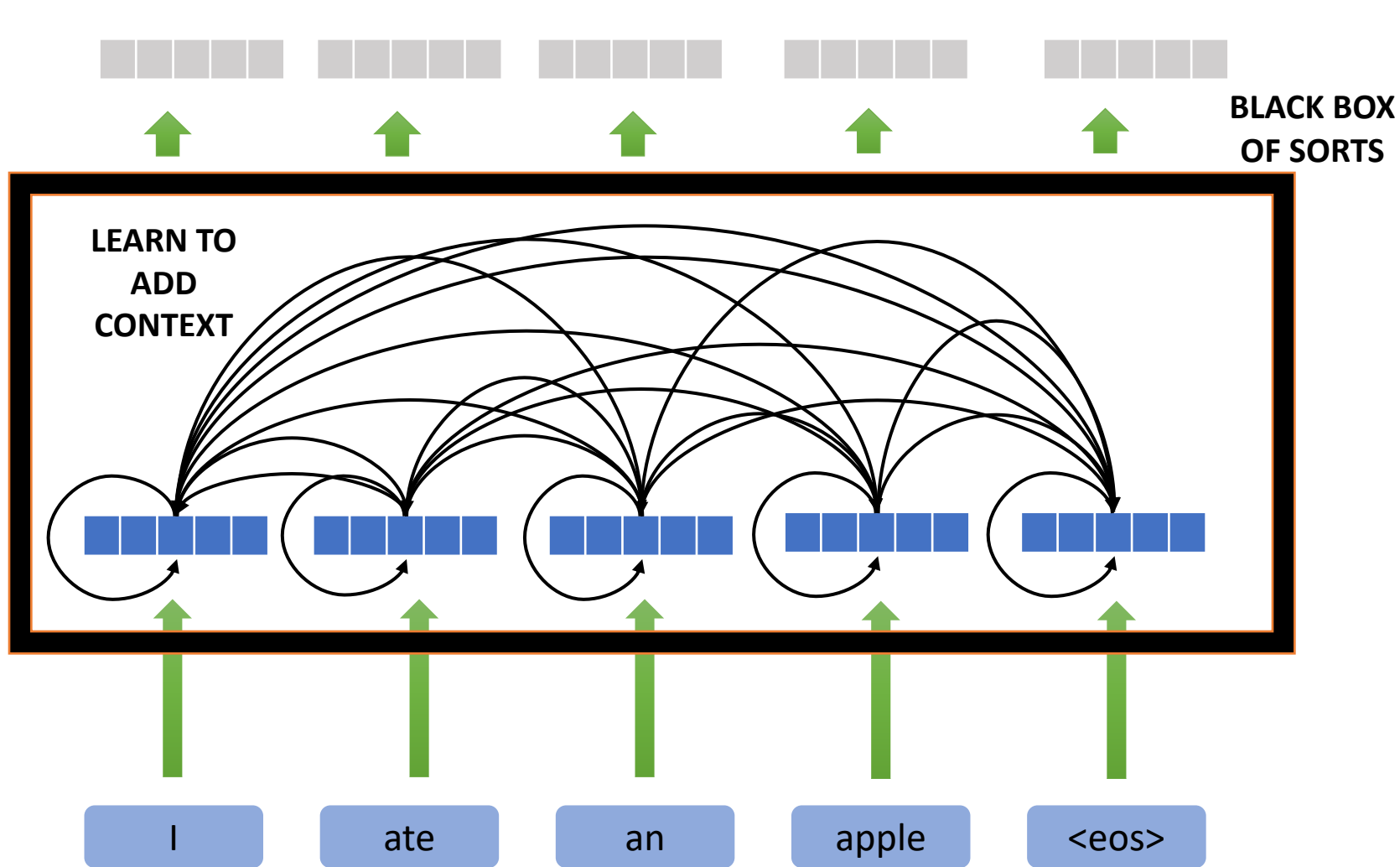


Encoder



Encoder

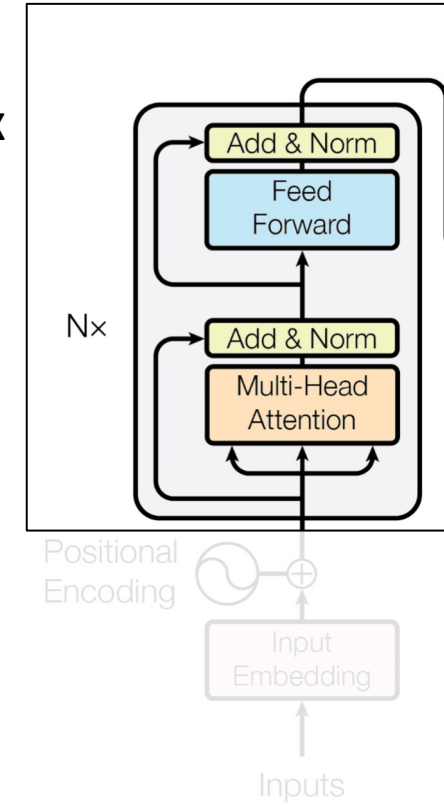
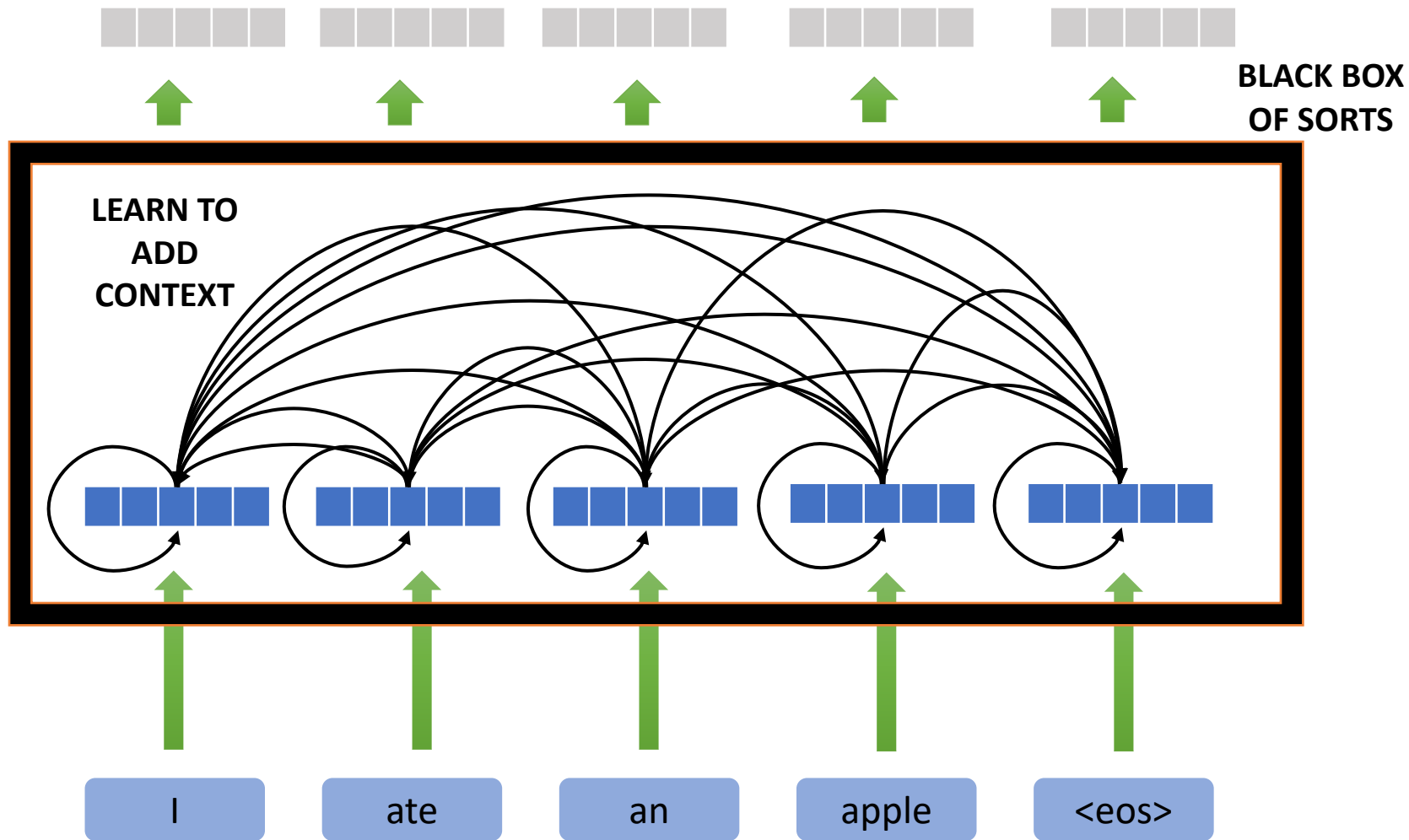
CONTEXTUALLY RICH EMBEDDINGS



Encoder

$\alpha_{[ij]}$?

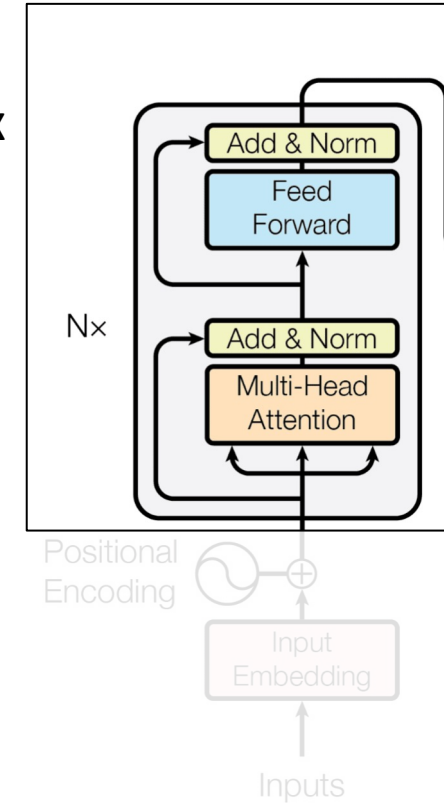
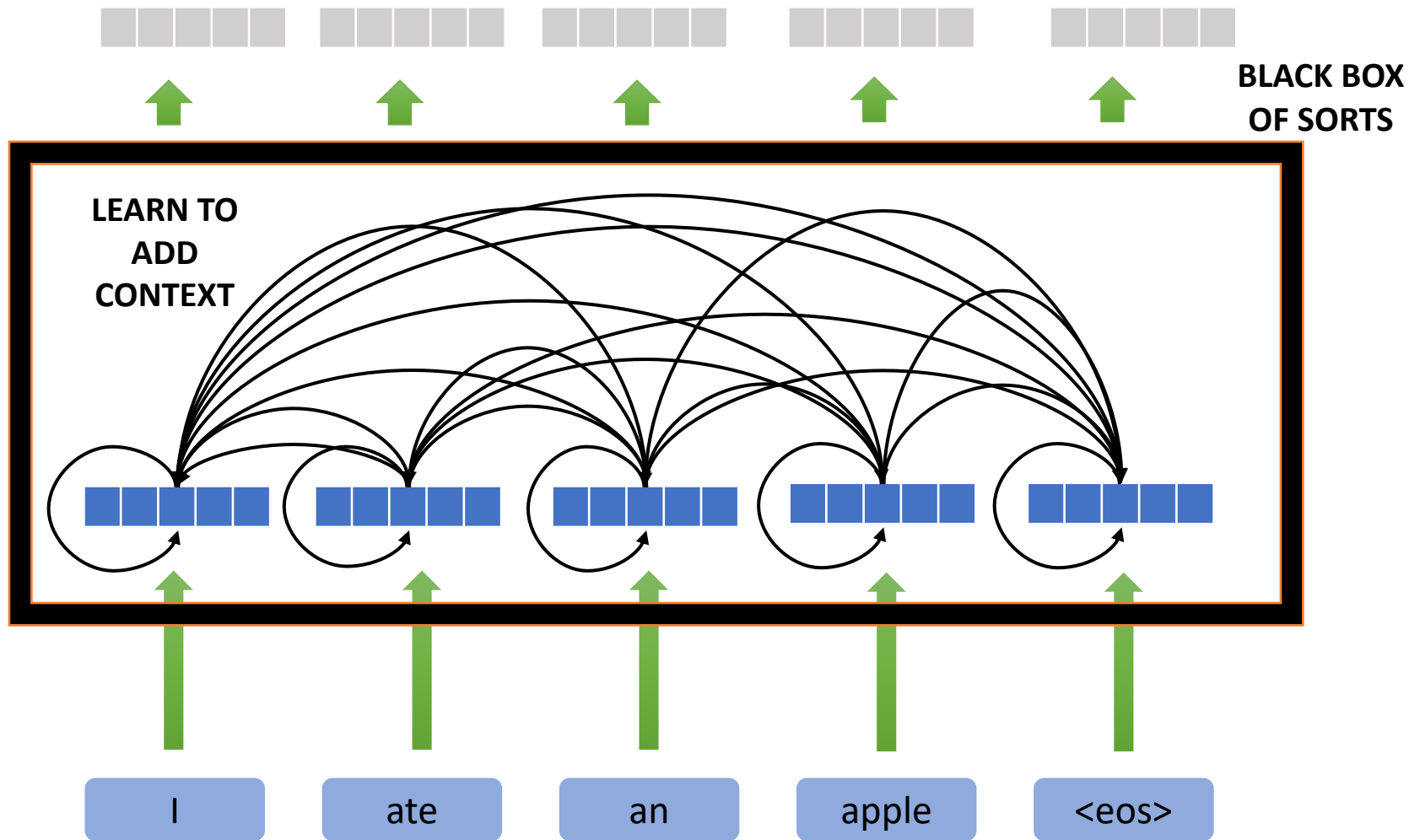
CONTEXTUALLY RICH EMBEDDINGS



Encoder

$$\alpha_{[ij]} \quad ? \quad \Sigma \quad \Pi \quad ?$$

CONTEXTUALLY RICH EMBEDDINGS

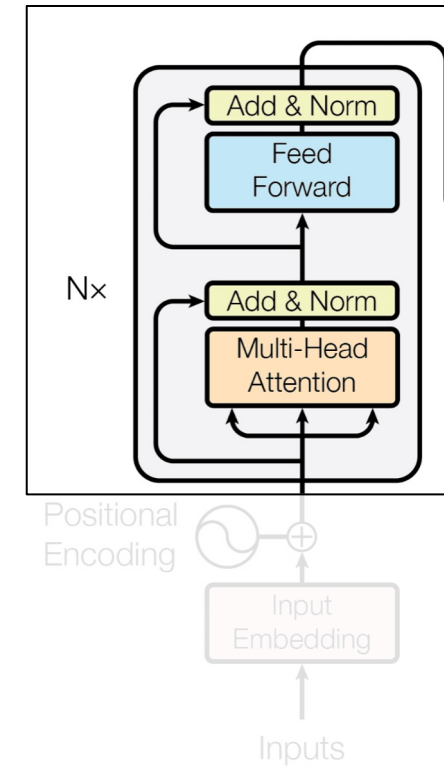


Attention

$$\alpha_{[ij]} \quad ?$$

From lecture 18:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$



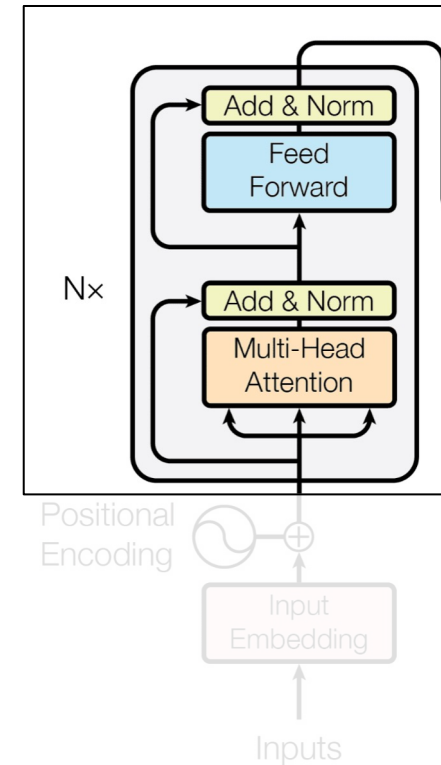
Attention

$$\alpha_{[ij]} \quad ?$$

From lecture 18:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

- Query
- Key
- Value



Query, Key & Value

Database

{Key, Value store}

```
{ "order_100": { "items": "a1", "delivery_date": "a2", ... },  
  "order_101": { "items": "b1", "delivery_date": "b2", ... },  
  "order_102": { "items": "c1", "delivery_date": "c2", ... },  
  "order_103": { "items": "d1", "delivery_date": "d2", ... },  
  "order_104": { "items": "e1", "delivery_date": "e2", ... },  
  "order_105": { "items": "f1", "delivery_date": "f2", ... },  
  "order_106": { "items": "g1", "delivery_date": "g2", ... },  
  "order_107": { "items": "h1", "delivery_date": "h2", ... },  
  "order_108": { "items": "i1", "delivery_date": "i2", ... },  
  "order_109": { "items": "j1", "delivery_date": "j2", ... },  
  "order_110": { "items": "k1", "delivery_date": "k2", ... }
```


Query, Key & Value

Database

{Key, Value store}

{Query: "Order details of order_104"}

OR

{Query: "Order details of order_106"}

```
{ "order_100": { "items": "a1", "delivery_date": "a2", ... },  
  "order_101": { "items": "b1", "delivery_date": "b2", ... },  
  "order_102": { "items": "c1", "delivery_date": "c2", ... },  
  "order_103": { "items": "d1", "delivery_date": "d2", ... },  
  "order_104": { "items": "e1", "delivery_date": "e2", ... },  
  "order_105": { "items": "f1", "delivery_date": "f2", ... },  
  "order_106": { "items": "g1", "delivery_date": "g2", ... },  
  "order_107": { "items": "h1", "delivery_date": "h2", ... },  
  "order_108": { "items": "i1", "delivery_date": "i2", ... },  
  "order_109": { "items": "j1", "delivery_date": "j2", ... },  
  "order_110": { "items": "k1", "delivery_date": "k2", ... }
```

Query, Key & Value

{Key, Value store}

{Query: "Order details of order_104"}

OR

{Query: "Order details of order_106"}

```
{ "order_100": { "items": "a1", "delivery_date": "a2", ... },  
  "order_101": { "items": "b1", "delivery_date": "b2", ... },  
  "order_102": { "items": "c1", "delivery_date": "c2", ... },  
  "order_103": { "items": "d1", "delivery_date": "d2", ... },  
  "order_104": { "items": "e1", "delivery_date": "e2", ... },  
  "order_105": { "items": "f1", "delivery_date": "f2", ... },  
  "order_106": { "items": "g1", "delivery_date": "g2", ... },  
  "order_107": { "items": "h1", "delivery_date": "h2", ... },  
  "order_108": { "items": "i1", "delivery_date": "i2", ... },  
  "order_109": { "items": "j1", "delivery_date": "j2", ... },  
  "order_110": { "items": "k1", "delivery_date": "k2", ... }
```

Query, Key & Value

{Key, Value store}

{Query: "Order details of order_104"}

OR

{Query: "Order details of order_106"}

```
{
  "order_100": {
    "items": "a1",
    "delivery_date": "a2",
    ...
  },
  "order_101": {
    "items": "b1",
    "delivery_date": "b2",
    ...
  },
  "order_102": {
    "items": "c1",
    "delivery_date": "c2",
    ...
  },
  "order_103": {
    "items": "d1",
    "delivery_date": "d2",
    ...
  },
  "order_104": {
    "items": "e1",
    "delivery_date": "e2",
    ...
  },
  "order_105": {
    "items": "f1",
    "delivery_date": "f2",
    ...
  },
  "order_106": {
    "items": "g1",
    "delivery_date": "g2",
    ...
  },
  "order_107": {
    "items": "h1",
    "delivery_date": "h2",
    ...
  },
  "order_108": {
    "items": "i1",
    "delivery_date": "i2",
    ...
  },
  "order_109": {
    "items": "j1",
    "delivery_date": "j2",
    ...
  },
  "order_110": {
    "items": "k1",
    "delivery_date": "k2",
    ...
  }
}
```

Query, Key & Value

{Key, Value store}

{Query: "Order details of order_104"}

OR

{Query: "Order details of order_106"}

```
{
  "order_100": {
    "items": "a1",
    "delivery_date": "a2",
    ...
  },
  "order_101": {
    "items": "b1",
    "delivery_date": "b2",
    ...
  },
  "order_102": {
    "items": "c1",
    "delivery_date": "c2",
    ...
  },
  "order_103": {
    "items": "d1",
    "delivery_date": "d2",
    ...
  },
  "order_104": {
    "items": "e1",
    "delivery_date": "e2",
    ...
  },
  "order_105": {
    "items": "f1",
    "delivery_date": "f2",
    ...
  },
  "order_106": {
    "items": "g1",
    "delivery_date": "g2",
    ...
  },
  "order_107": {
    "items": "h1",
    "delivery_date": "h2",
    ...
  },
  "order_108": {
    "items": "i1",
    "delivery_date": "i2",
    ...
  },
  "order_109": {
    "items": "j1",
    "delivery_date": "j2",
    ...
  },
  "order_110": {
    "items": "k1",
    "delivery_date": "k2",
    ...
  }
}
```

Query, Key & Value

Done at the same time !!

{Query: "Order details of order_104"}

OR

{Query: "Order details of order_106"}

{Key, Value store}

```
{ "order_100": { "items": "a1", "delivery_date": "a2", ... },  
  "order_101": { "items": "b1", "delivery_date": "b2", ... },  
  "order_102": { "items": "c1", "delivery_date": "c2", ... },  
  "order_103": { "items": "d1", "delivery_date": "d2", ... },  
  "order_104": { "items": "e1", "delivery_date": "e2", ... },  
  "order_105": { "items": "f1", "delivery_date": "f2", ... },  
  "order_106": { "items": "g1", "delivery_date": "g2", ... },  
  "order_107": { "items": "h1", "delivery_date": "h2", ... },  
  "order_108": { "items": "i1", "delivery_date": "i2", ... },  
  "order_109": { "items": "j1", "delivery_date": "j2", ... },  
  "order_110": { "items": "k1", "delivery_date": "k2", ... }
```

Query, Key & Value

{Query: "Order details of order_104"}

OR

{Query: "Order details of order_106"}

```
{ "order_100": { "items": "a1", "delivery_date": "a2", ... } },  
{ "order_101": { "items": "b1", "delivery_date": "b2", ... } },  
{ "order_102": { "items": "c1", "delivery_date": "c2", ... } },  
{ "order_103": { "items": "d1", "delivery_date": "d2", ... } },  
{ "order_104": { "items": "e1", "delivery_date": "e2", ... } },  
{ "order_105": { "items": "f1", "delivery_date": "f2", ... } },  
{ "order_106": { "items": "g1", "delivery_date": "g2", ... } },  
{ "order_107": { "items": "h1", "delivery_date": "h2", ... } },  
{ "order_108": { "items": "i1", "delivery_date": "i2", ... } },  
{ "order_109": { "items": "j1", "delivery_date": "j2", ... } },  
{ "order_110": { "items": "k1", "delivery_date": "k2", ... } }
```

Query

1. Search for info

Key

1. Interacts directly with Queries
2. Distinguishes one object from another
3. Identify which object is the most relevant and by how much

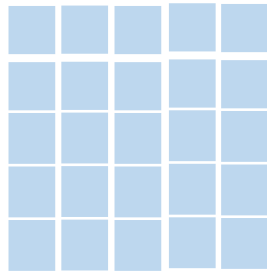
Value

1. Actual details of the object
2. More fine grained

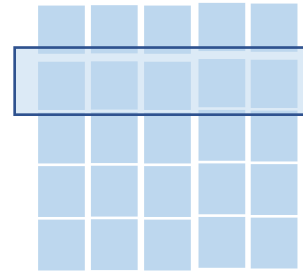
Attention



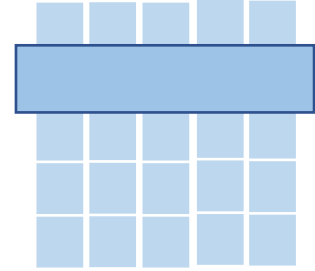
Query



Key Value Store

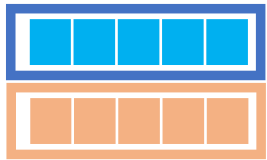


Key

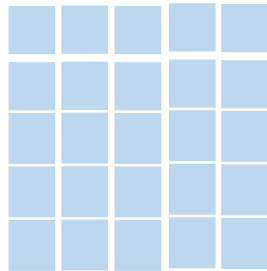


Value

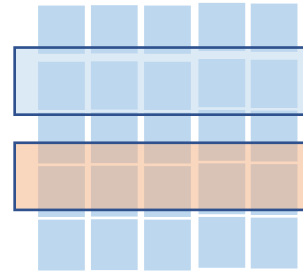
Attention



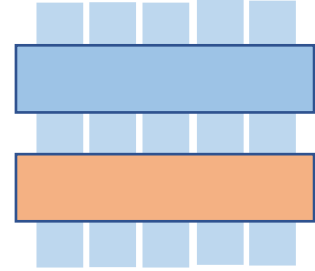
Query



Key Value Store



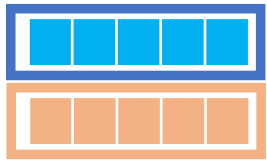
Key



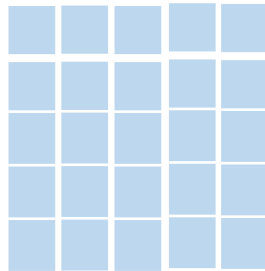
Value

Attention

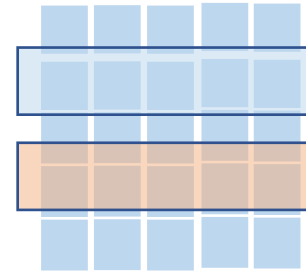
Done at the same time !!



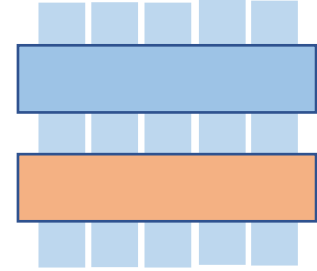
Query



Key Value Store



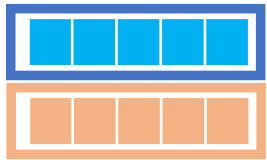
Key



Value

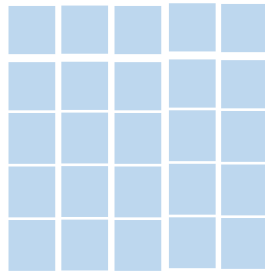
Attention

Parallelizable !!!



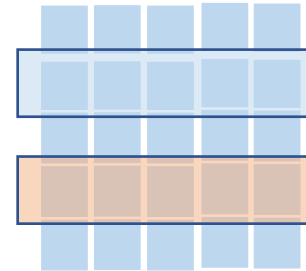
Query

Q



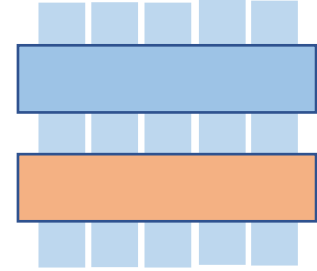
Key Value Store

QK^T



Key

$\text{softmax}(\frac{QK^T}{\sqrt{d}})$



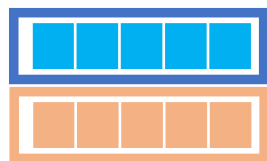
Value

$\text{softmax}(\frac{QK^T}{\sqrt{d}})V$

Attention

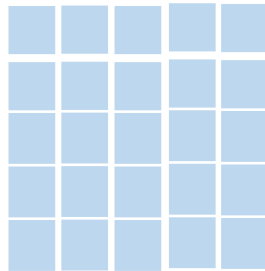
Parallelizable !!!

Attention Filter



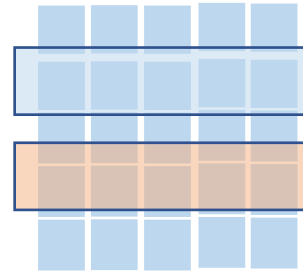
Query

Q



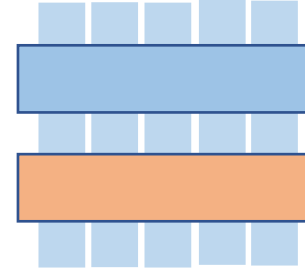
Key Value Store

QK^T



Key

$\text{softmax}(\frac{QK^T}{\sqrt{d}})$



Value

$\text{softmax}(\frac{QK^T}{\sqrt{d}})V$

Attention



l_1

I



l_2

ate



l_3

an



l_4

apple

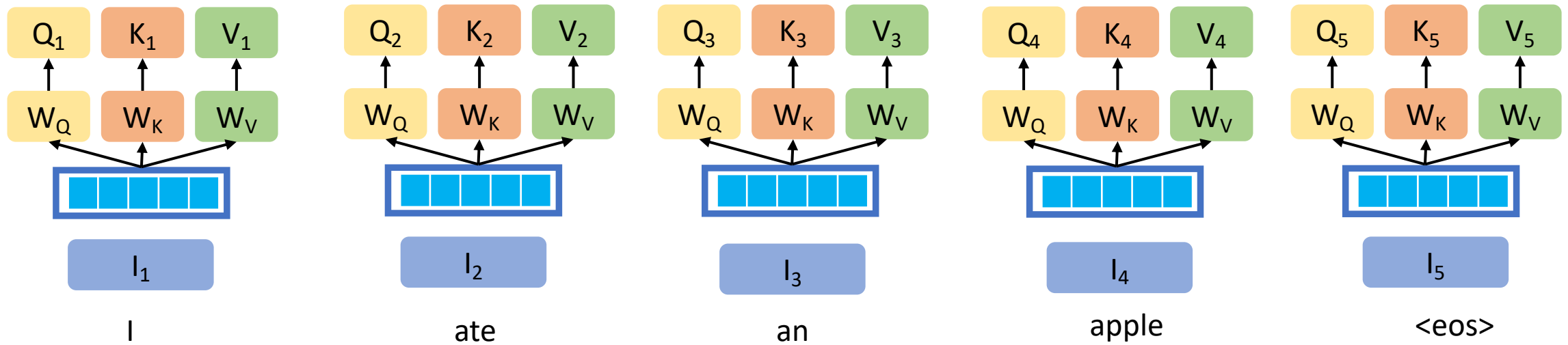


l_5

<eos>

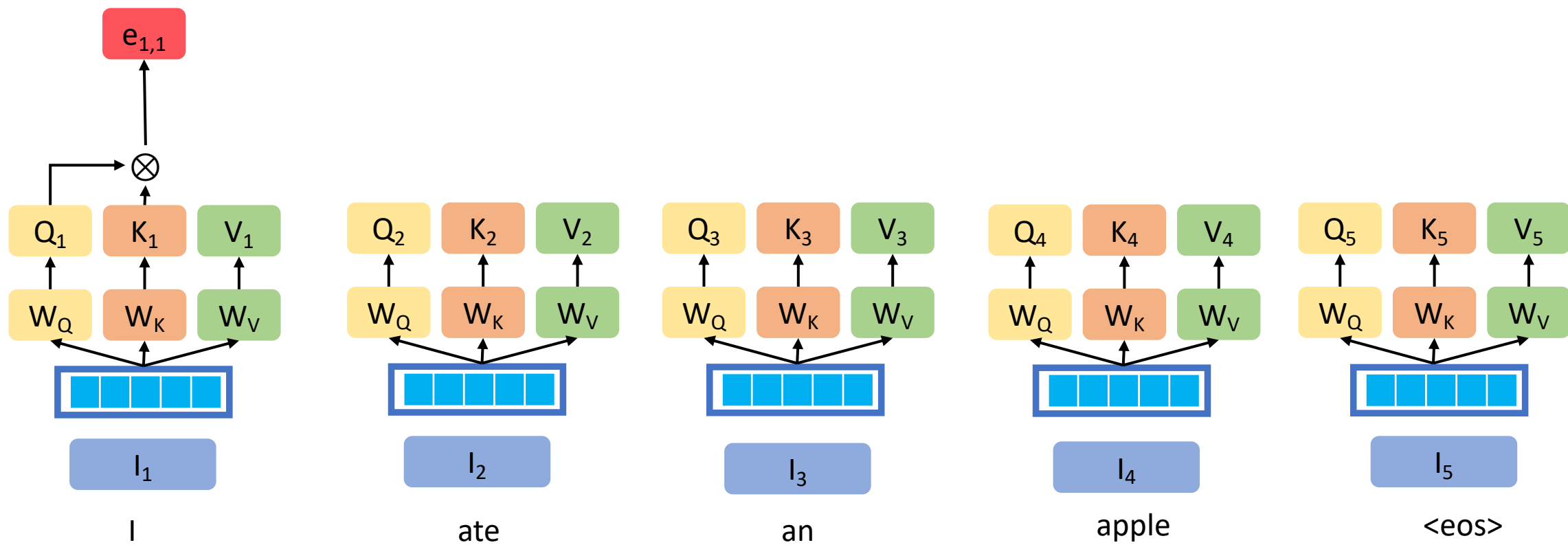
Dimensions across QKV have been dropped for brevity

Attention



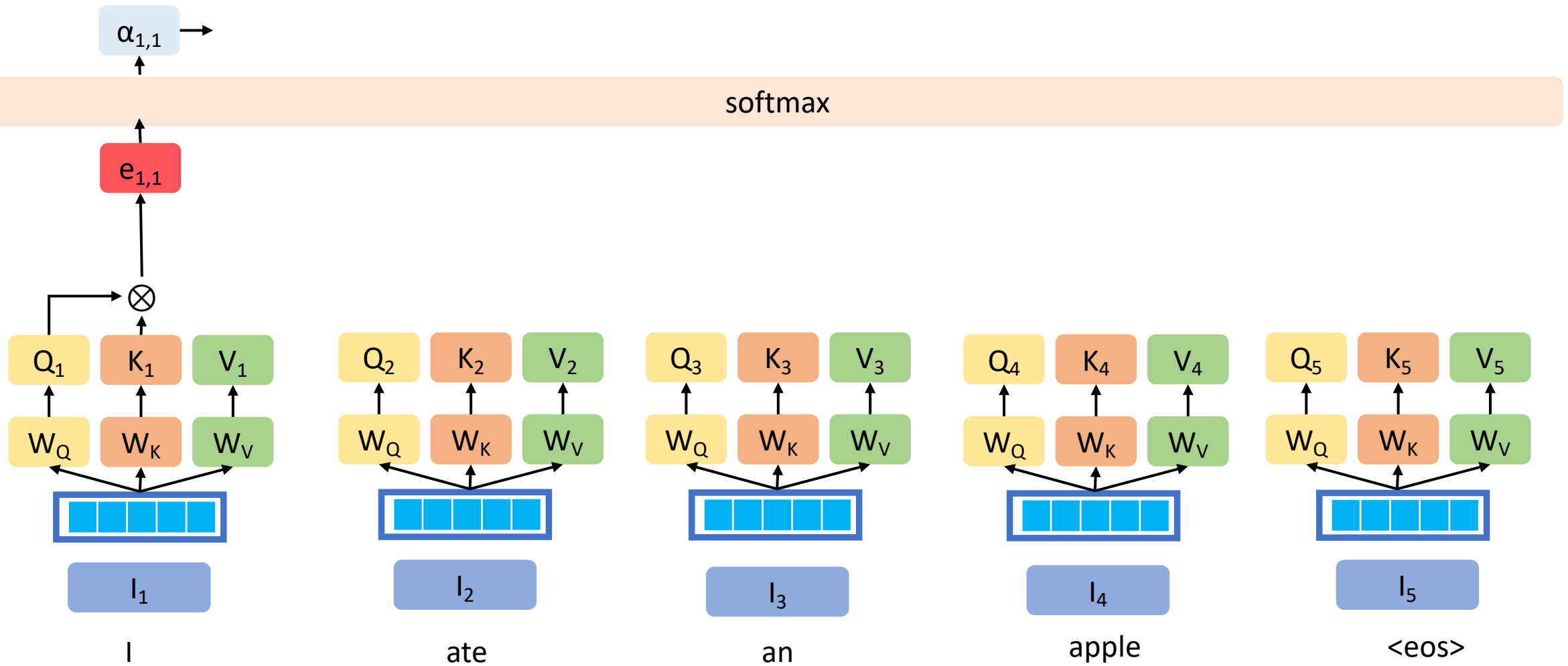
Dimensions across QKV have been dropped for brevity

Attention



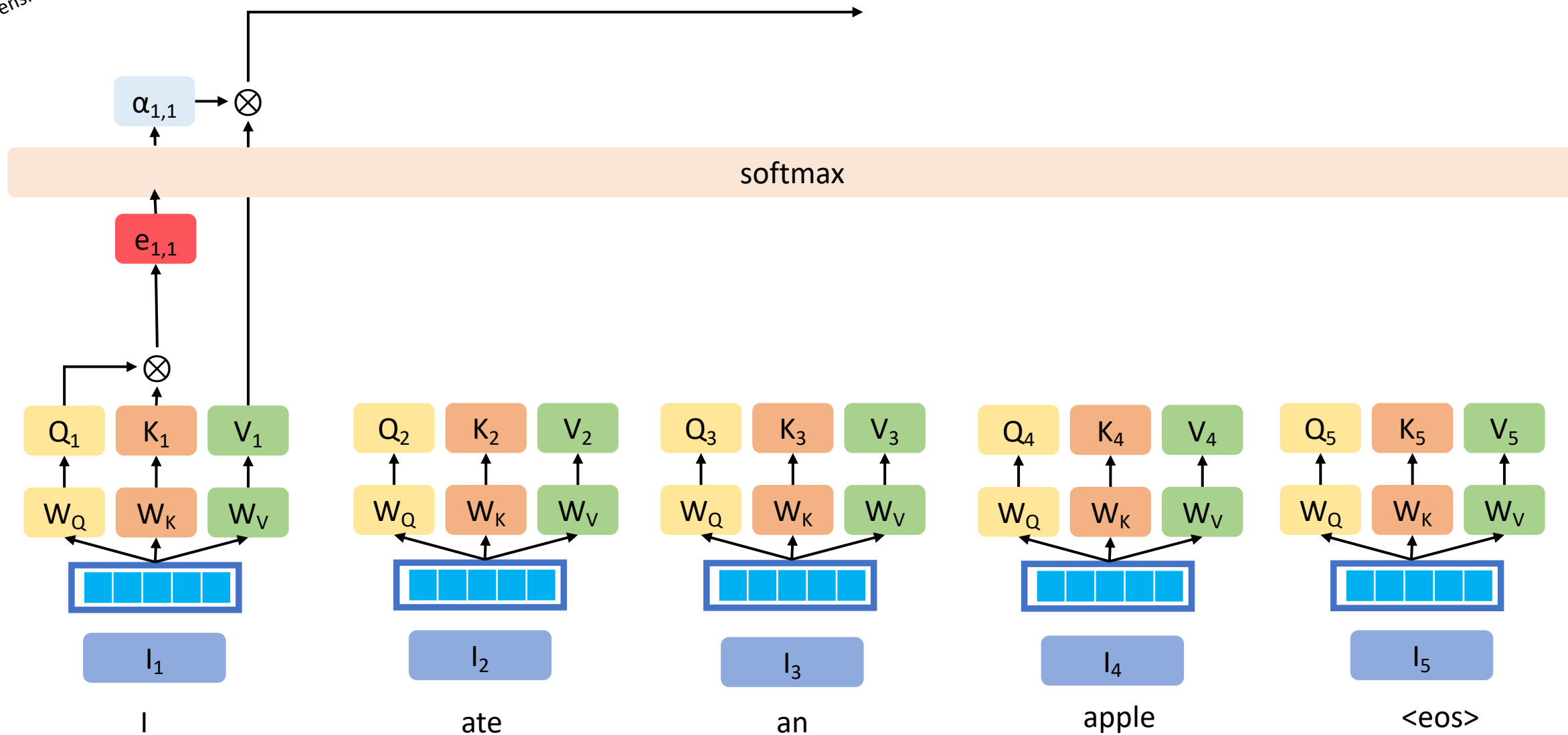
Dimensions across QKV have been dropped for brevity

Attention



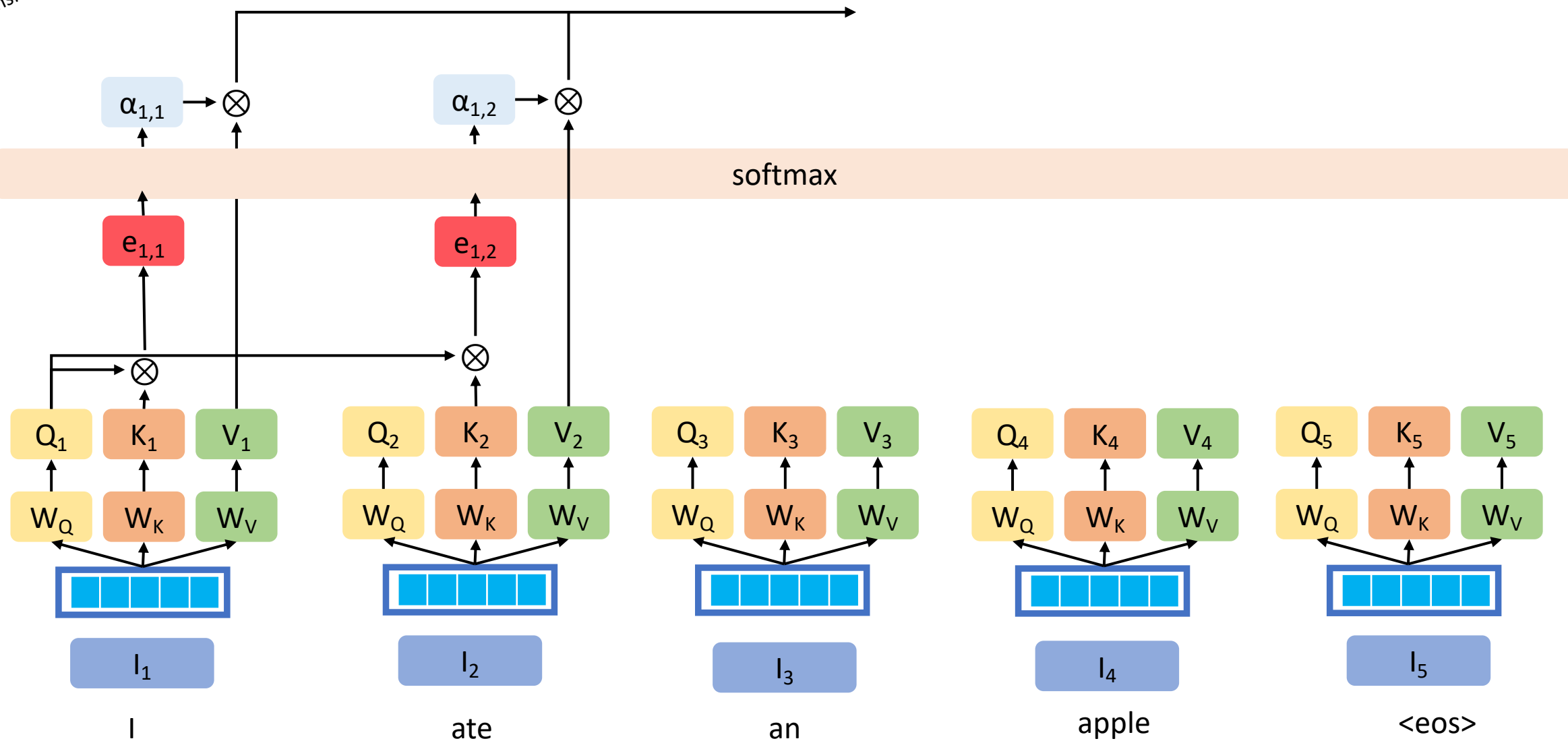
Dimensions across QKV have been dropped for brevity

Attention



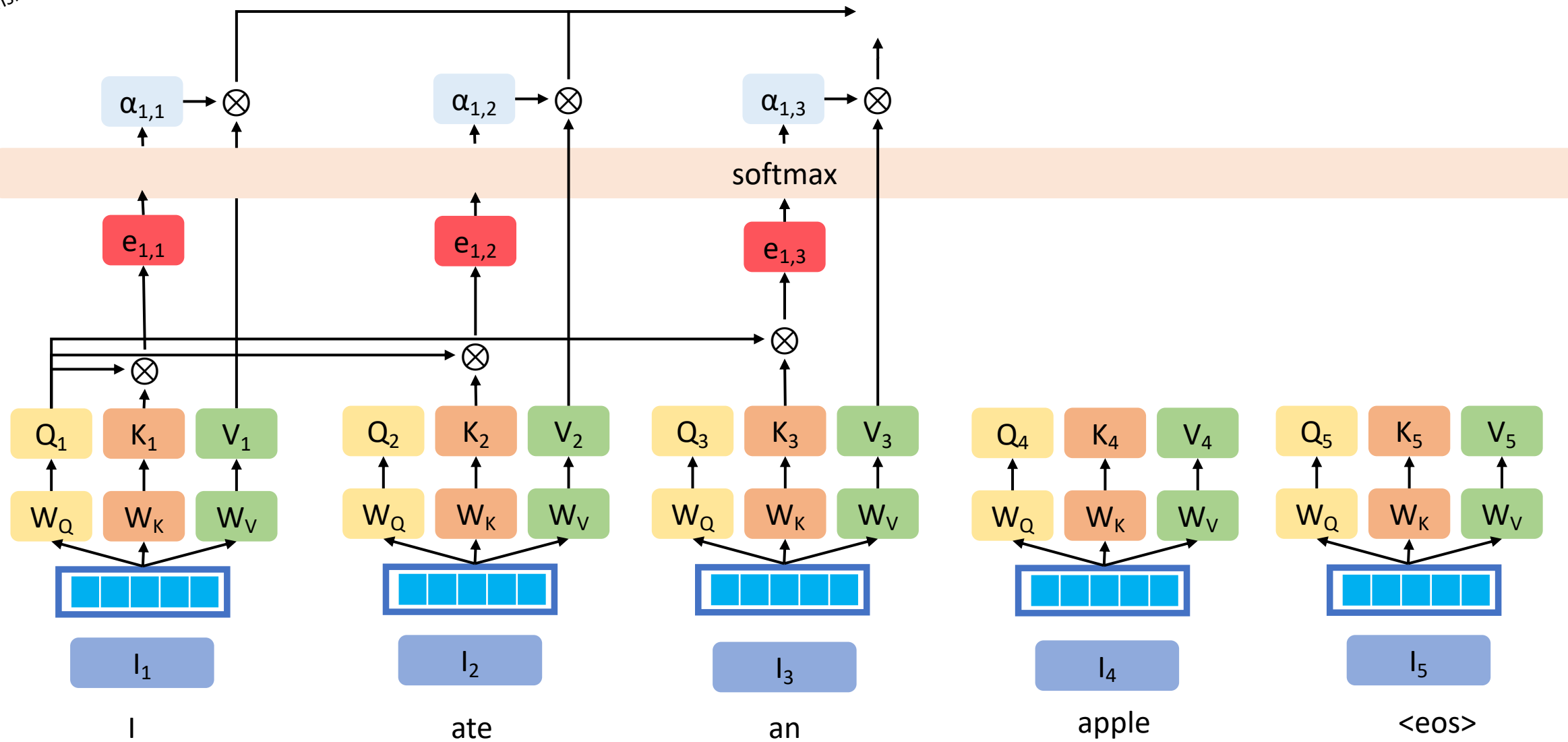
Dimensions across QKV have been dropped for brevity

Attention



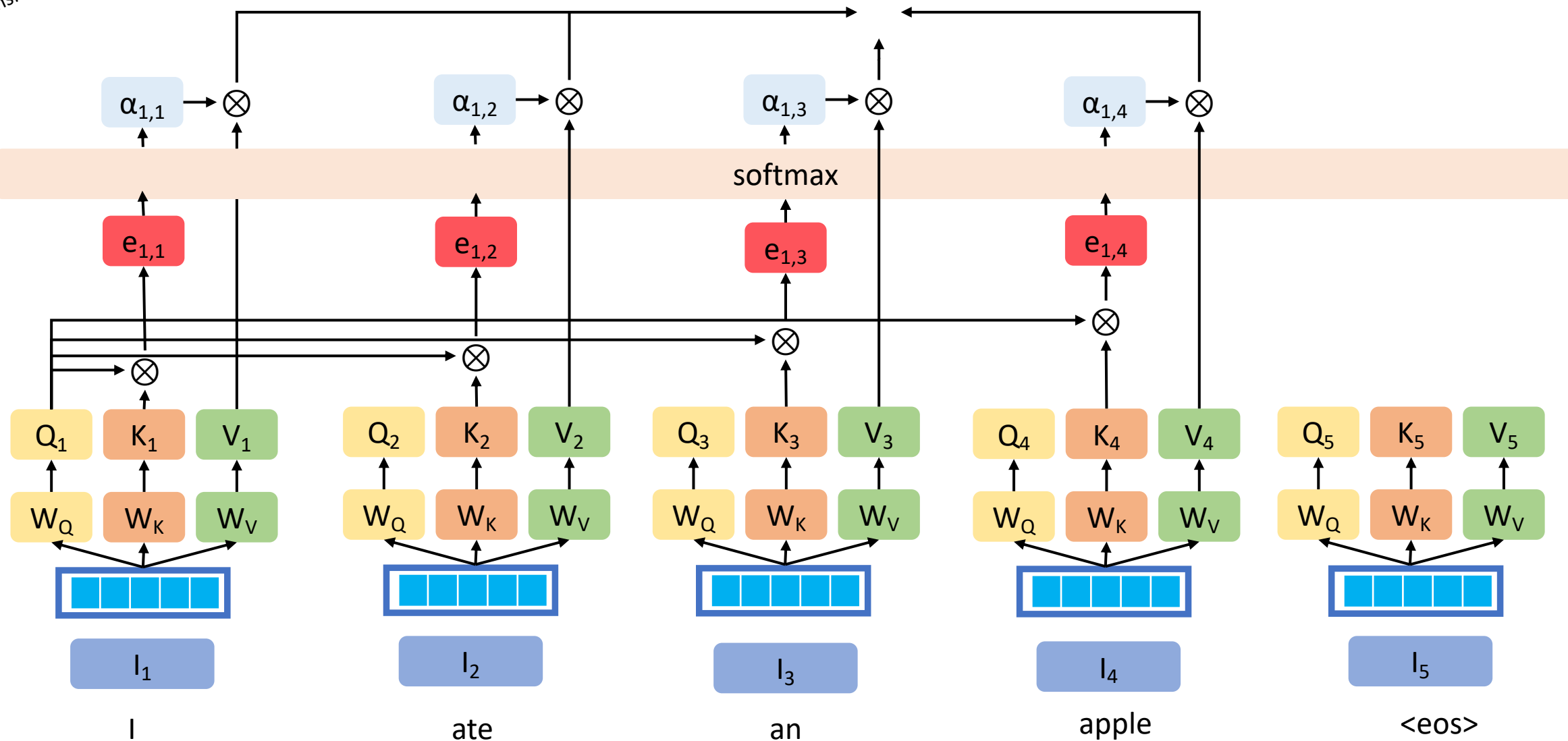
Dimensions across QKV have been dropped for brevity

Attention



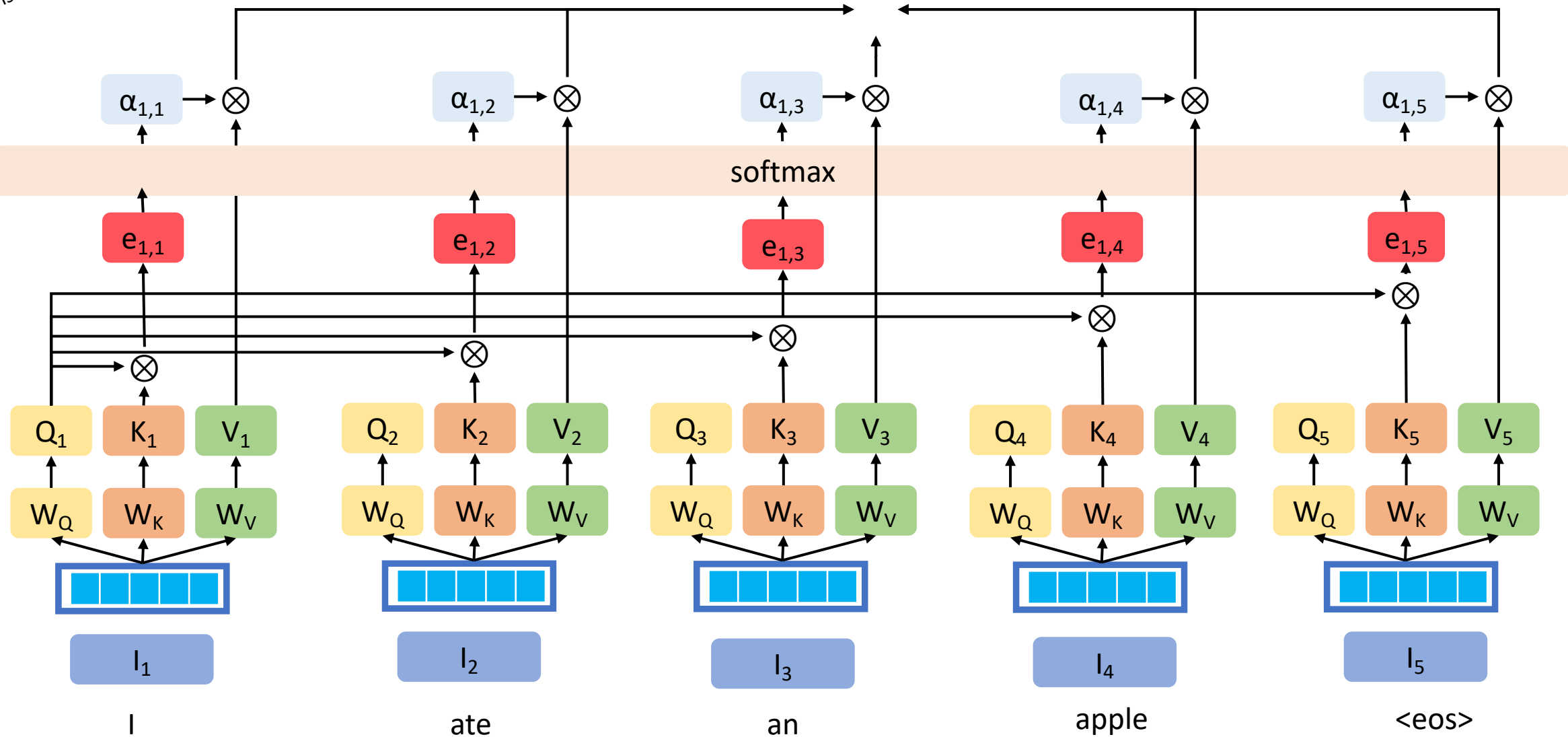
Dimensions across QKV have been dropped for brevity

Attention



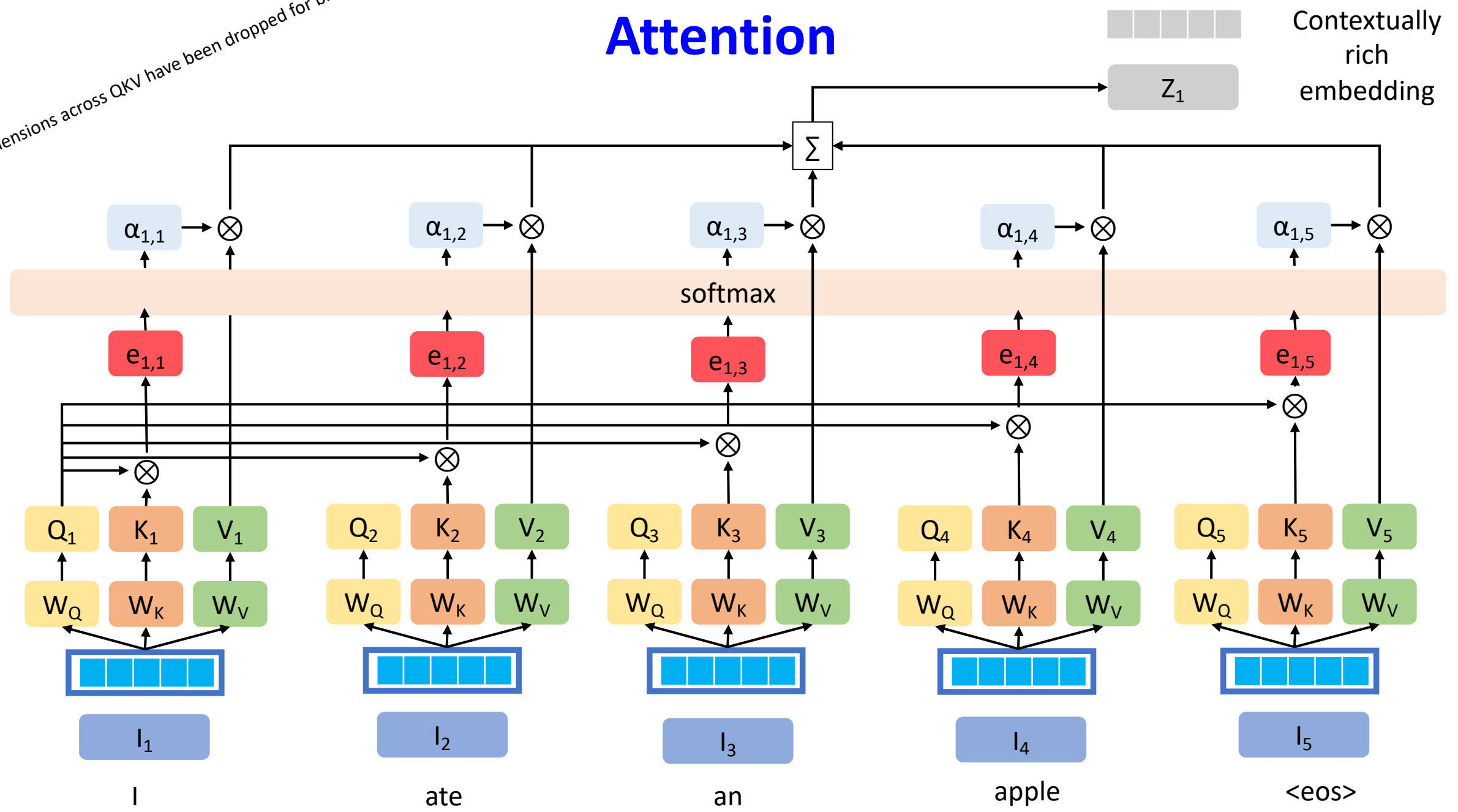
Dimensions across QKV have been dropped for brevity

Attention



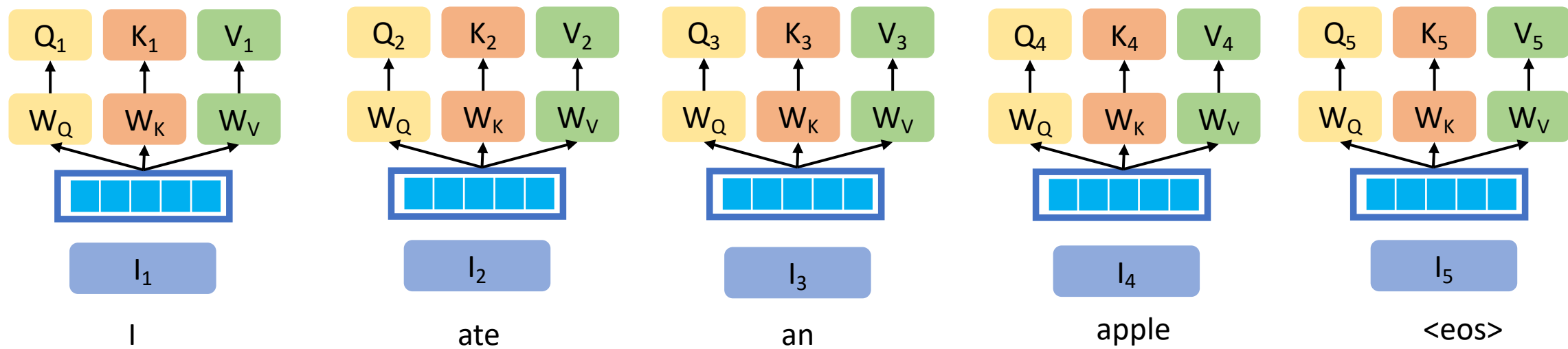
Dimensions across QKV have been dropped for brevity

Attention



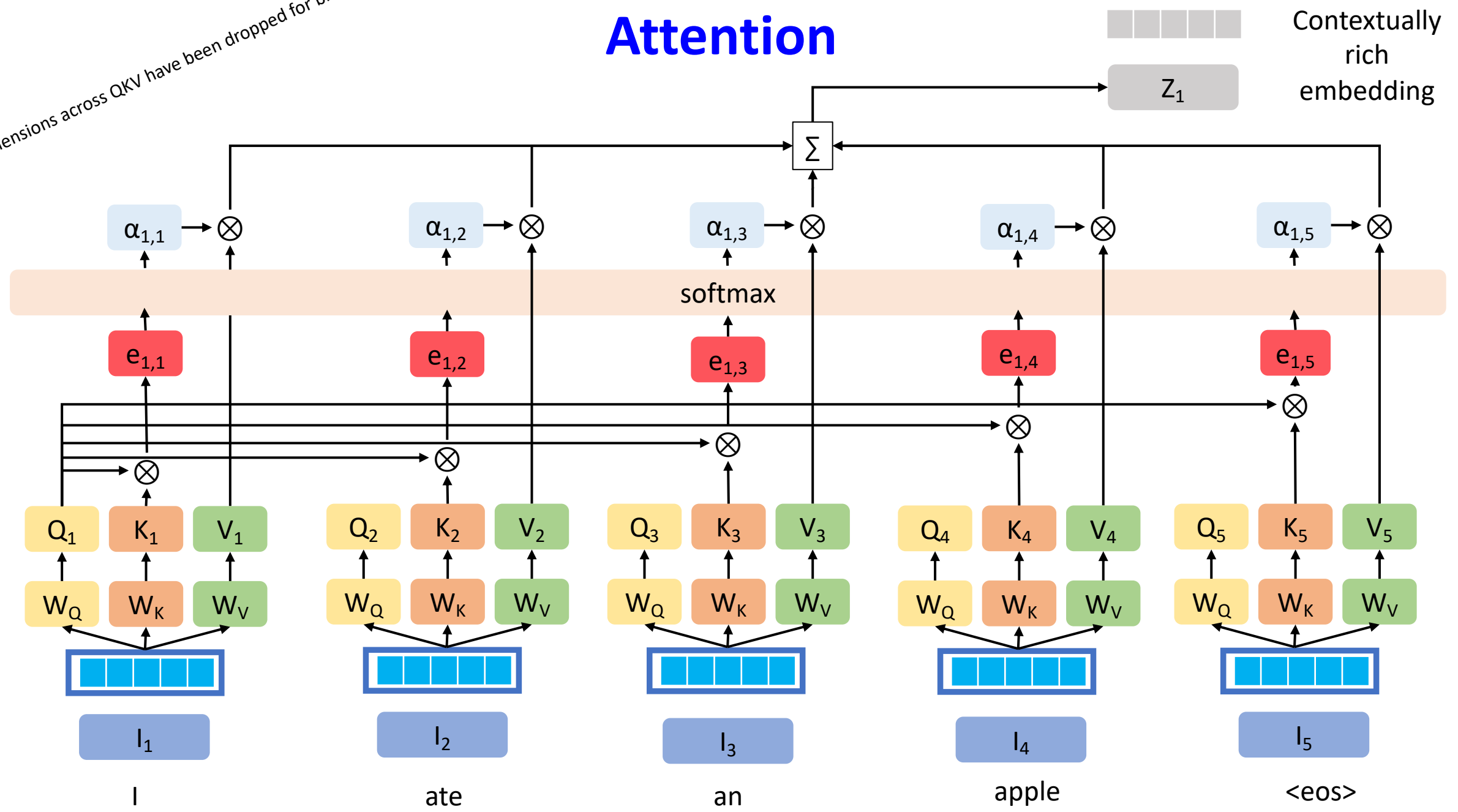
Dimensions across QKV have been dropped for brevity

Attention



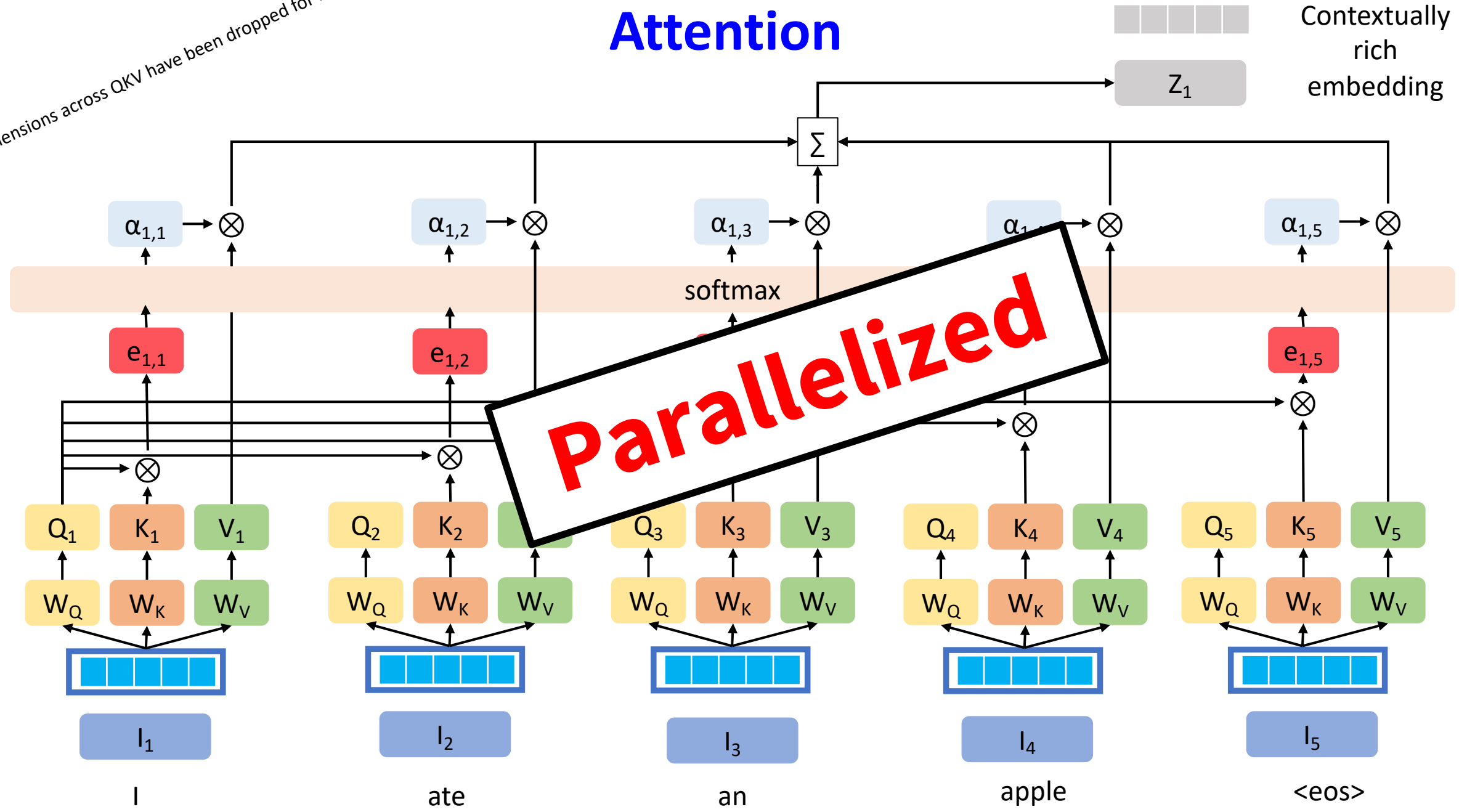
Dimensions across QKV have been dropped for brevity

Attention



Dimensions across QKV have been dropped for brevity

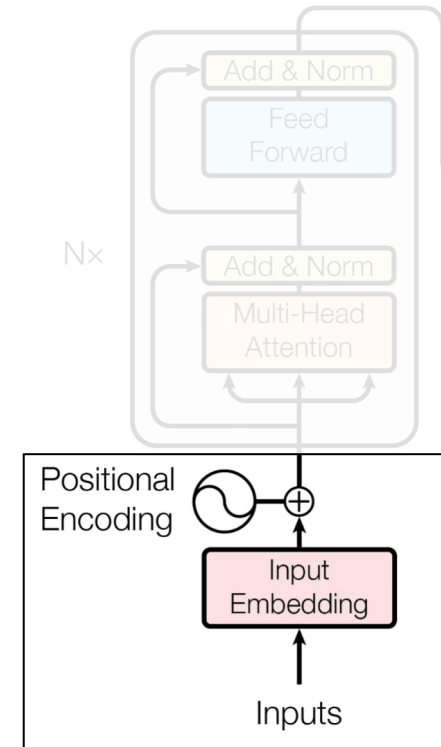
Attention



Poll 1 @1296

Which of the following are true about attention? (Select all that apply)

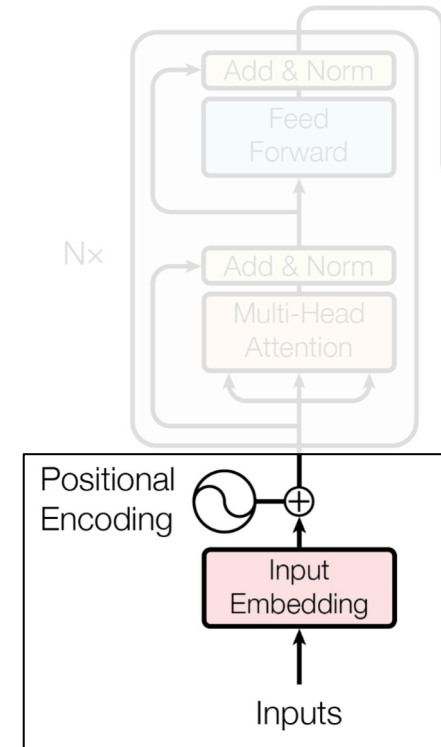
- a. To calculate attention weights for input I_2 , you would use key k_2 , and all queries
- b. To calculate attention weights for input I_2 , you would use query q_2 , and all keys
- c. We scale the QK^T product to bring attention weights in the range of $[0,1]$
- d. We scale the QK^T product to allow for numerical stability



Poll 1 @1296

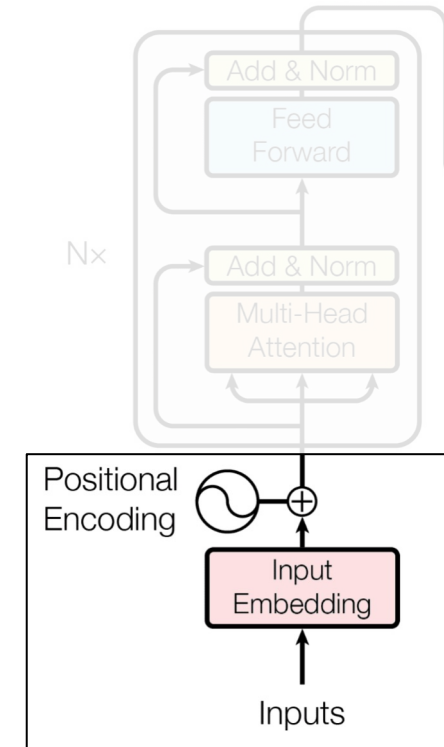
Which of the following are true about attention? (Select all that apply)

- a. To calculate attention weights for input I_2 , you would use key k_2 , and all queries
- b. To calculate attention weights for input I_2 , you would use query q_2 , and all keys**
- c. We scale the QK^T product to bring attention weights in the range of $[0,1]$
- d. We scale the QK^T product to allow for numerical stability**

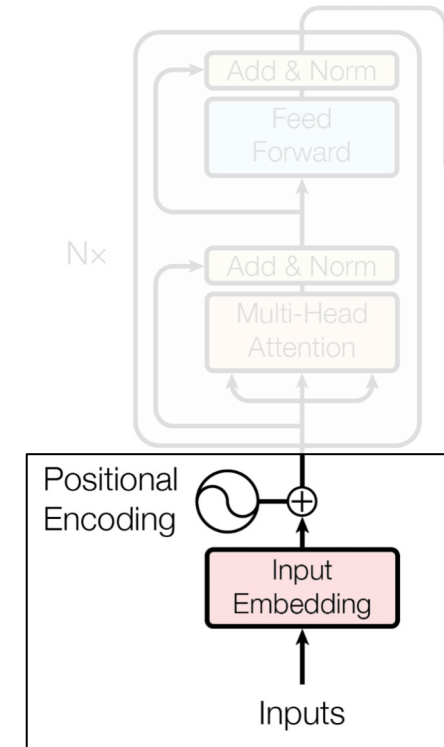
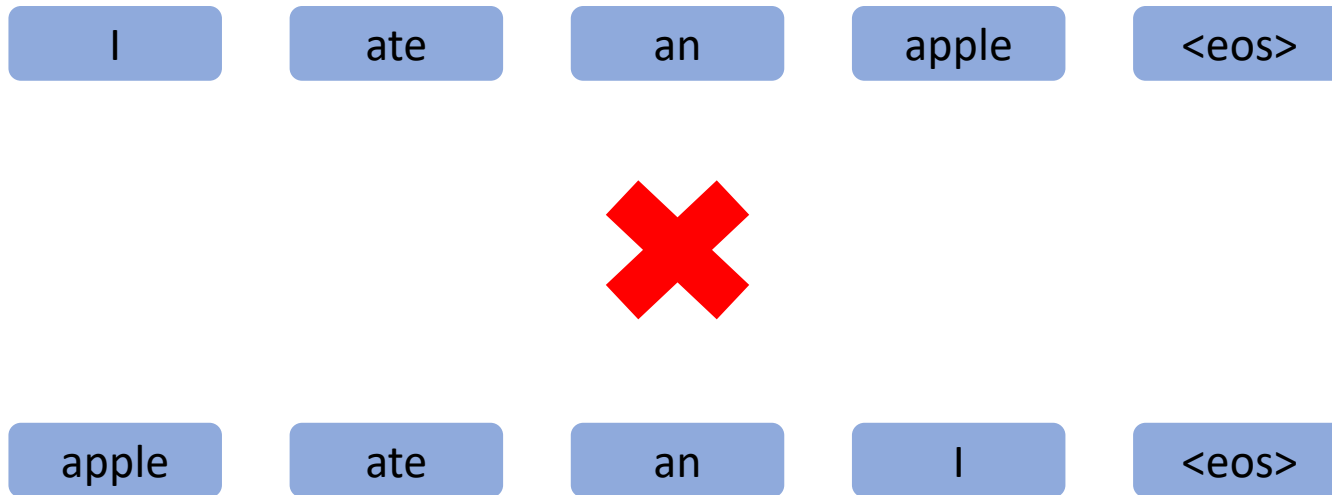


Positional Encoding

I ate an apple <eos>



Positional Encoding

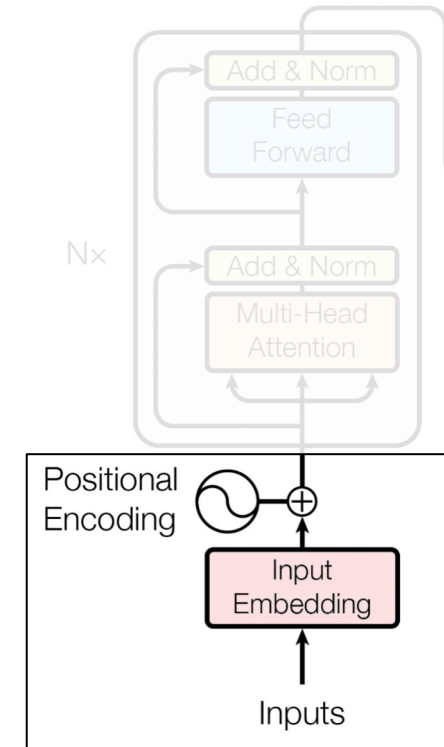


Positional Encoding

Positional Encoding

Requirements for Positional Encodings

- Some representation of time ? (like **seq2seq** ?)
- Should be unique for each position – not cyclic



Positional Encoding

Positional Encoding

Requirements for Positional Encodings

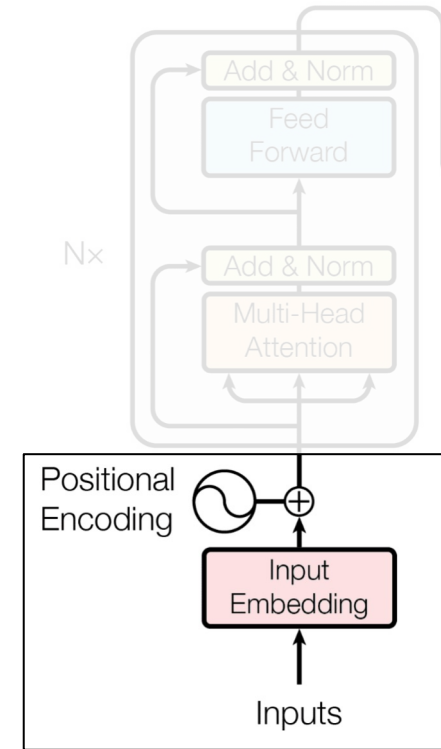
- Some representation of time ? (like **seq2seq** ?)
- Should be unique for each position – not cyclic

Possible Candidates :

$$P_{t+1} = P_t + \Delta c$$

$$P_{t+1} = e^{P_t \Delta c}$$

$$P_{t+1} = P_t^{t \Delta c}$$



Positional Encoding

Positional Encoding

Requirements for Positional Encodings

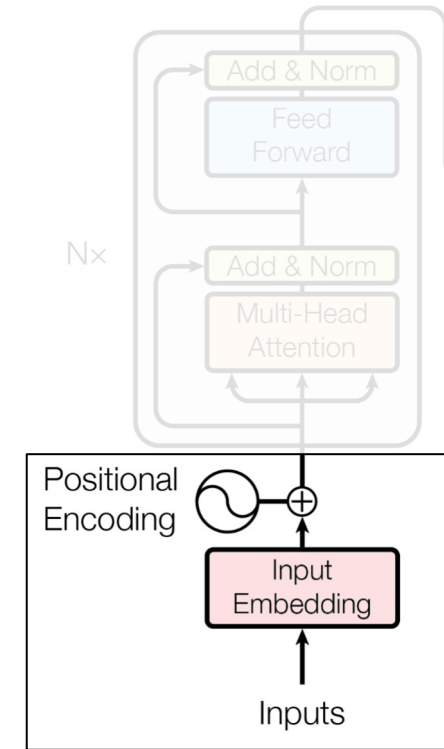
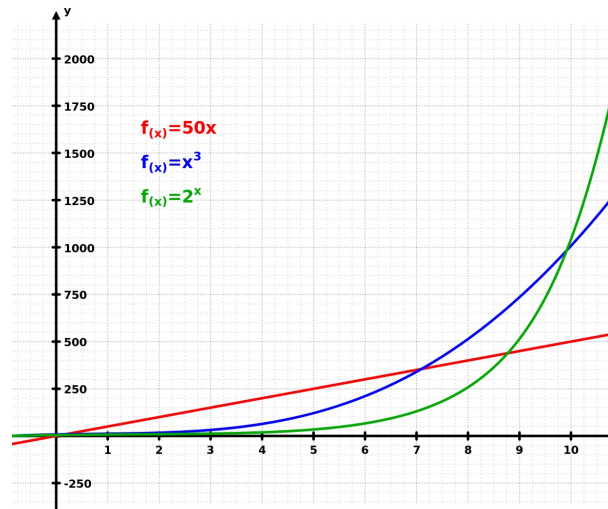
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Positional Encoding

Positional Encoding

Requirements for Positional Encodings

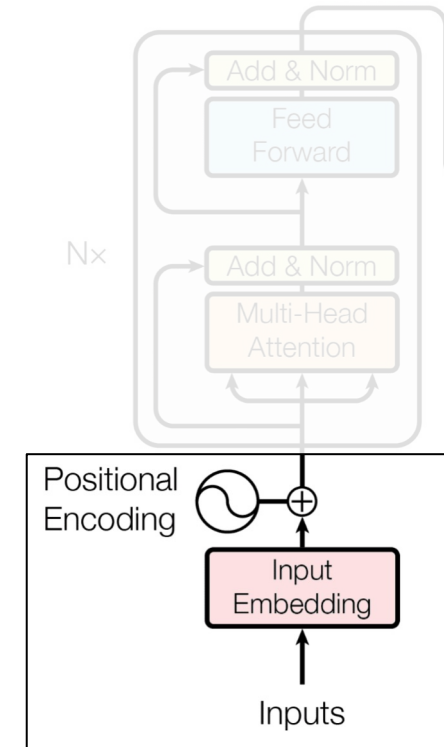
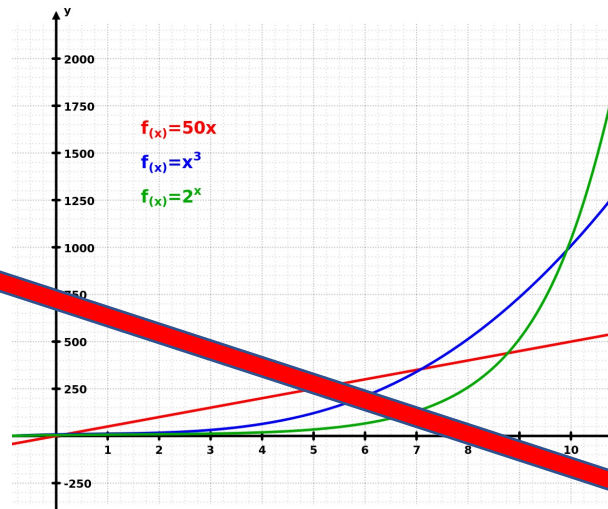
- Some representation of time ? (like **seq2seq** ?)
- Should be unique for each position – not cyclic
- **Bounded**

Possible Candidates :

$$P_{t+1} = P_t + \Delta c$$

$$P_{t+1} = e^{P_t \Delta c}$$

$$P_{t+1} = P_t^{t \Delta c}$$



Positional Encoding

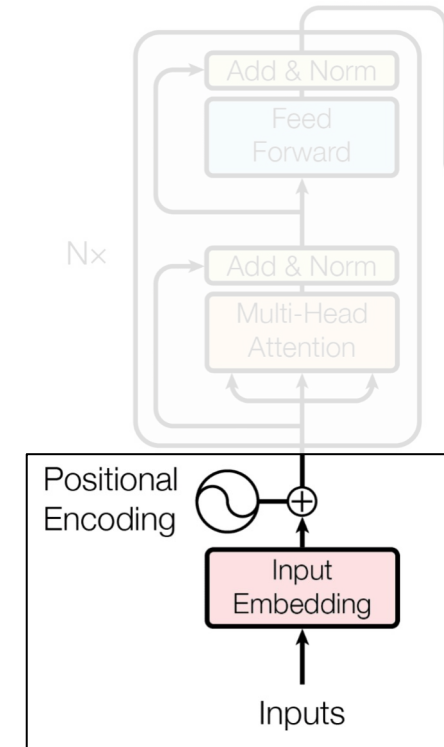
Positional Encoding

Requirements for Positional Encodings

- Some representation of time ? (like **seq2seq** ?)
- Should be unique for each position – not cyclic
- **Bounded**

Possible Candidates :

$$P(t + t') = M^{t'} \times P(t)$$



Positional Encoding

Positional Encoding

Requirements for Positional Encodings

- Some representation of time ? (like **seq2seq** ?)
- Should be unique for each position – not cyclic
- **Bounded**

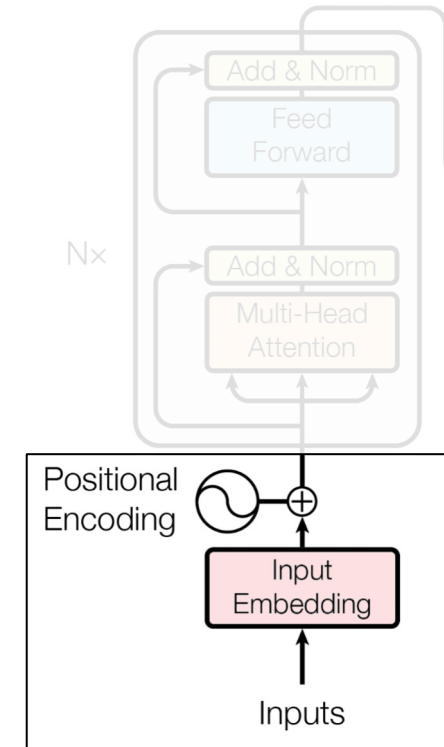
Possible Candidates :

$$P(t + t') = M^{t'} \times P(t)$$

M ?

1. Should be a unitary matrix
2. Magnitudes of eigen value should be 1 -> norm preserving

Positional Encoding



Positional Encoding

Requirements for Positional Encodings

- Some representation of time ? (like **seq2seq** ?)
- Should be unique for each position – not cyclic
- **Bounded**

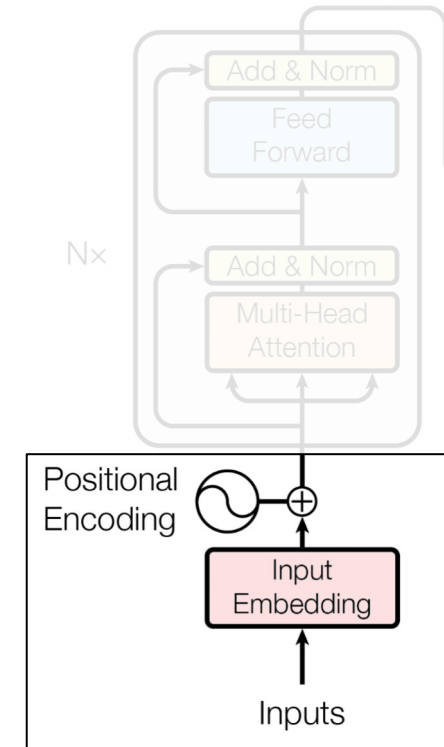
Possible Candidates :

$$P(t + t') = M^{t'} \times P(t)$$

M

1. The matrix can be learnt
2. Produces unique rotated embeddings each time

Positional Encoding



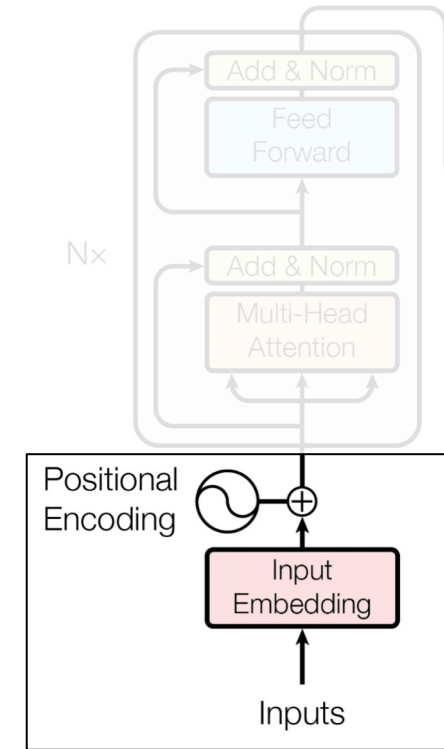
Rotary Positional Embedding

RoFORMER: ENHANCED TRANSFORMER WITH ROTARY POSITION EMBEDDING

$$f_{\{q,k\}}(\mathbf{x}_m, m) = \begin{pmatrix} \cos m\theta & -\sin m\theta \\ \sin m\theta & \cos m\theta \end{pmatrix} \begin{pmatrix} W_{\{q,k\}}^{(11)} & W_{\{q,k\}}^{(12)} \\ W_{\{q,k\}}^{(21)} & W_{\{q,k\}}^{(22)} \end{pmatrix} \begin{pmatrix} x_m^{(1)} \\ x_m^{(2)} \end{pmatrix}$$

Table 2: Comparing RoFormer and BERT by fine tuning on downstream GLEU tasks.

Model	MRPC	SST-2	QNLI	STS-B	QQP	MNLI(m/mm)
BERTDevlin et al. [2019]	88.9	93.5	90.5	85.8	71.2	84.6/83.4
RoFormer	89.5	90.7	88.0	87.0	86.4	80.2/79.8



[REF: Rotary Positional Embeddings](#) 

Positional Encoding

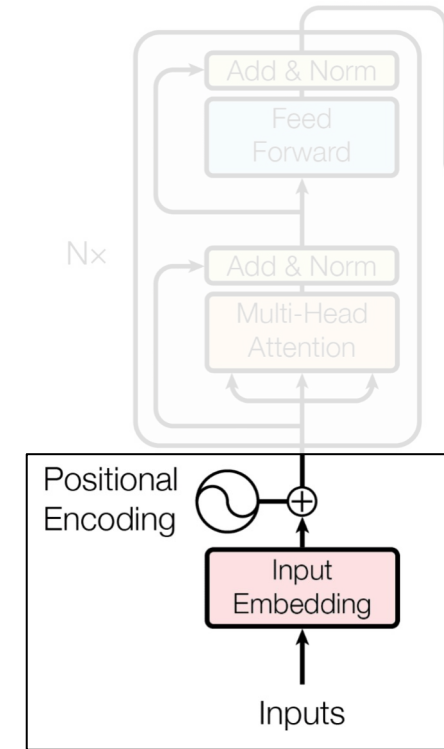
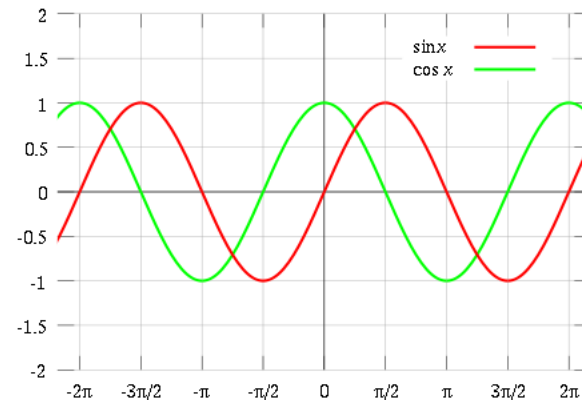
Requirements for Positional Encodings

- Some representation of time ? (like **seq2seq** ?)
- Should be unique for each position – **not cyclic**
- Bounded

Actual Candidates :

$\sin(g(t))$

$\cosine(g(t))$



Positional Encoding

Positional Encoding

Requirements for $g(t)$

- *Must have same dimensions as input embeddings*
- *Must produce overall unique encodings*

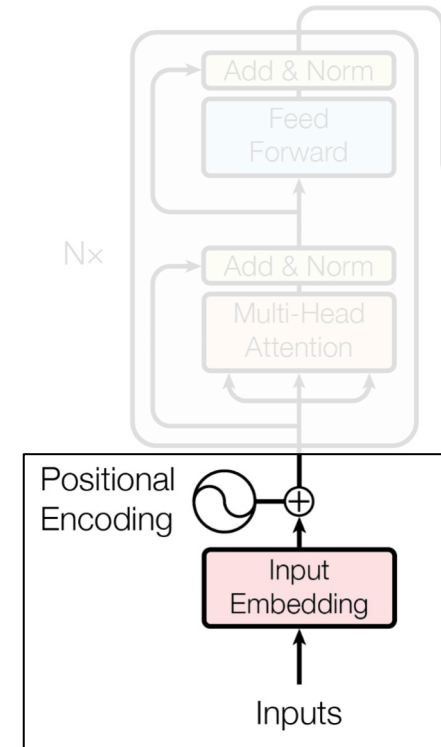
pos -> idx of the token in input sentence

i -> i^{th} dimension out of d

$$PE_{(pos, 2i)} = \sin(pos/10000^{2i/d_{model}})$$

$$PE_{(pos, 2i+1)} = \cos(pos/10000^{2i/d_{model}})$$

Positional Encoding



Positional Encoding

$$PE_{(pos, 2i)} = \sin(pos/10000^{2i/d_{model}})$$

$$PE_{(pos, 2i+1)} = \cos(pos/10000^{2i/d_{model}})$$

Requirements for $g(t)$

- Must have same dimensions as input embeddings
- Must produce overall unique encodings

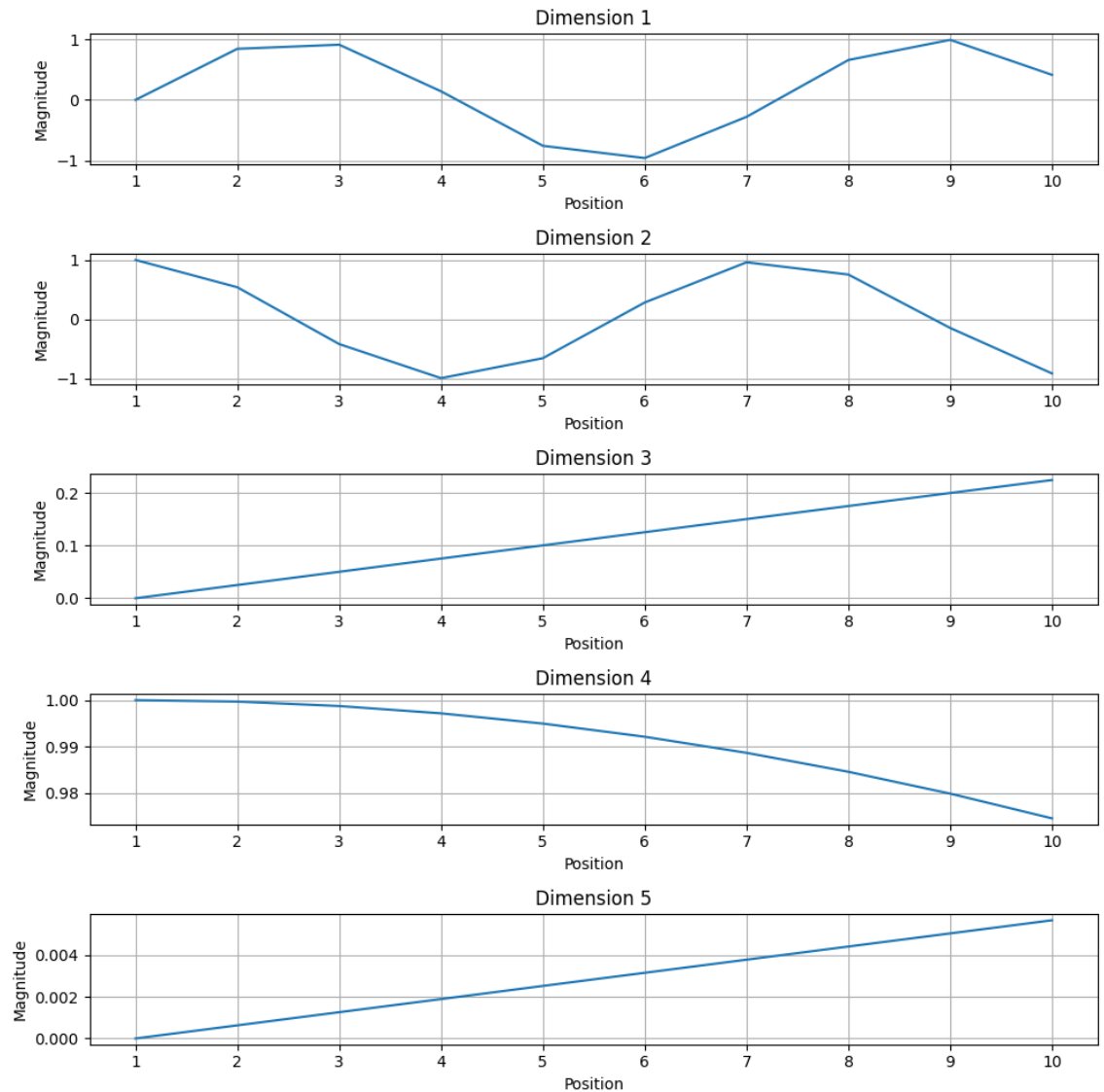
pos -> idx of the token in input sentence

i -> i^{th} dimension out of d

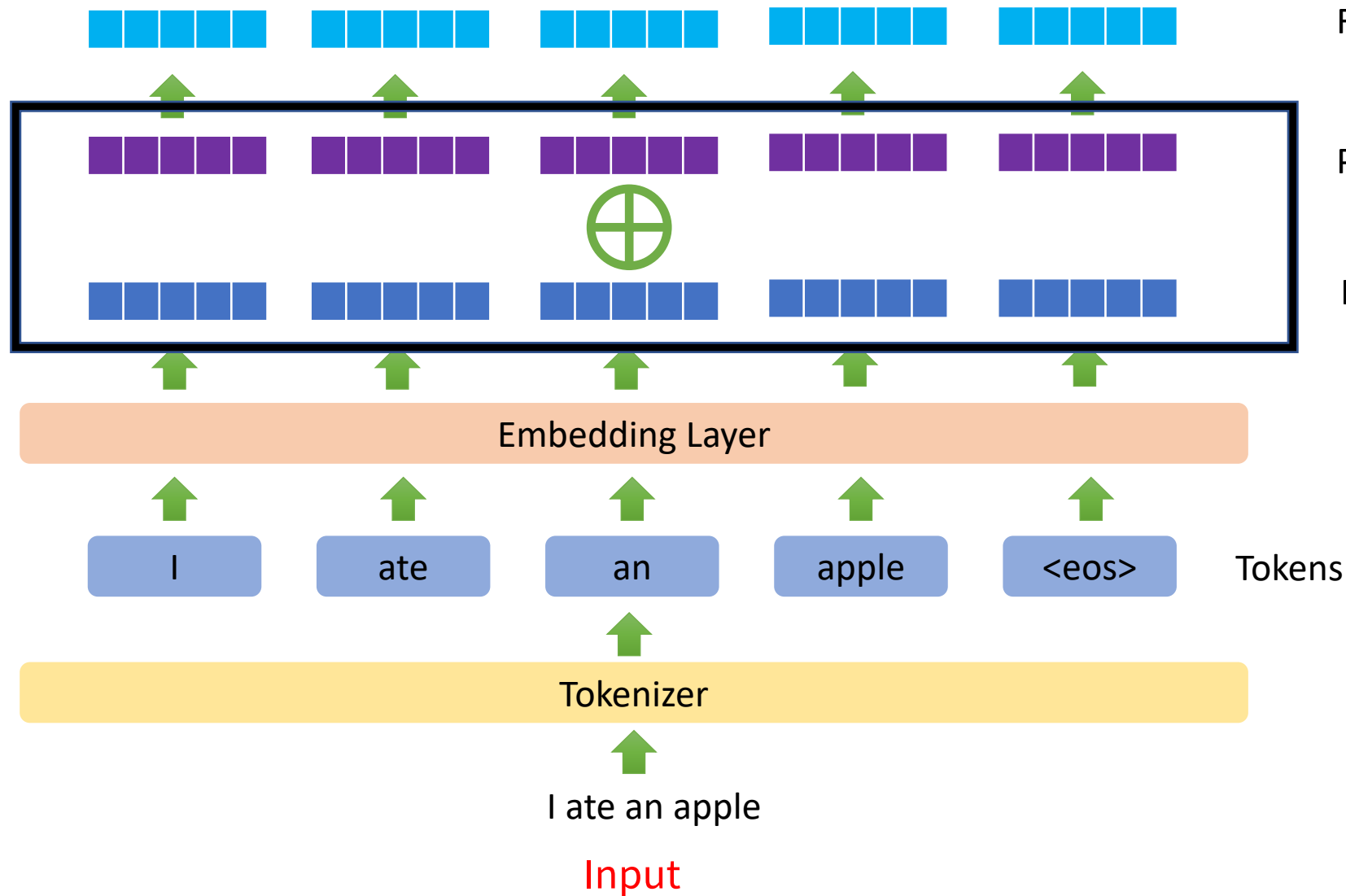


Positional Encoding:

	0	1	2	3	4
Dim 1	0.000	0.841	0.909	0.141	-0.757
Dim 2	1.000	0.540	-0.416	-0.990	-0.654
Dim 3	0.000	0.025	0.050	0.075	0.100
Dim 4	1.000	1.000	0.999	0.997	0.995
Dim 5	0.000	0.001	0.001	0.002	0.003



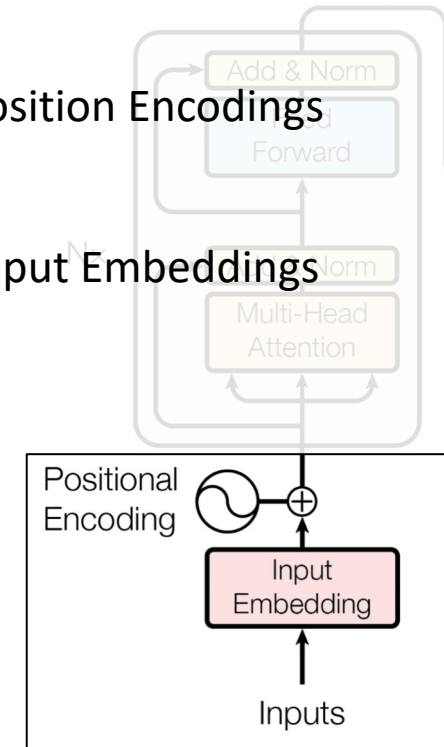
Positional Encoding



Final Input Embeddings

Position Encodings

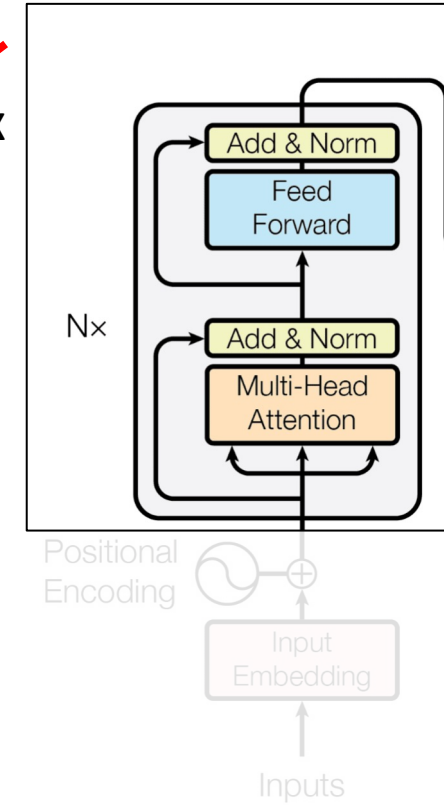
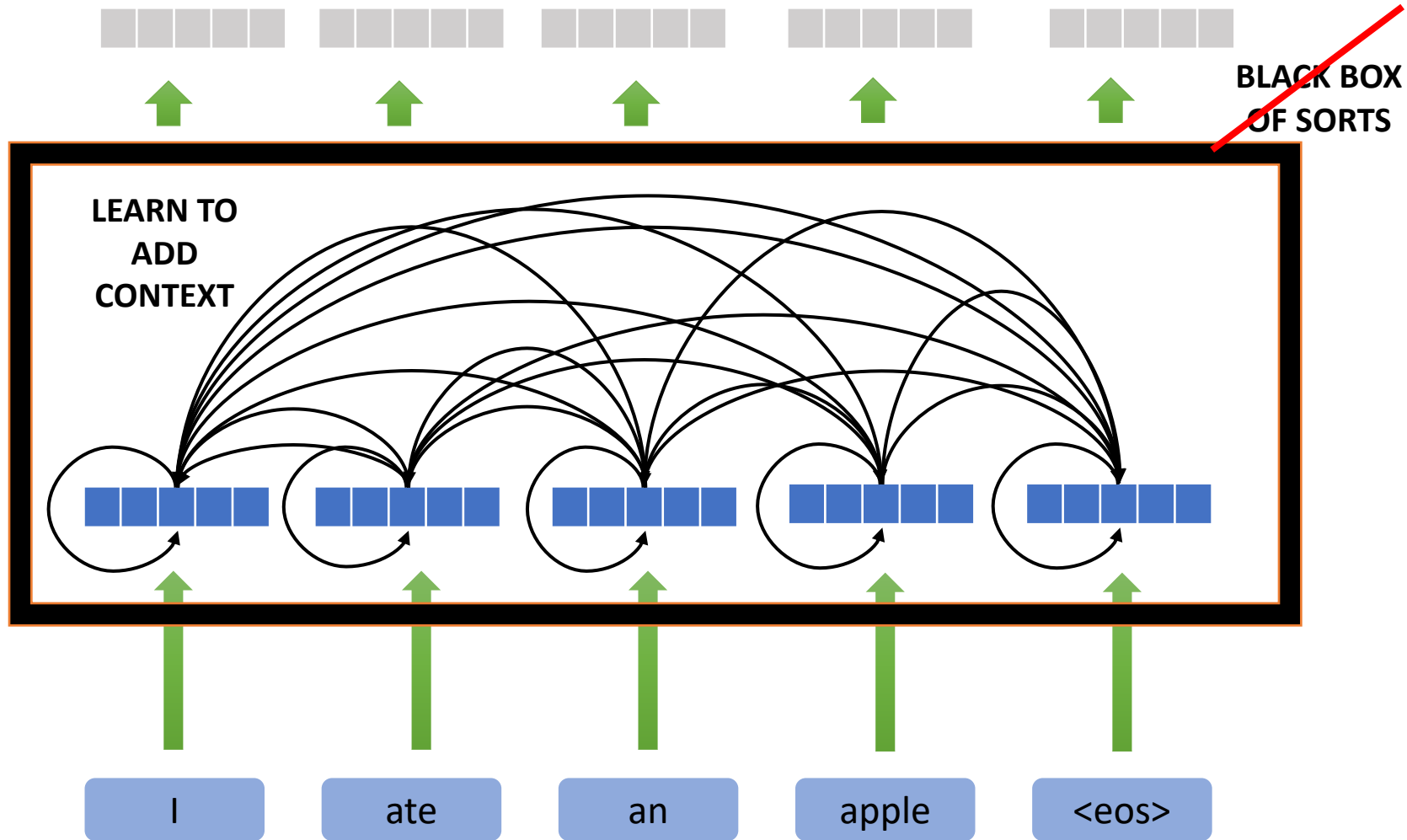
Input Embeddings



Encoder

 $\alpha_{[ij]}$ Σ

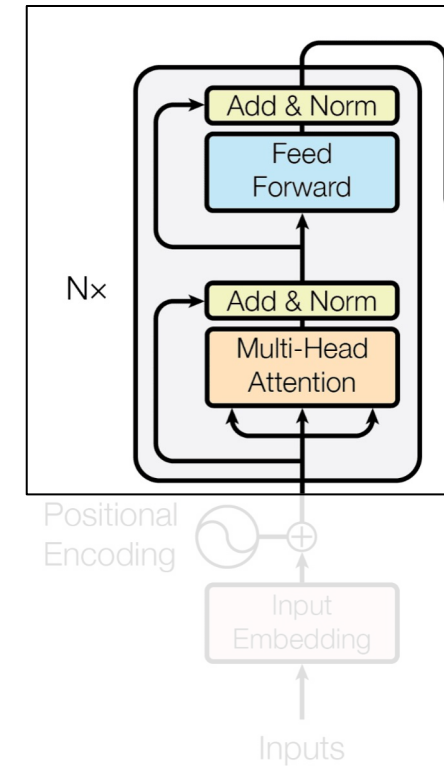
CONTEXTUALLY RICH EMBEDDINGS



Self Attention

From lecture 18:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$



Self Attention

The

animal

didn't

cross

the

street

because

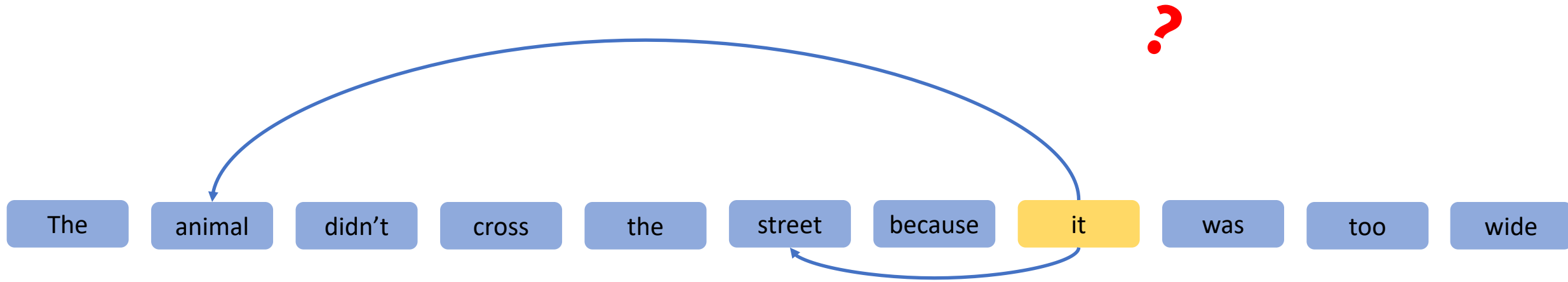
it

was

too

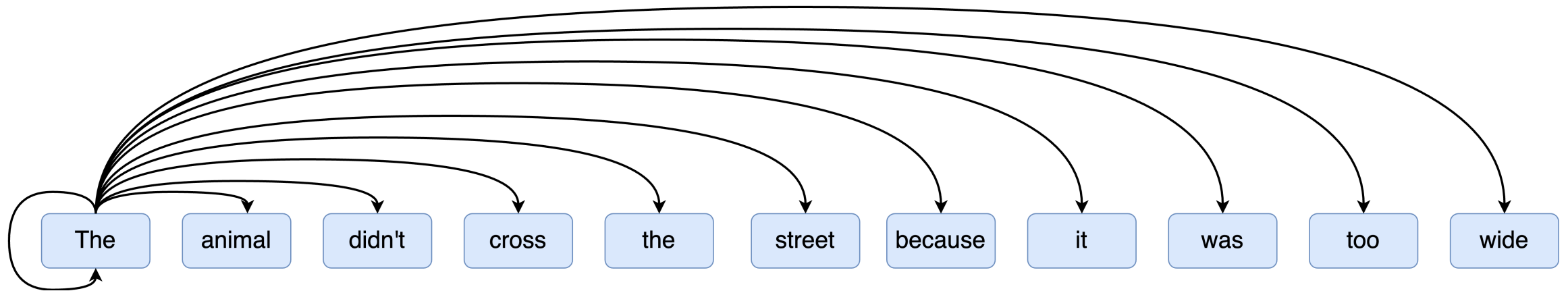
wide

Self Attention

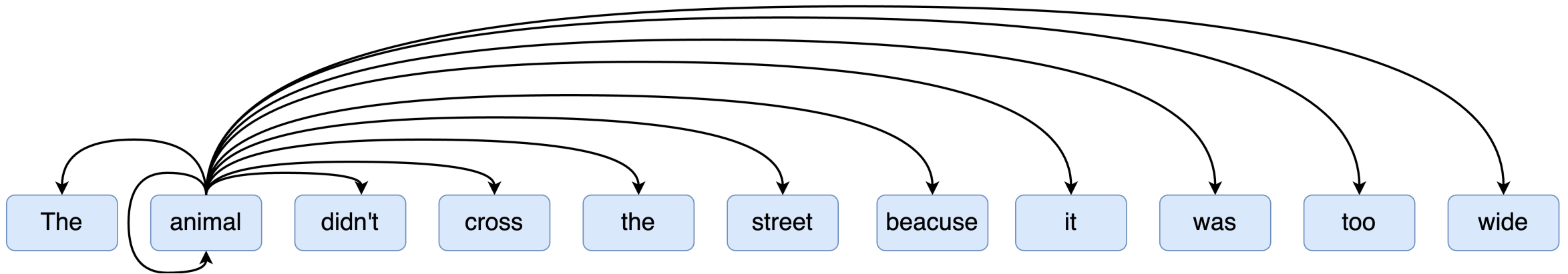


coreference resolution ?

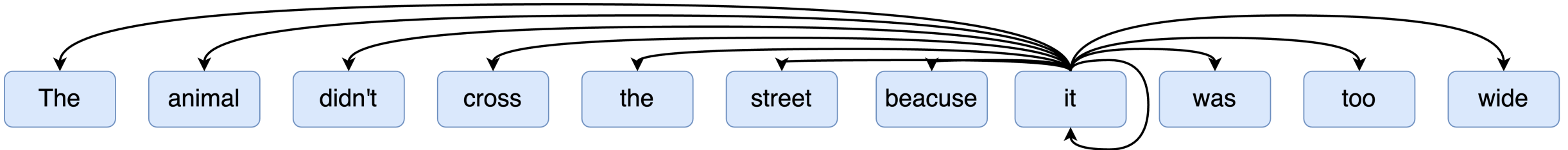
Self Attention



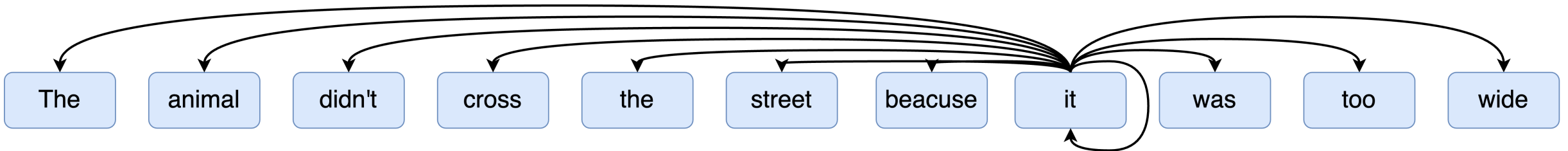
Self Attention



Self Attention



Self Attention



SELF

Query Inputs

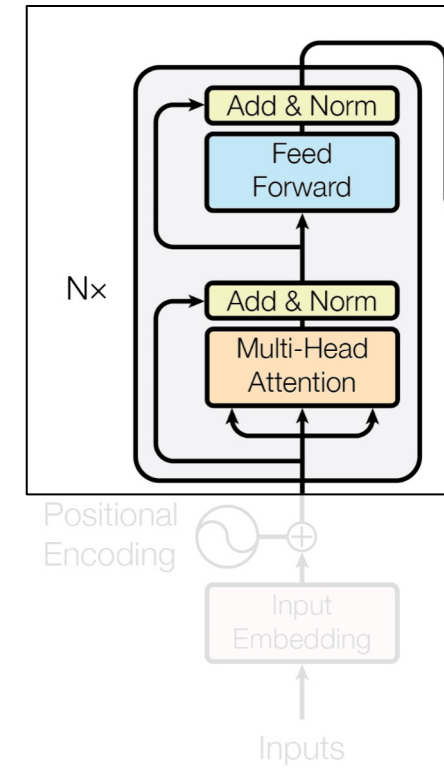
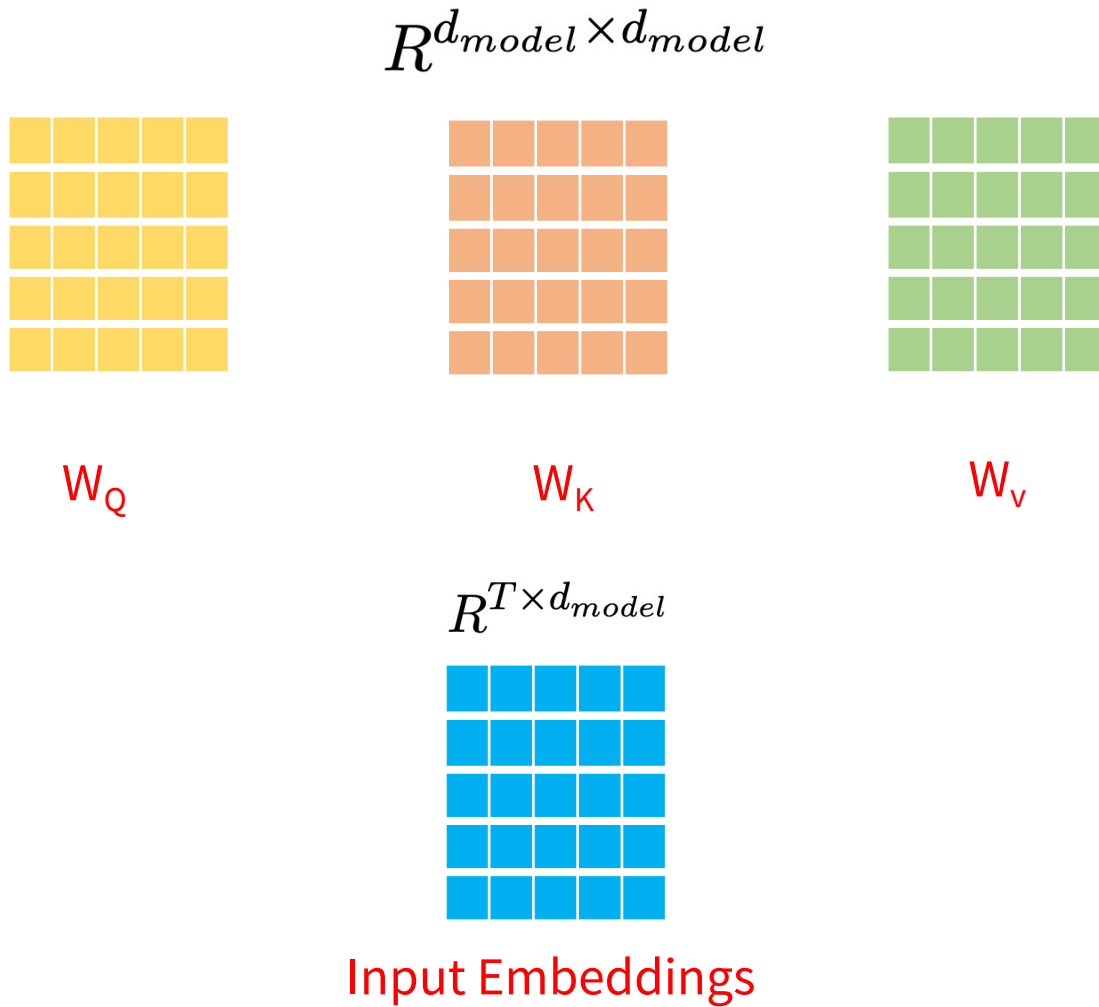
=

Key Inputs

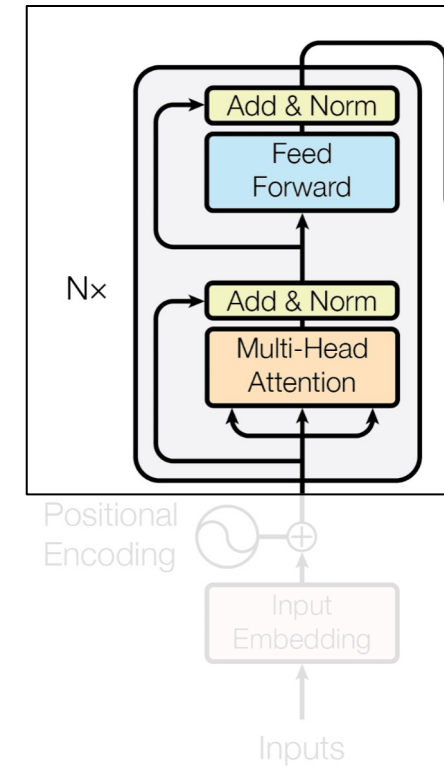
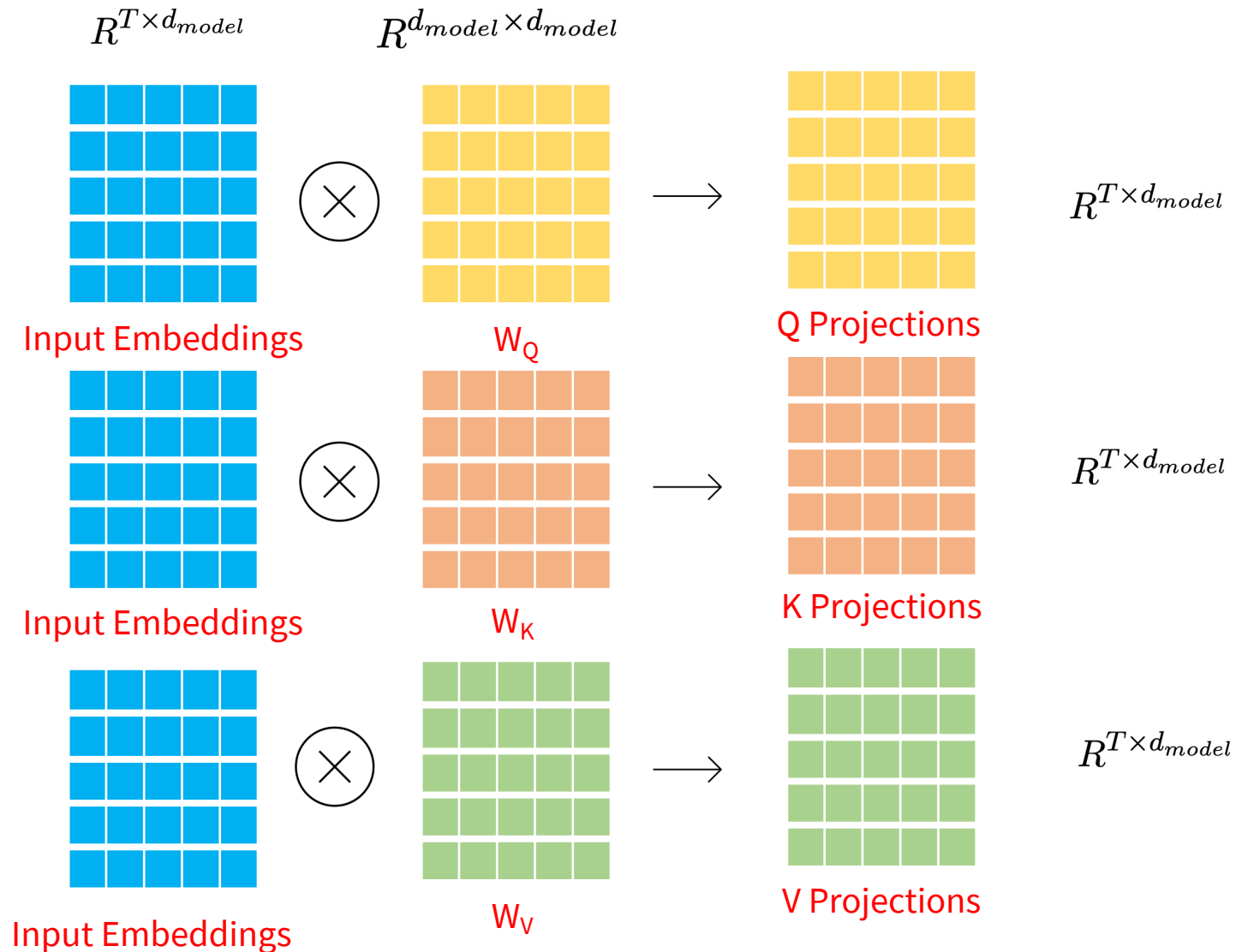
=

Value Inputs

Self Attention

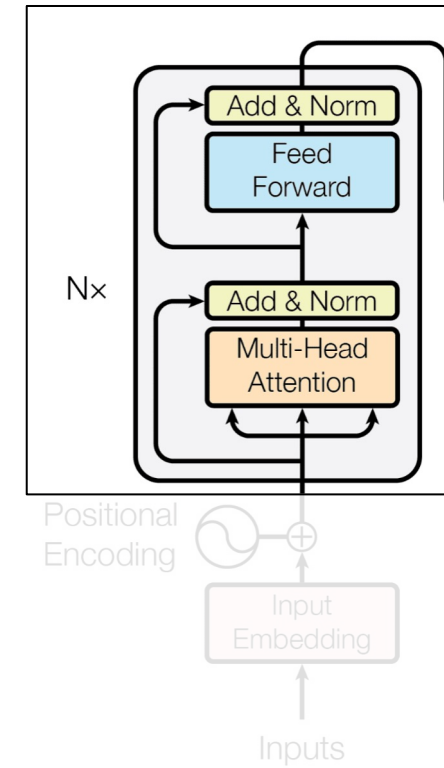
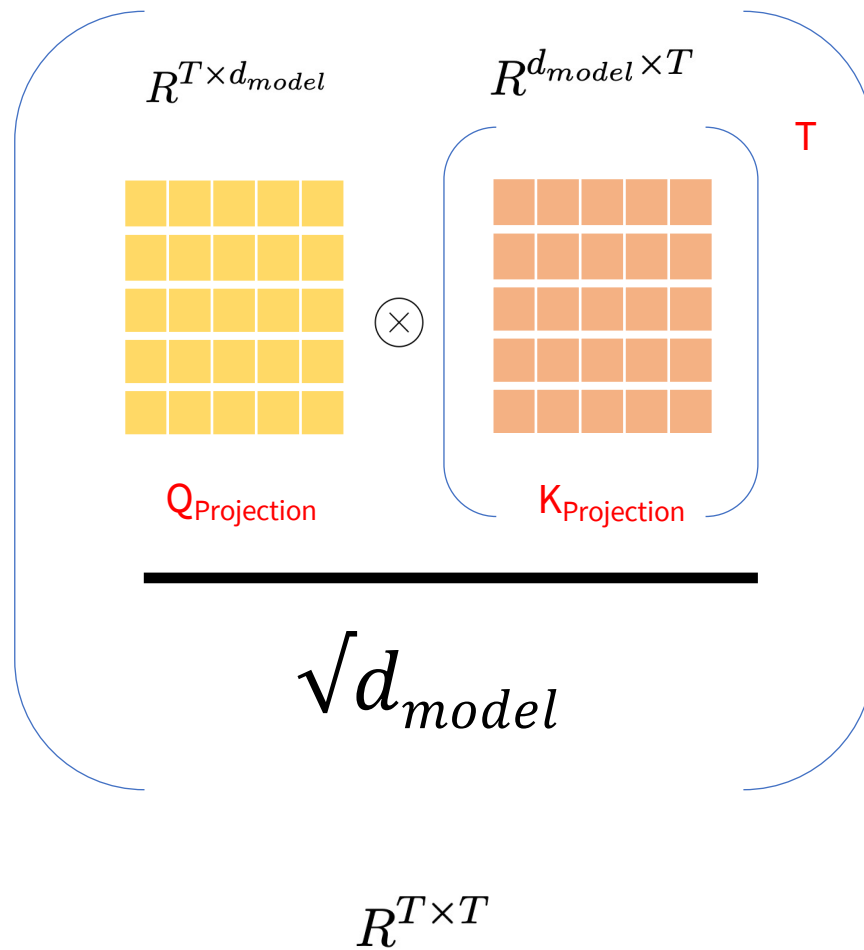


Self Attention

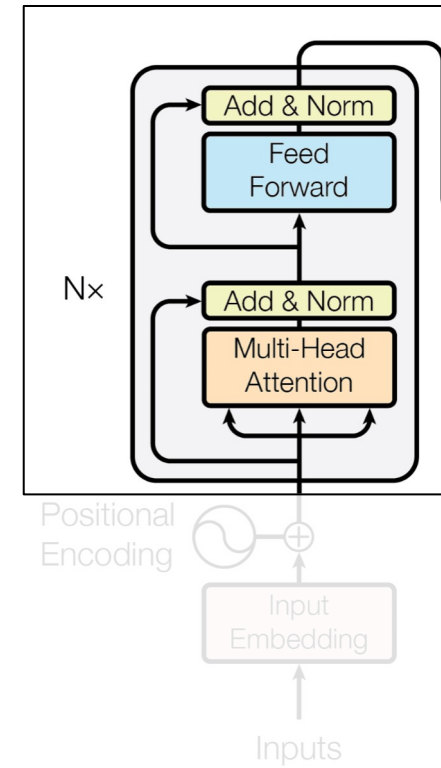
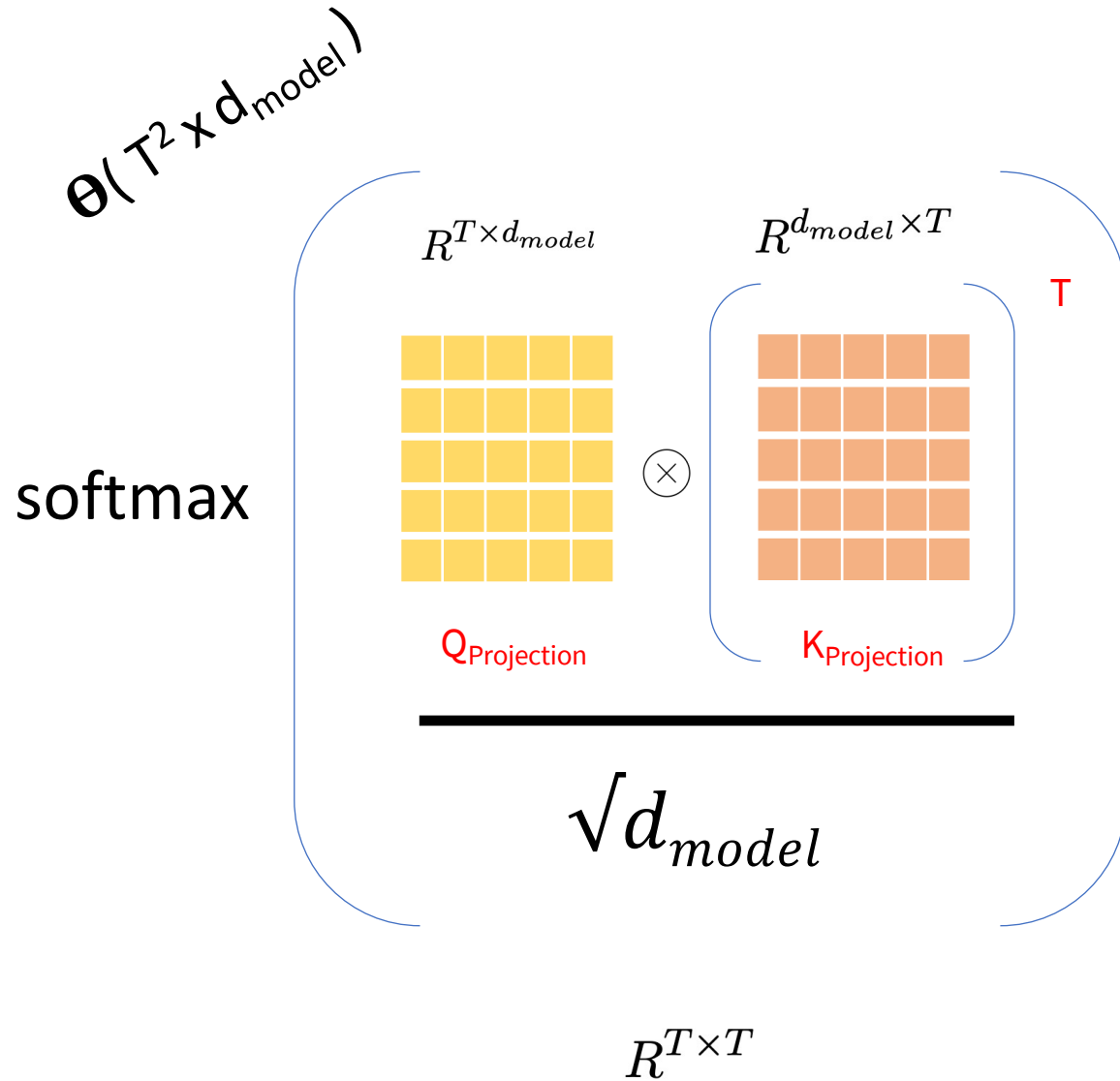


Self Attention

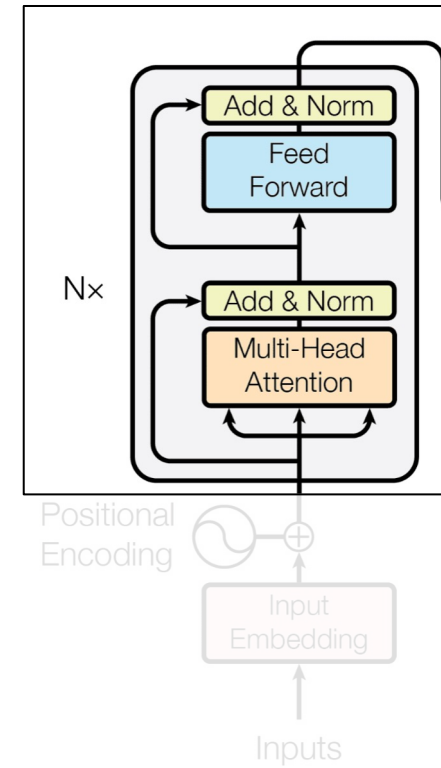
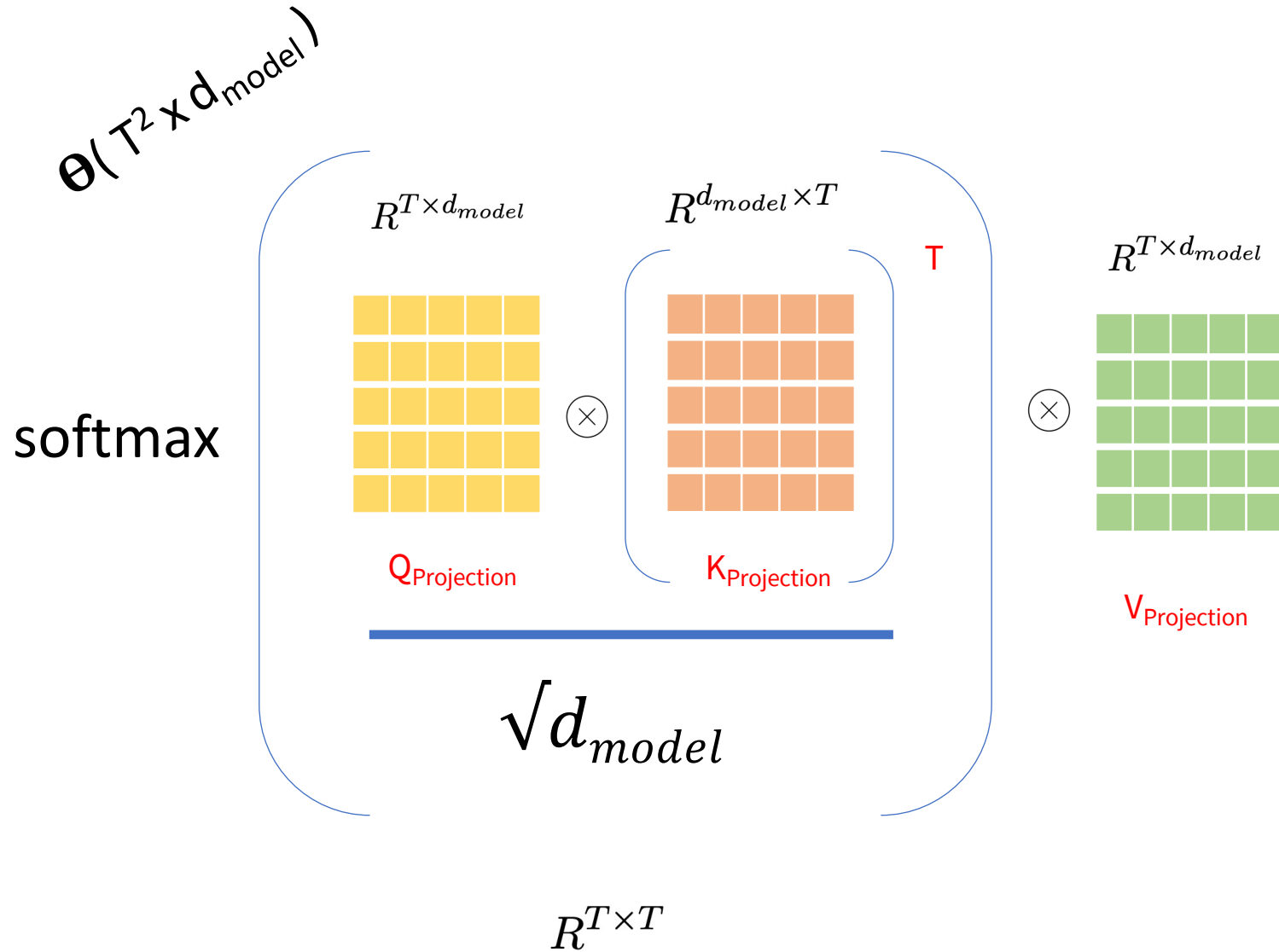
softmax



Self Attention

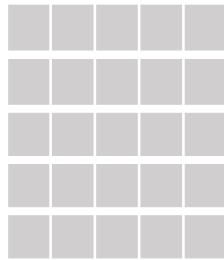


Self Attention

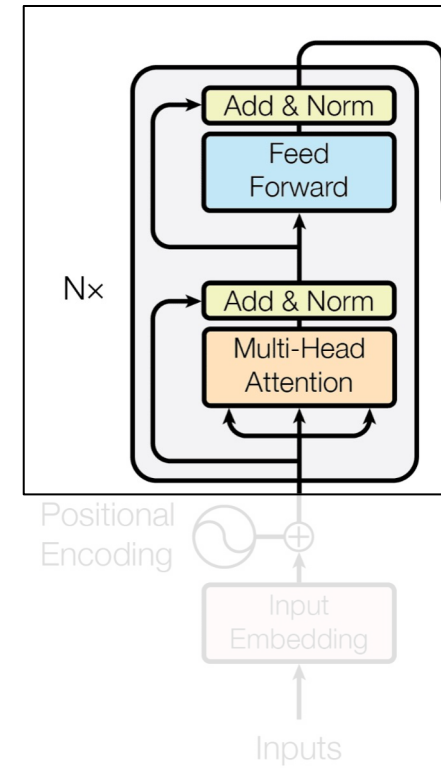


Self Attention

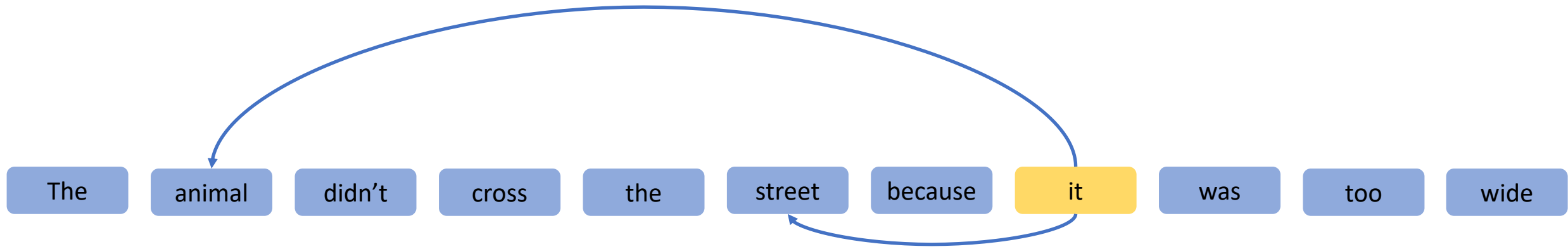
$$R^{T \times d_{model}}$$



Attention: Z



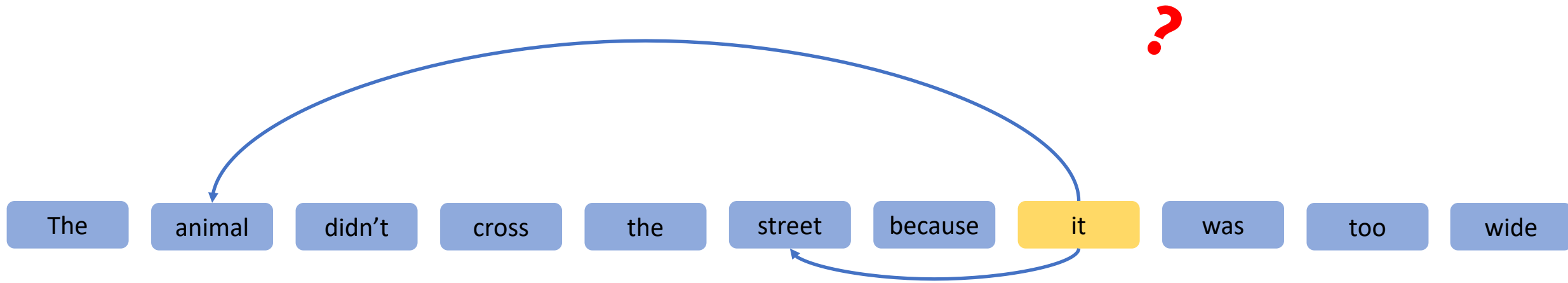
Self Attention



coreference resolution



Self Attention



Sentence boundaries ?

coreference resolution



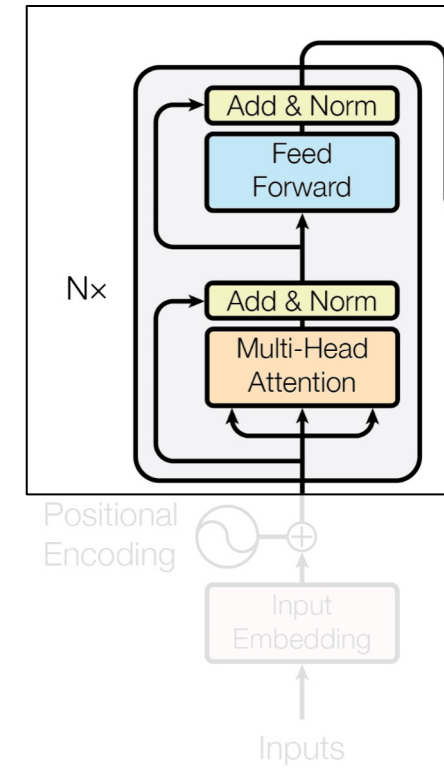
Context ?

Semantic relationships ?

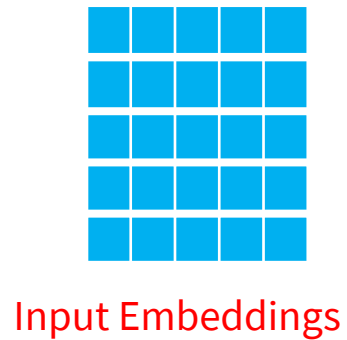
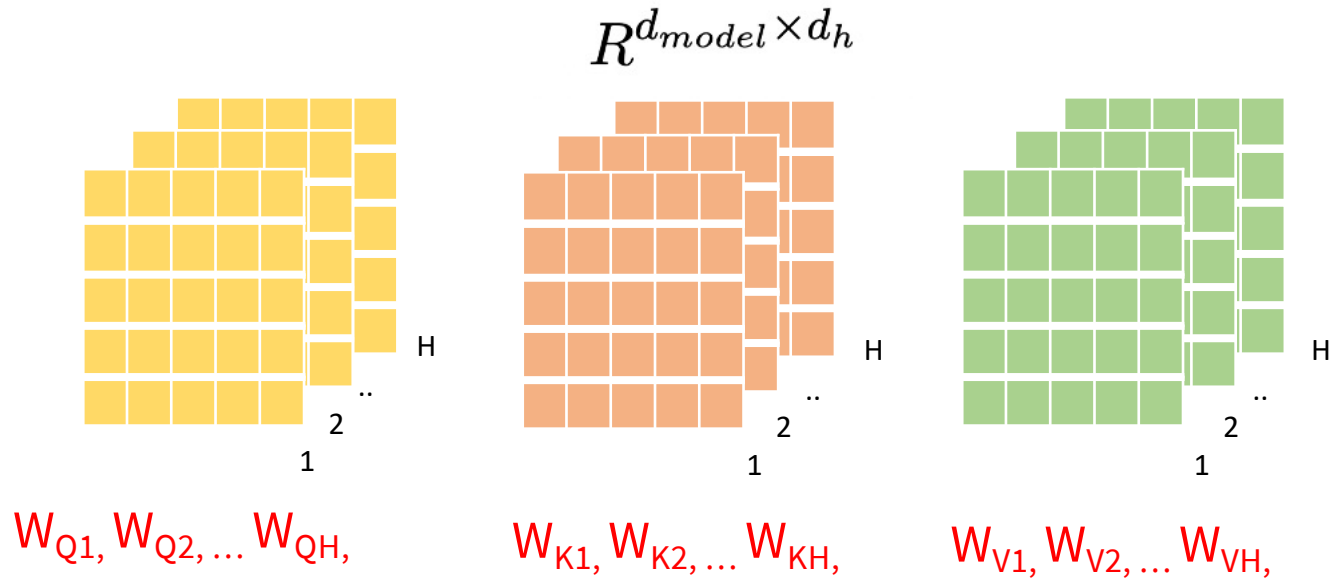
Part of Speech ?

Comparisons ?

Self Attention

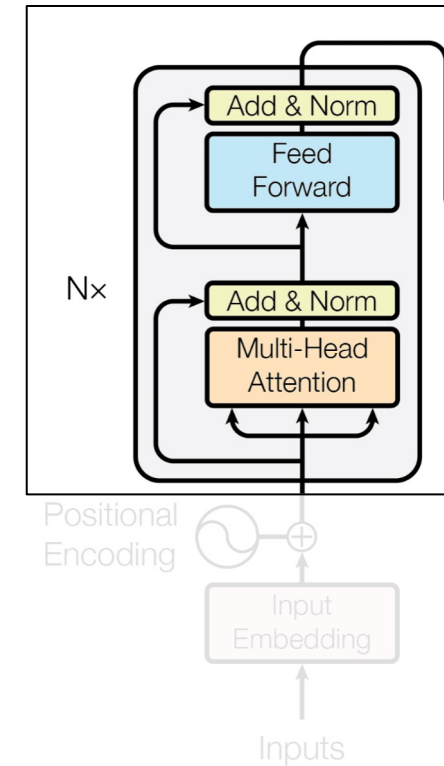


Multi-Head Attention

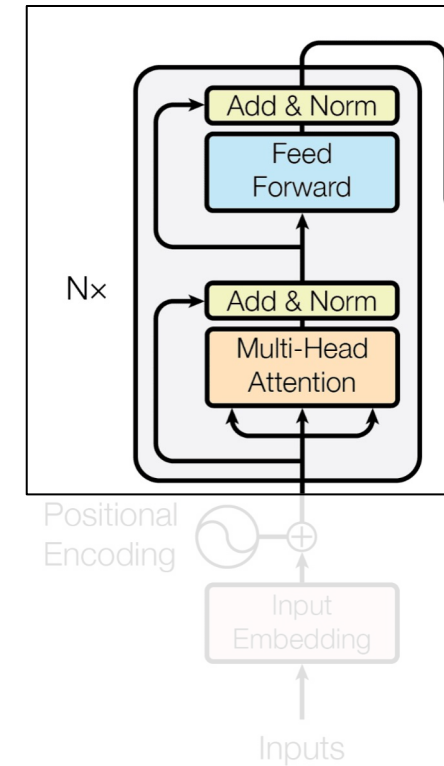
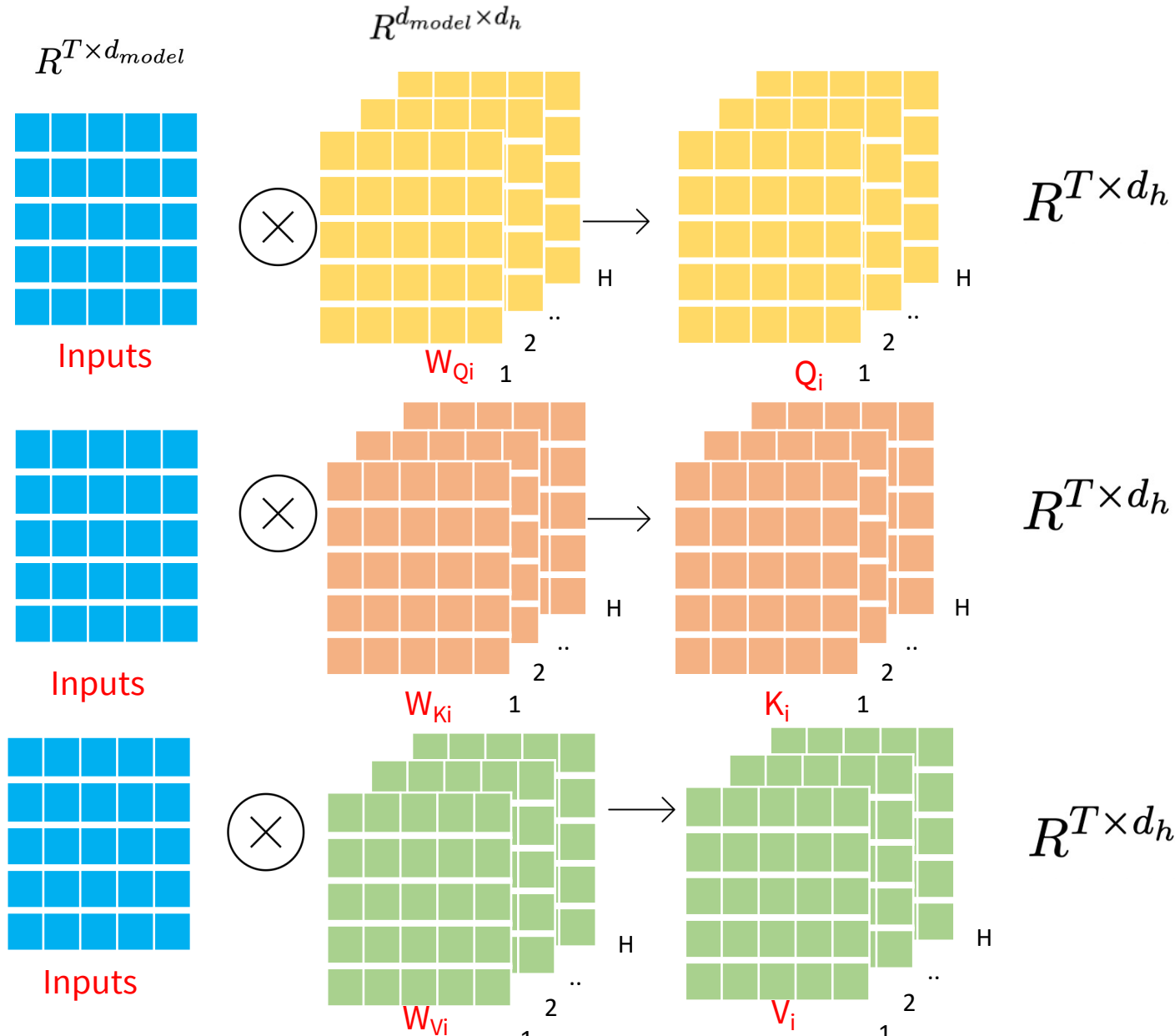


$$R^{T \times d_{model}}$$

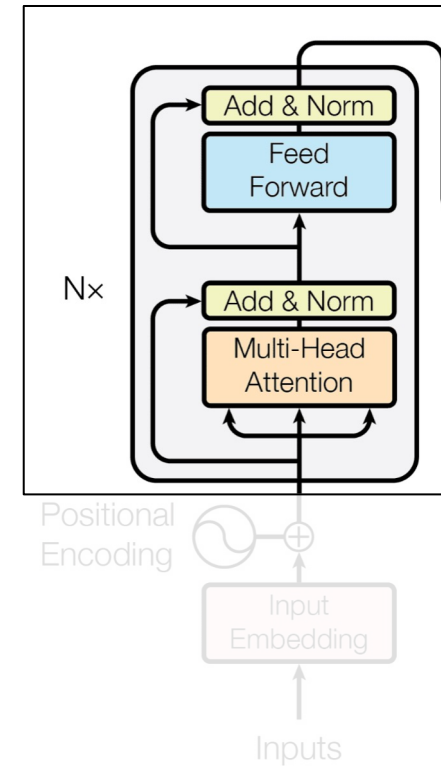
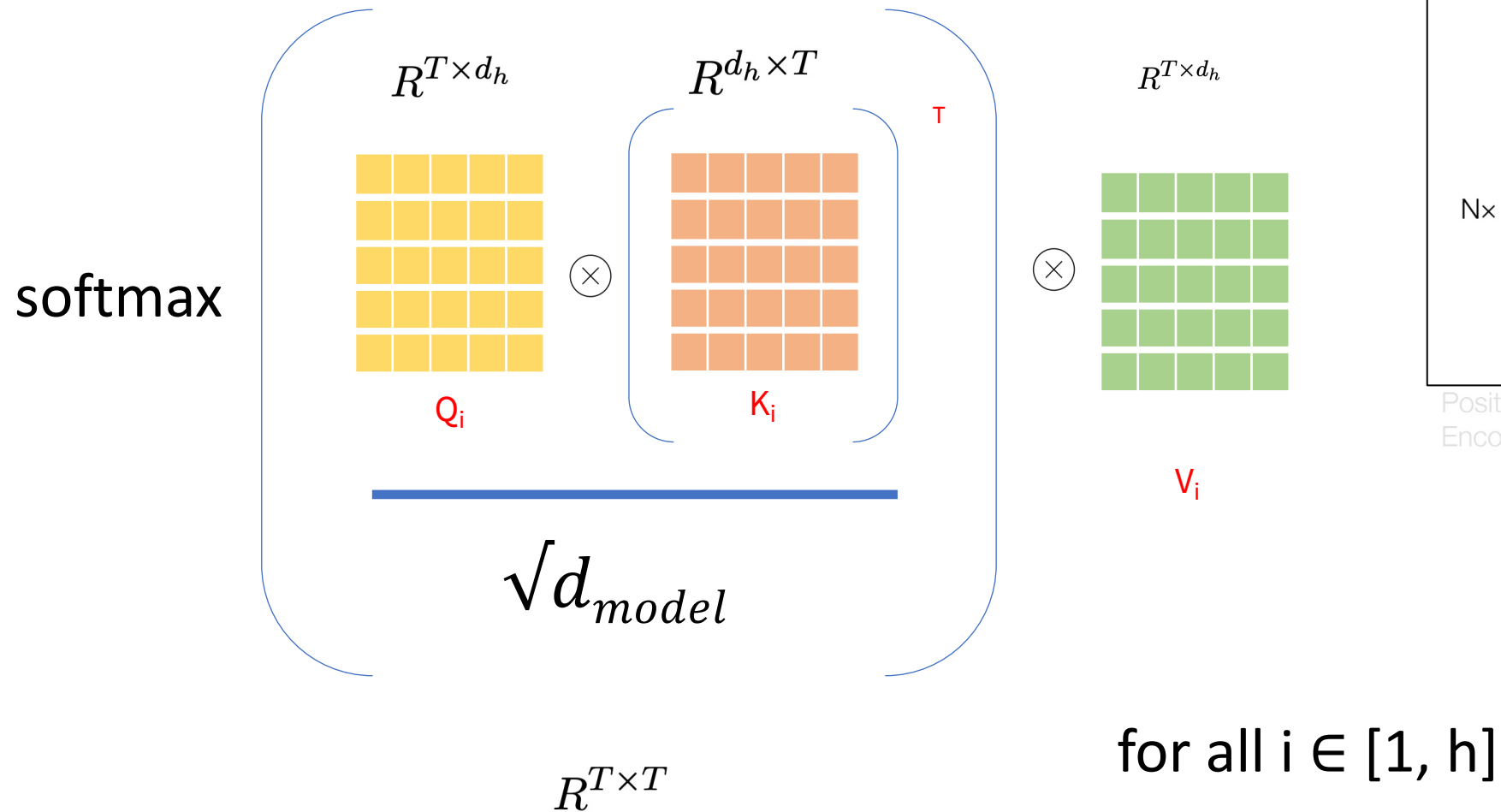
$$d_h = \frac{d_{model}}{h}$$



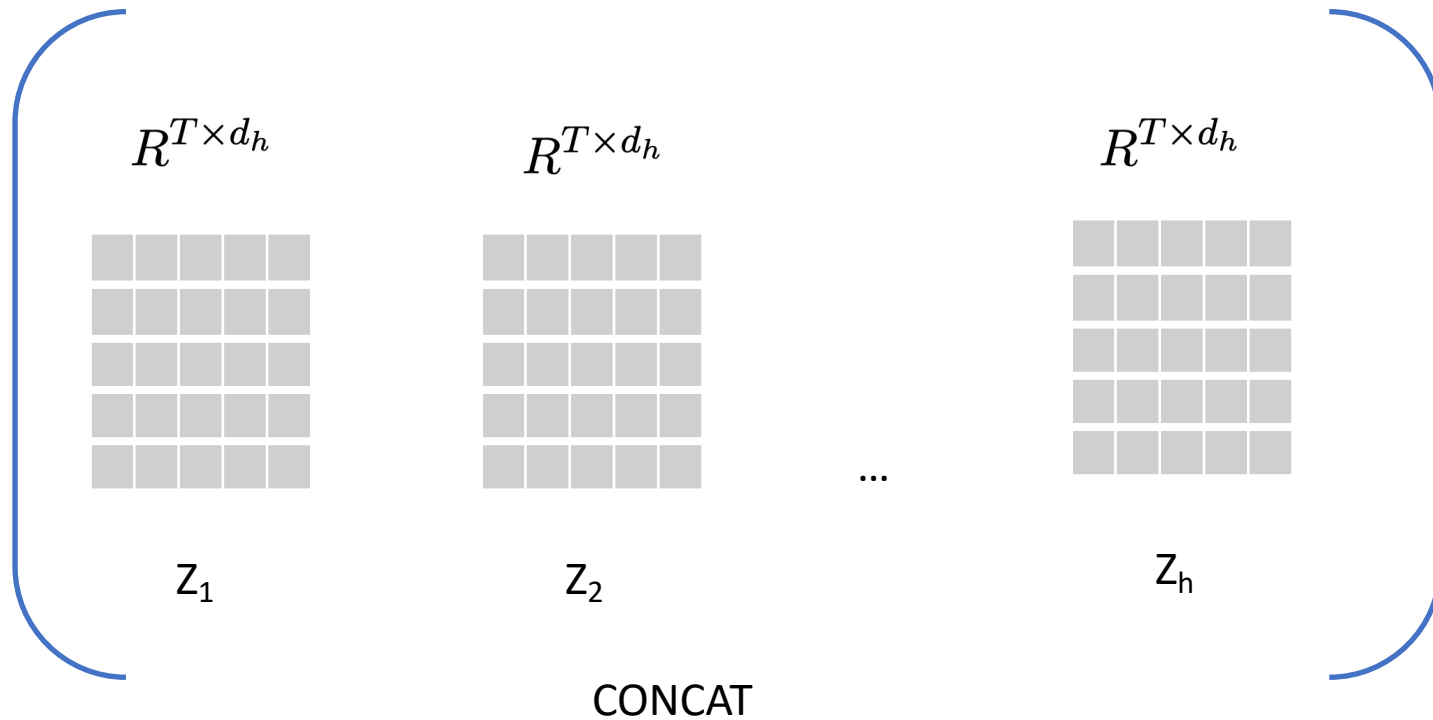
Multi-Head Attention



Multi-Head Attention



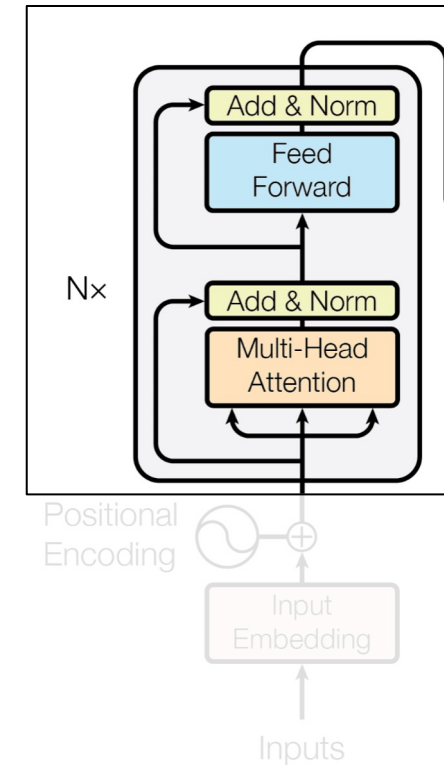
Multi-Head Attention



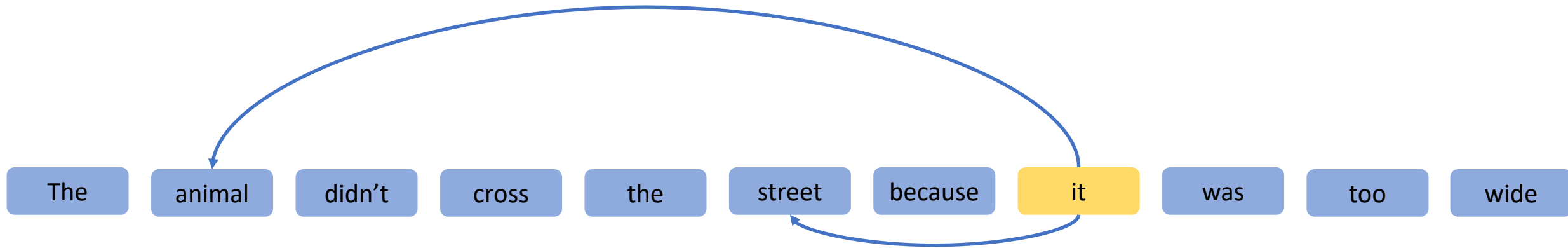
Multi Head Attention : Z

$$R^T \times d_{model}$$

$$d_h = \frac{d_{model}}{h}$$



Multi-Head Attention



Sentence boundaries ?



coreference resolution



Context ?



Semantic relationships ?



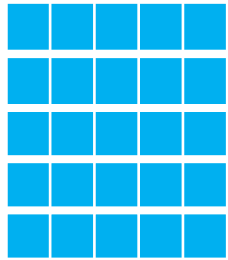
Part of Speech ?



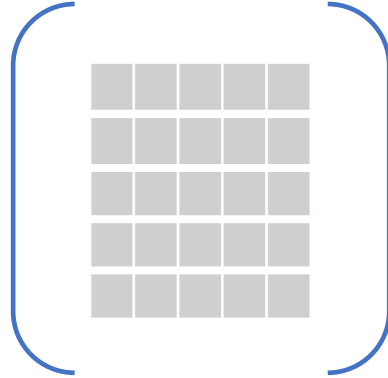
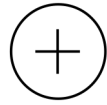
Comparisons ?



Add & Norm



Input



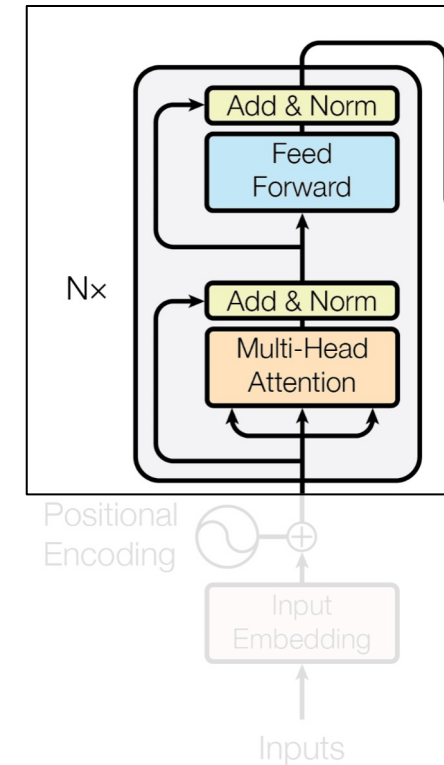
Norm(Z)

Normalization(Z)

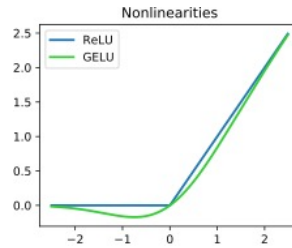
- Mean 0, Std dev 1
- Stabilizes training
- Regularization effect

Add -> Residuals

- Avoid vanishing gradients
- Train deeper networks

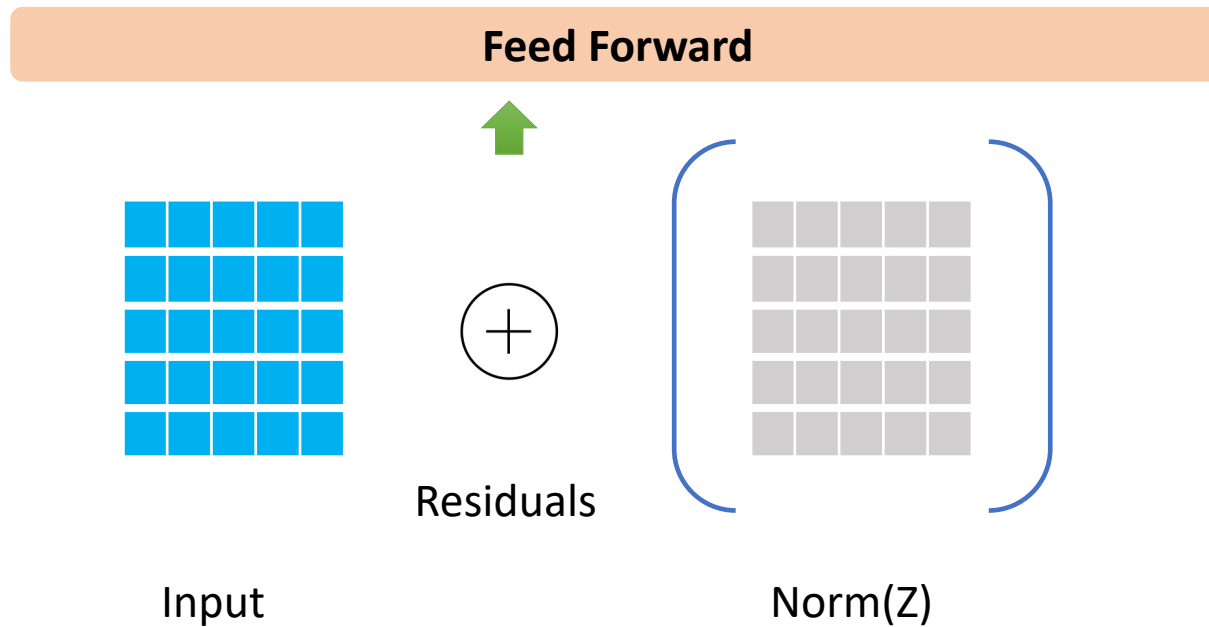
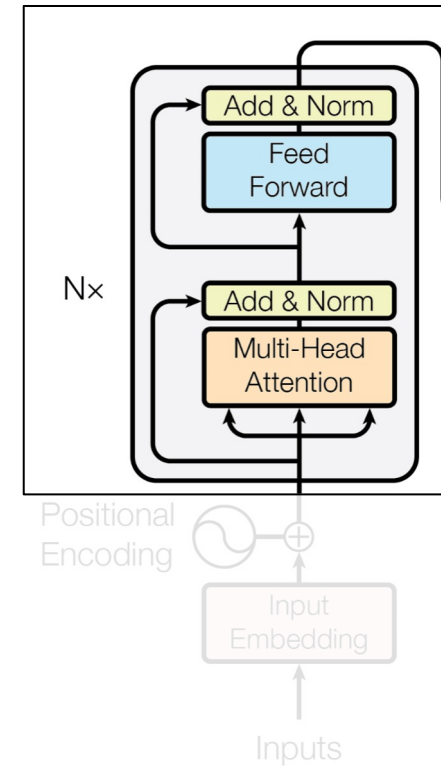


Feed Forward



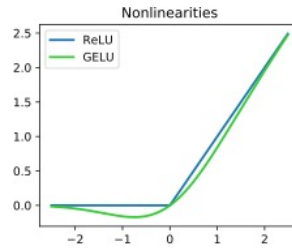
Feed Forward

- Non Linearity
- Complex Relationships
- Learn from each other

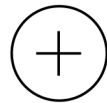
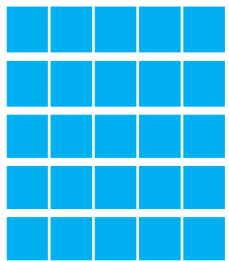


Add & Norm

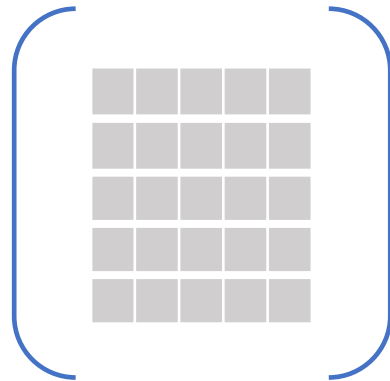
Add & Norm



Feed Forward

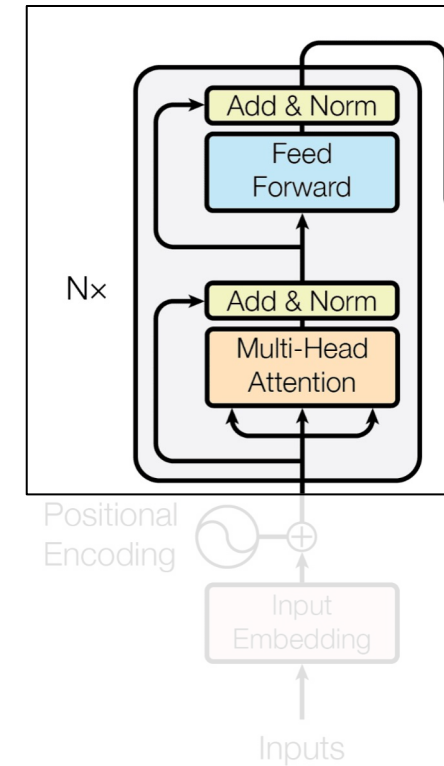


Residuals



Input

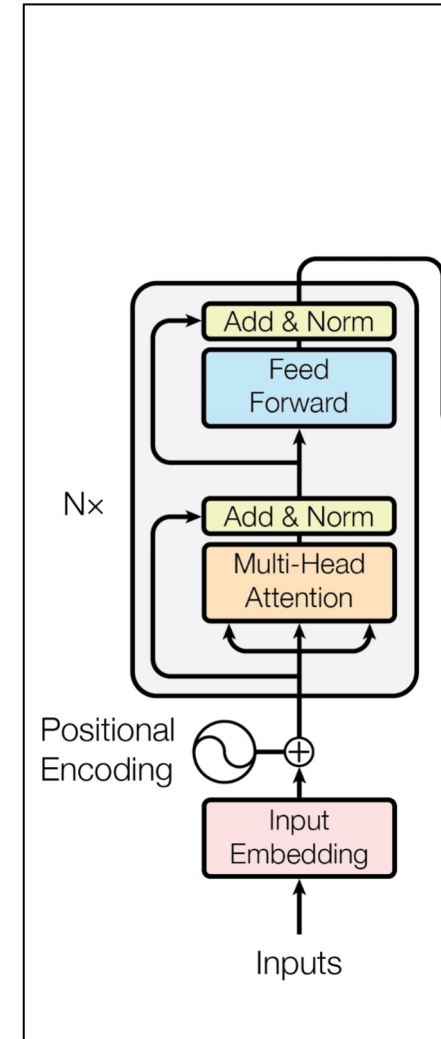
Norm(Z)



Encoders

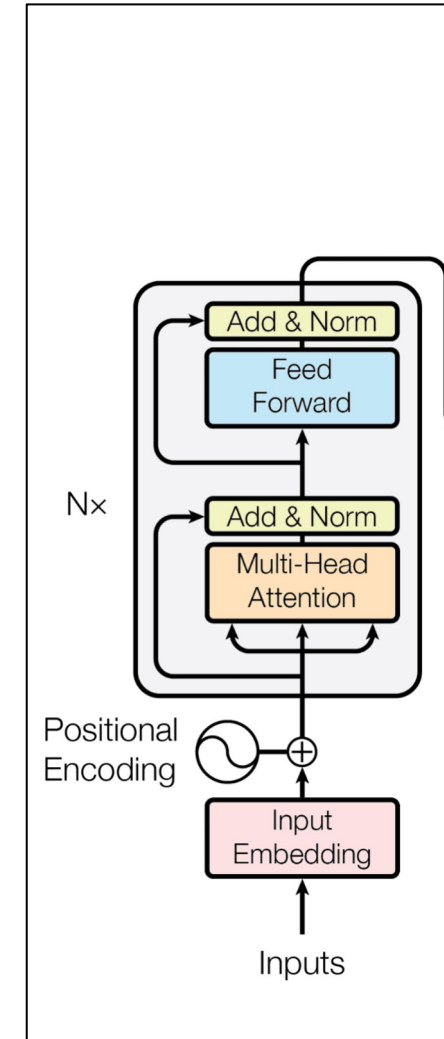
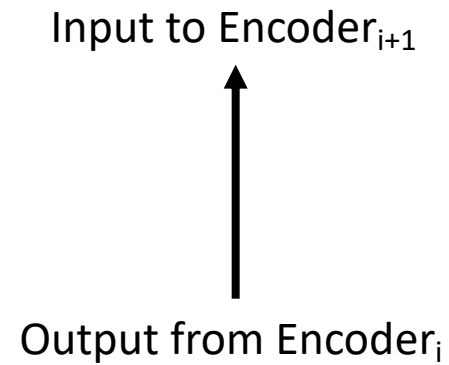
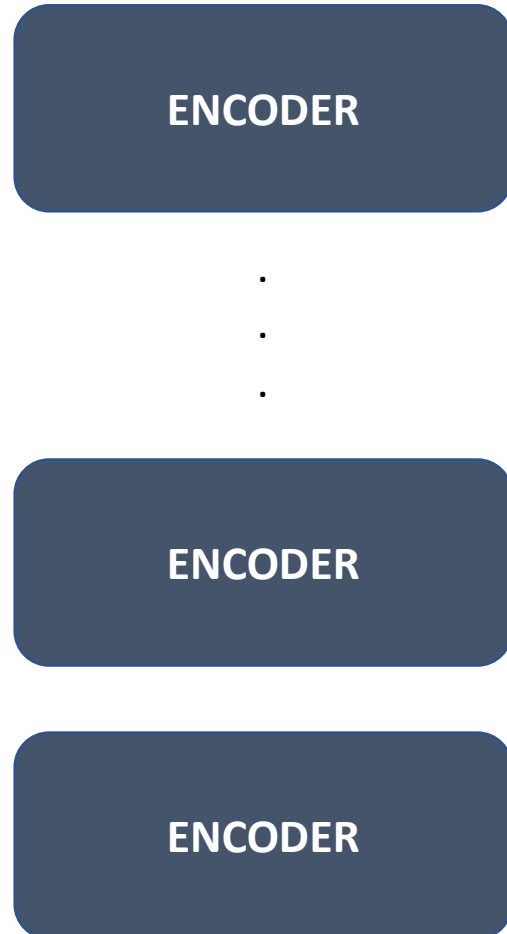
Encoder

ENCODER



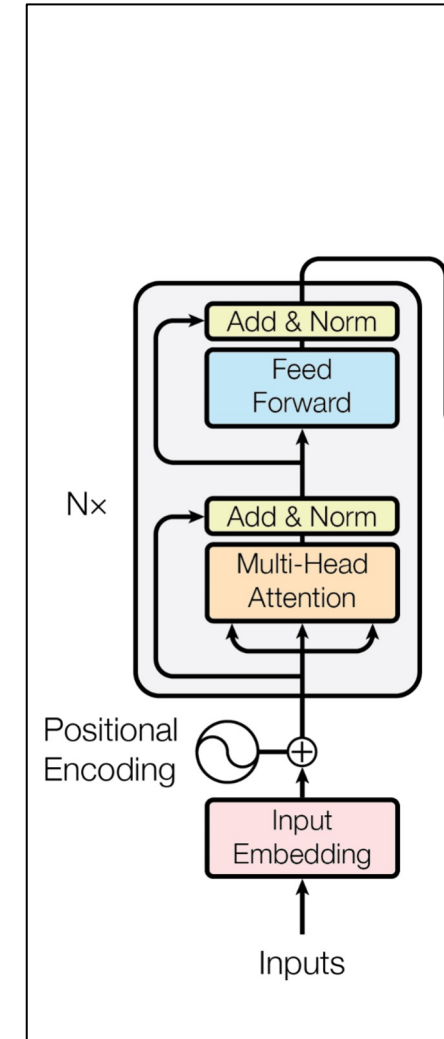
Encoders

Encoder



Transformers

- ✓ Tokenization
- ✓ Input Embeddings
- ✓ Position Encodings
- ✓ Residuals
- ✓ Query
- ✓ Key
- ✓ Value
- ✓ Add & Norm
- ✓ Encoder
- Decoder
- ✓ Attention
- ✓ Self Attention
- ✓ Multi Head Attention
- Masked Attention
- Encoder Decoder Attention
- Output Probabilities / Logits
- Softmax
- Encoder-Decoder models
- Decoder only models

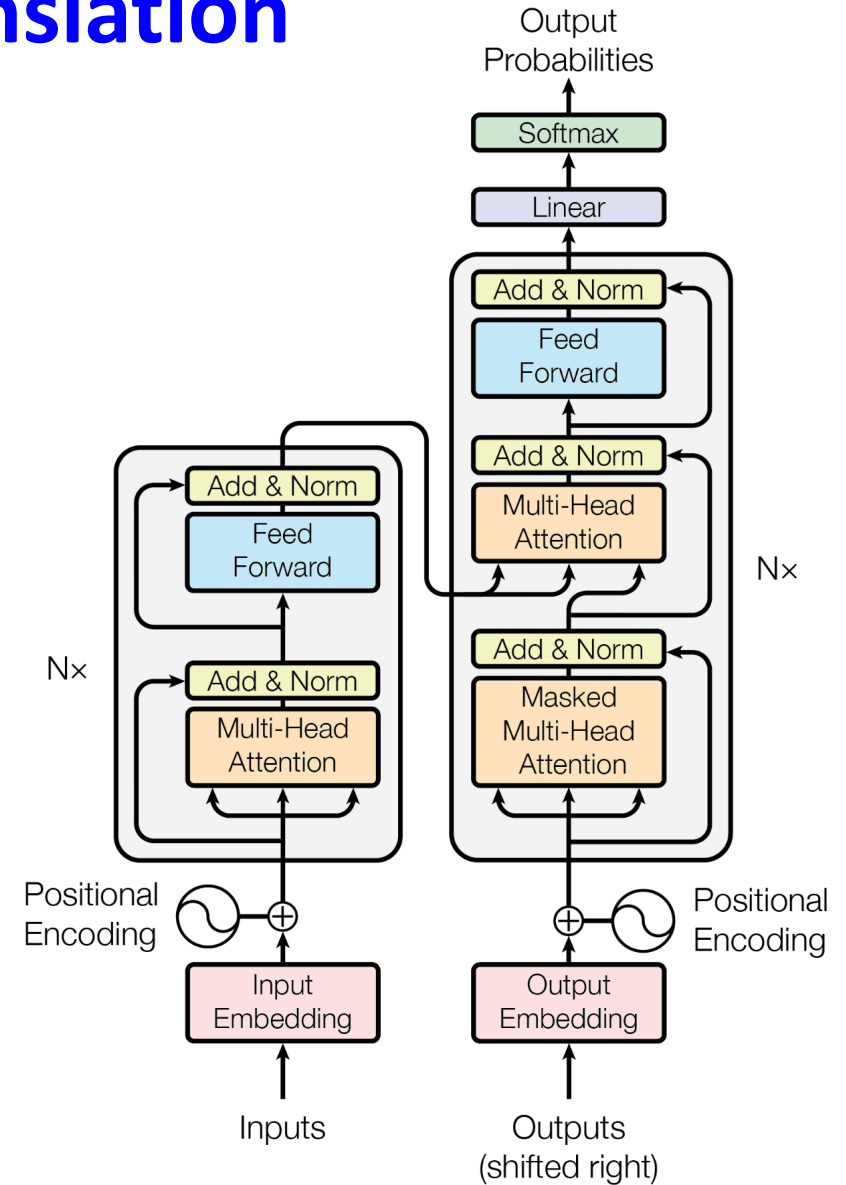


Machine Translation

Targets
Ich have einen apfel gegessen



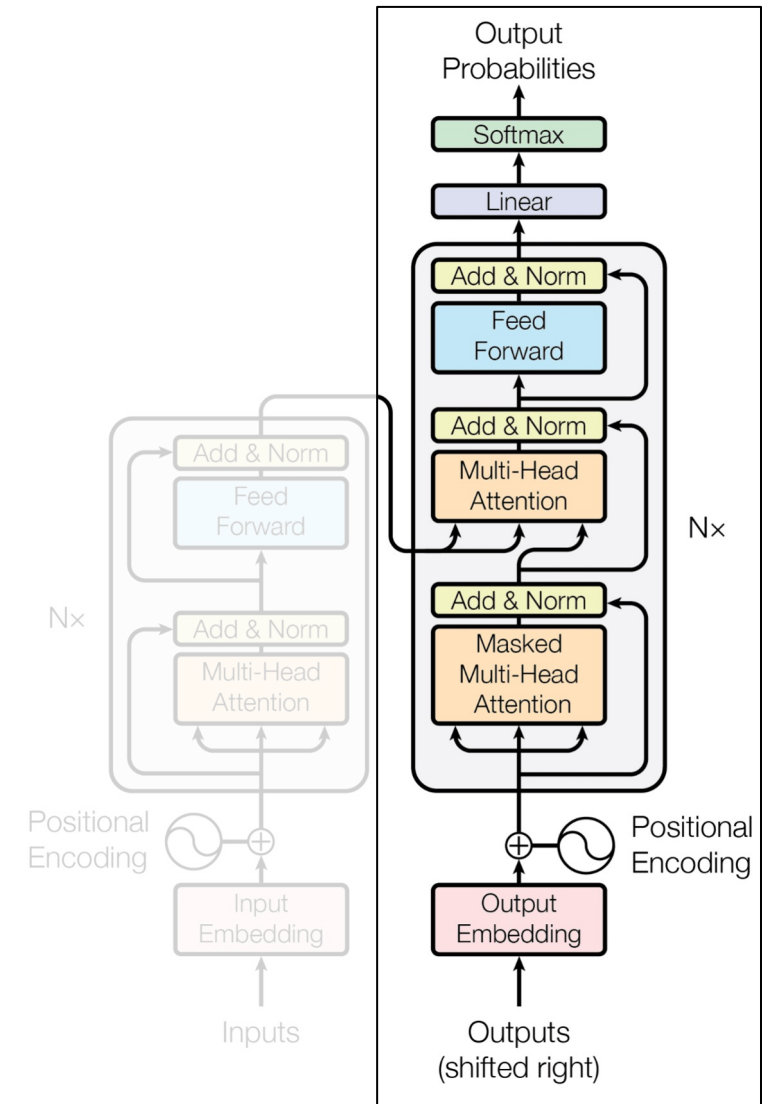
Inputs
I ate an apple



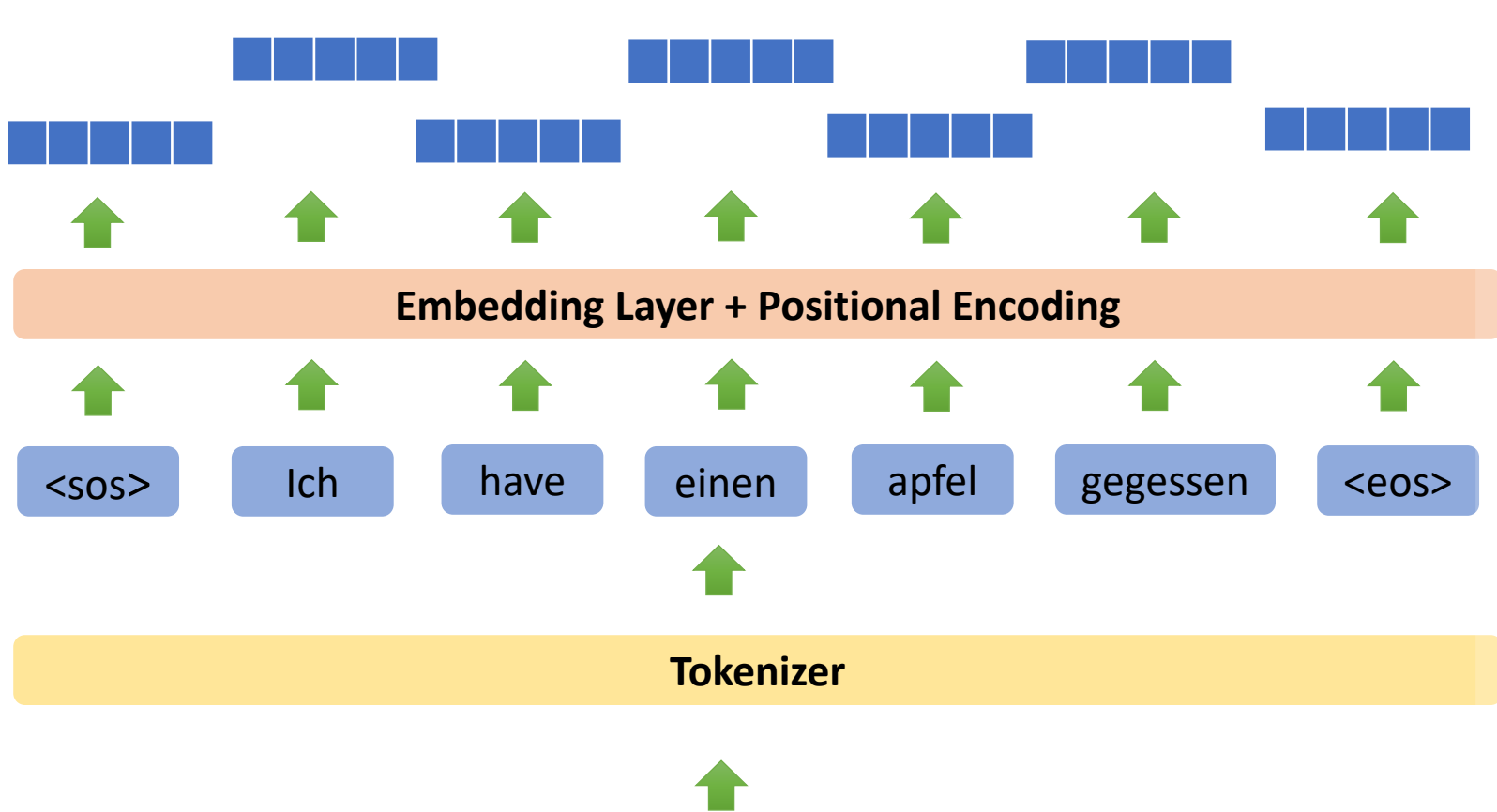
Targets

Targets

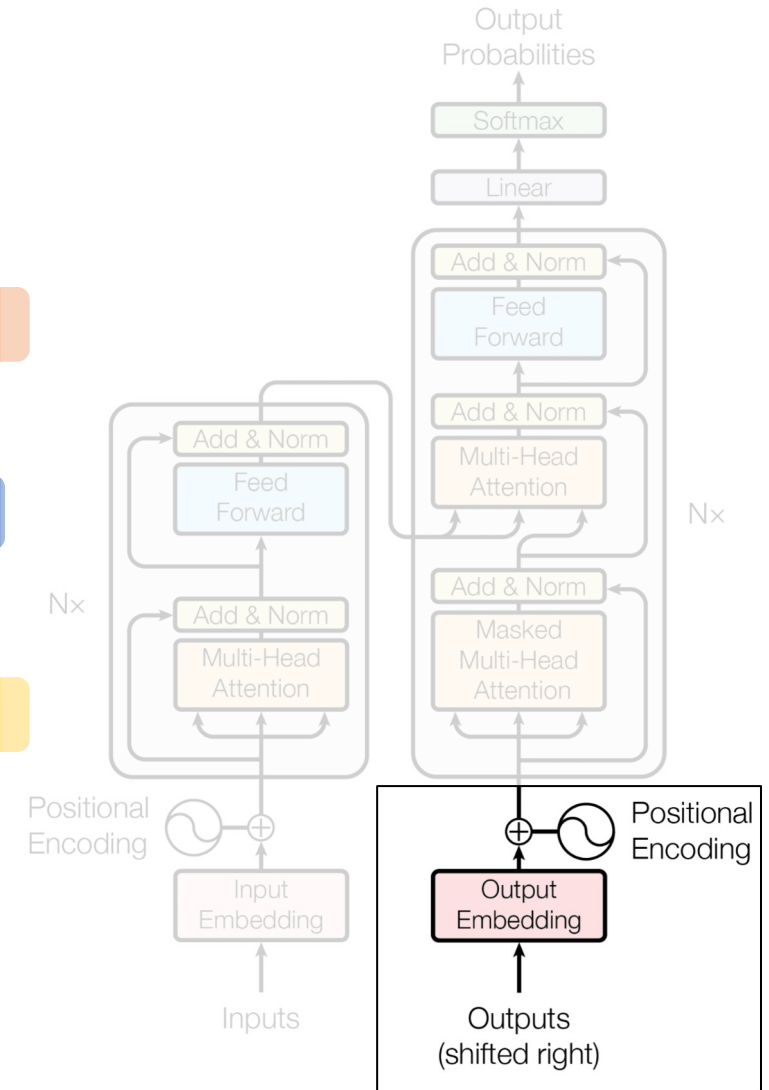
Ich have einen apfel gegessen



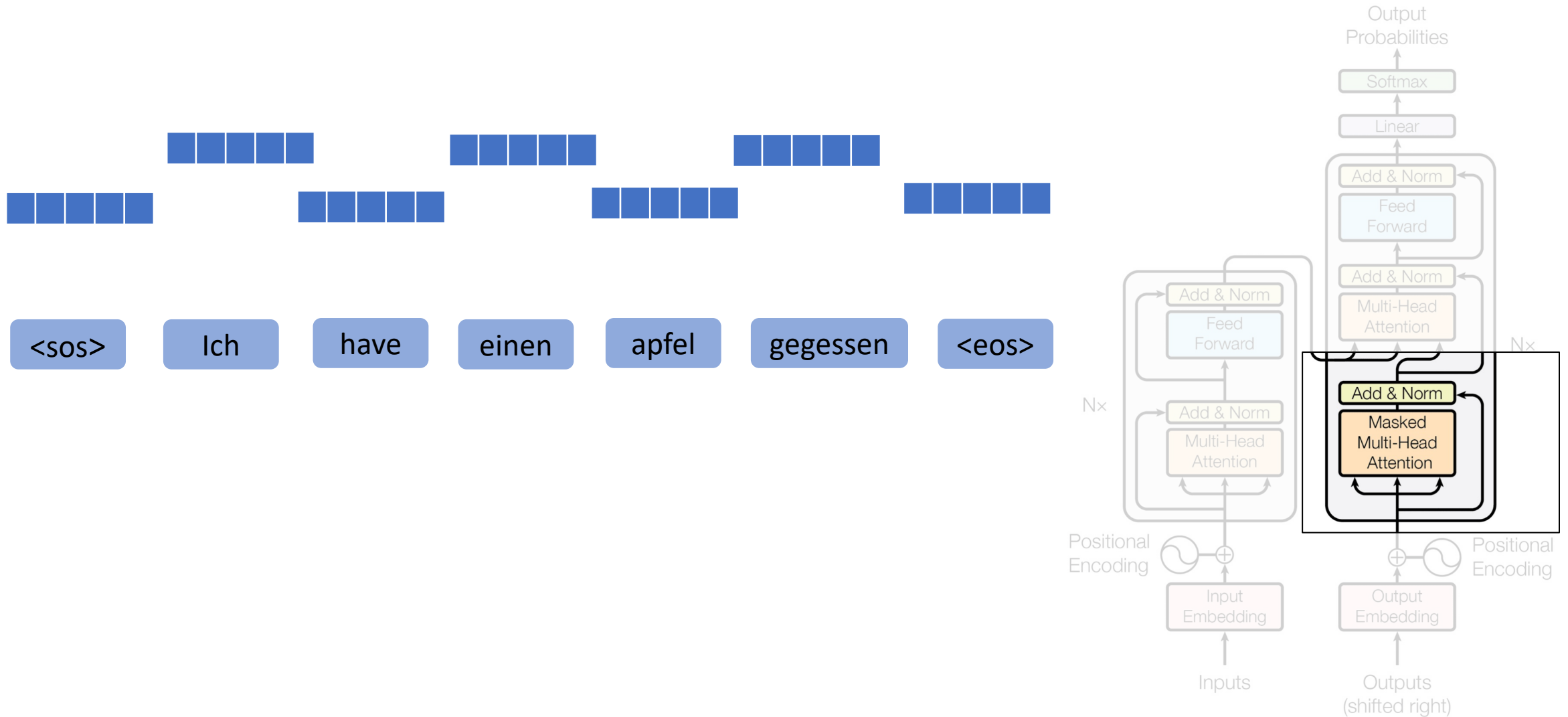
Targets



Generate Target Emebeddings

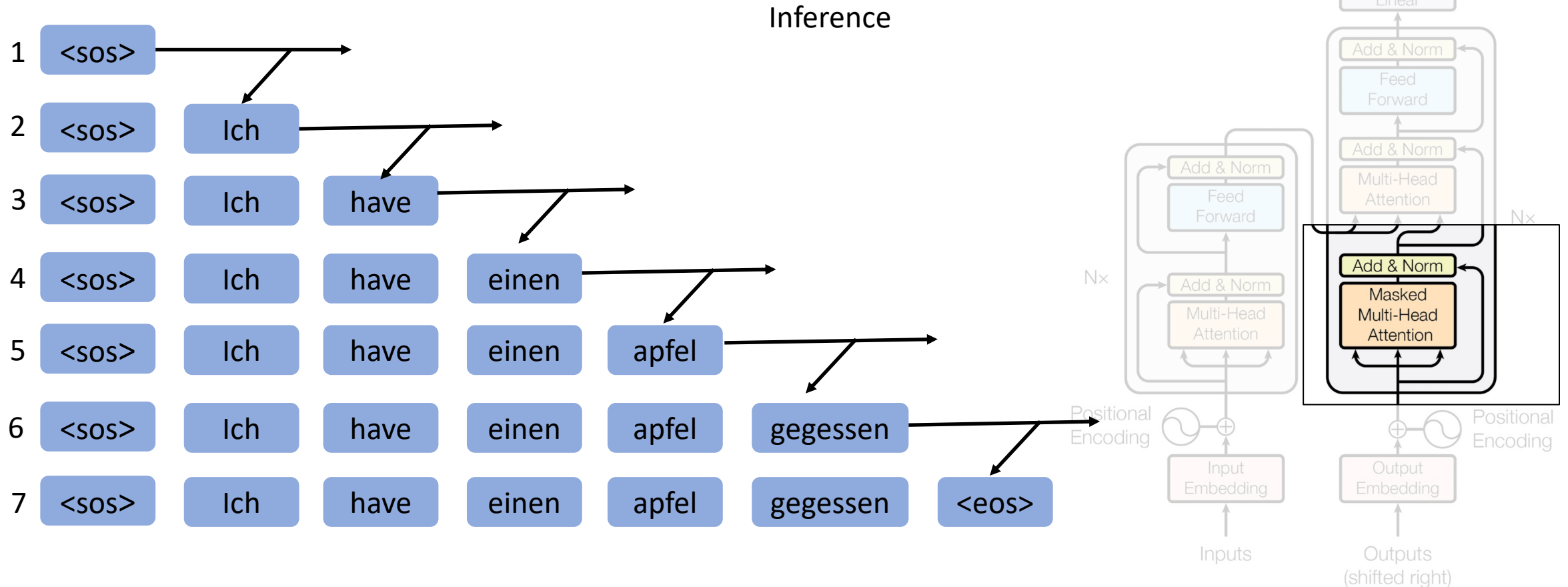


Masked Multi Head Attention



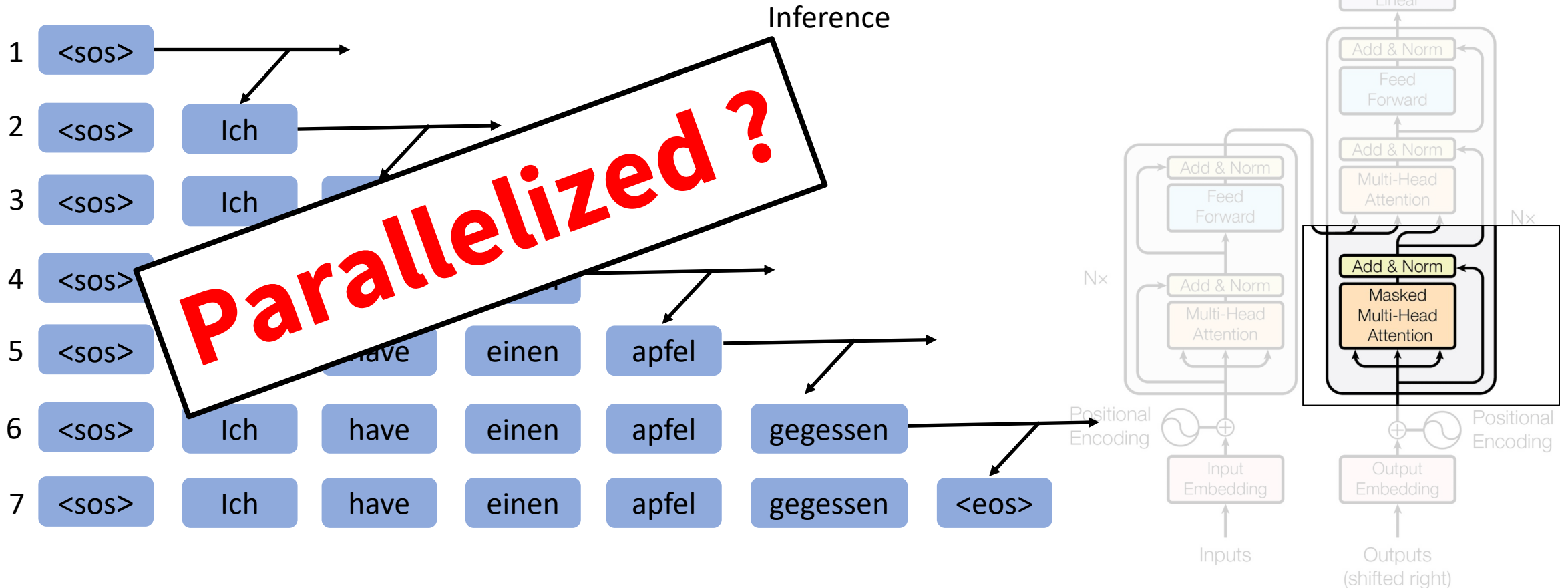
Masked Multi Head Attention

Decoding step by step (using Teacher Forcing)



Masked Multi Head Attention

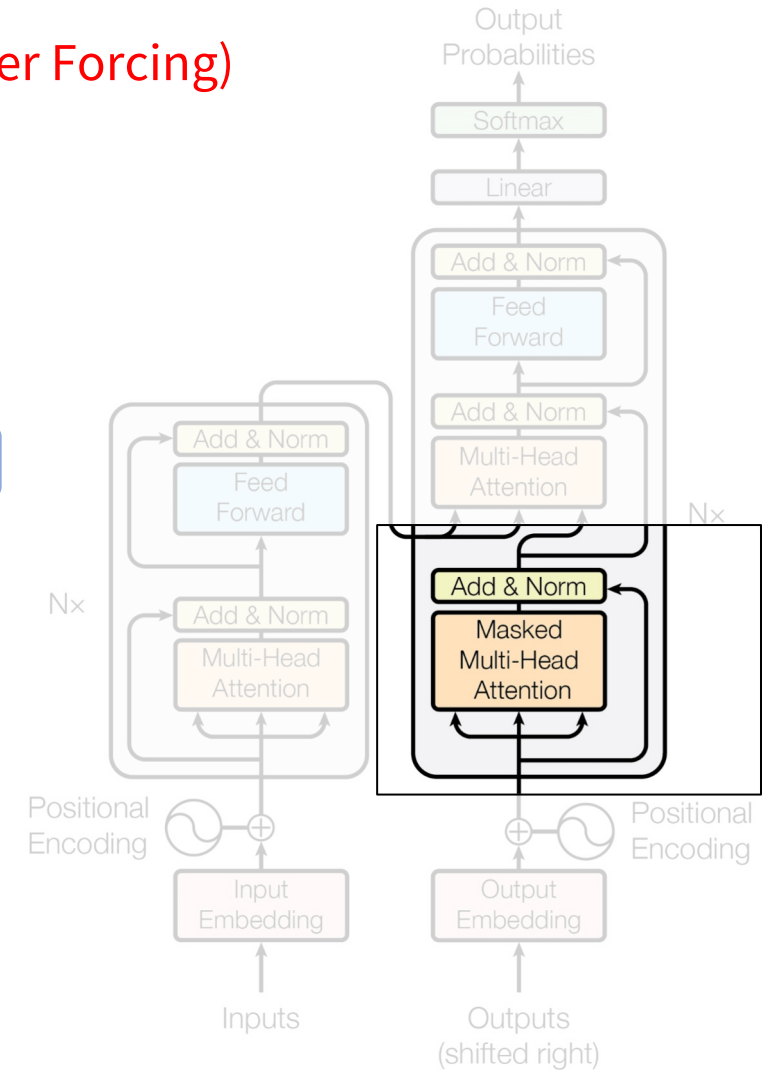
Decoding step by step (using Teacher Forcing)



Masked Multi Head Attention

Decoding step by step (using Teacher Forcing)

Training



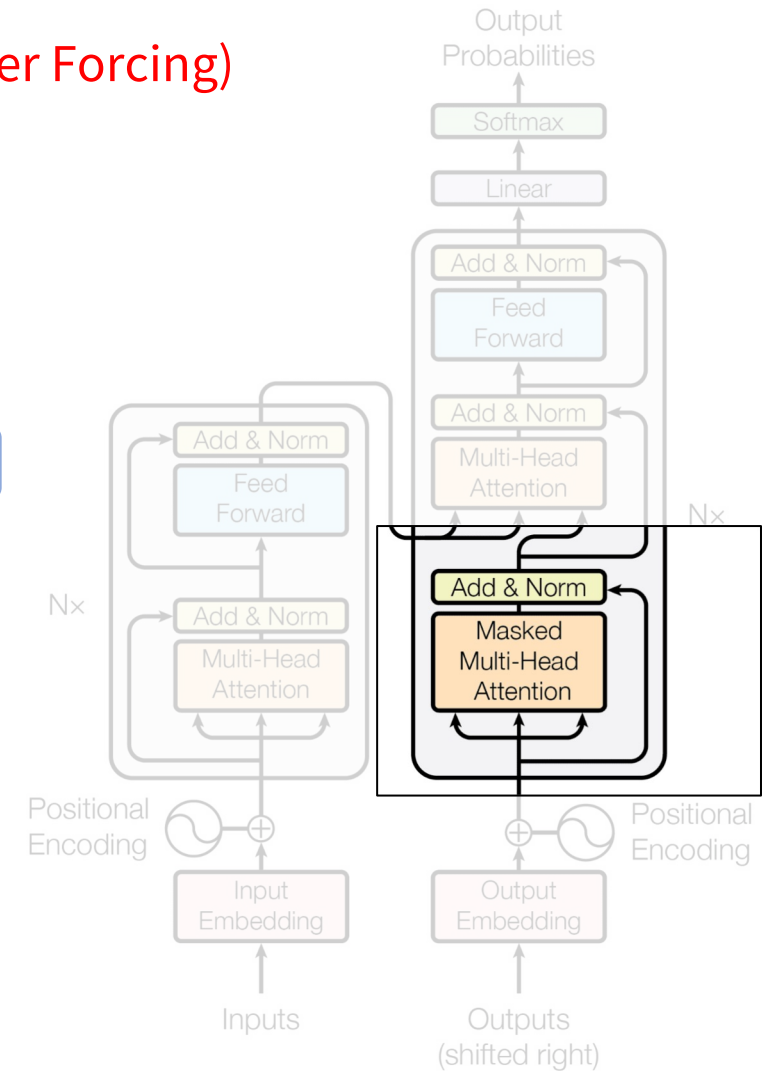
Masked Multi Head Attention

Decoding step by step (using Teacher Forcing)

Training

<sos> Ich have einen apfel gegessen <eos>

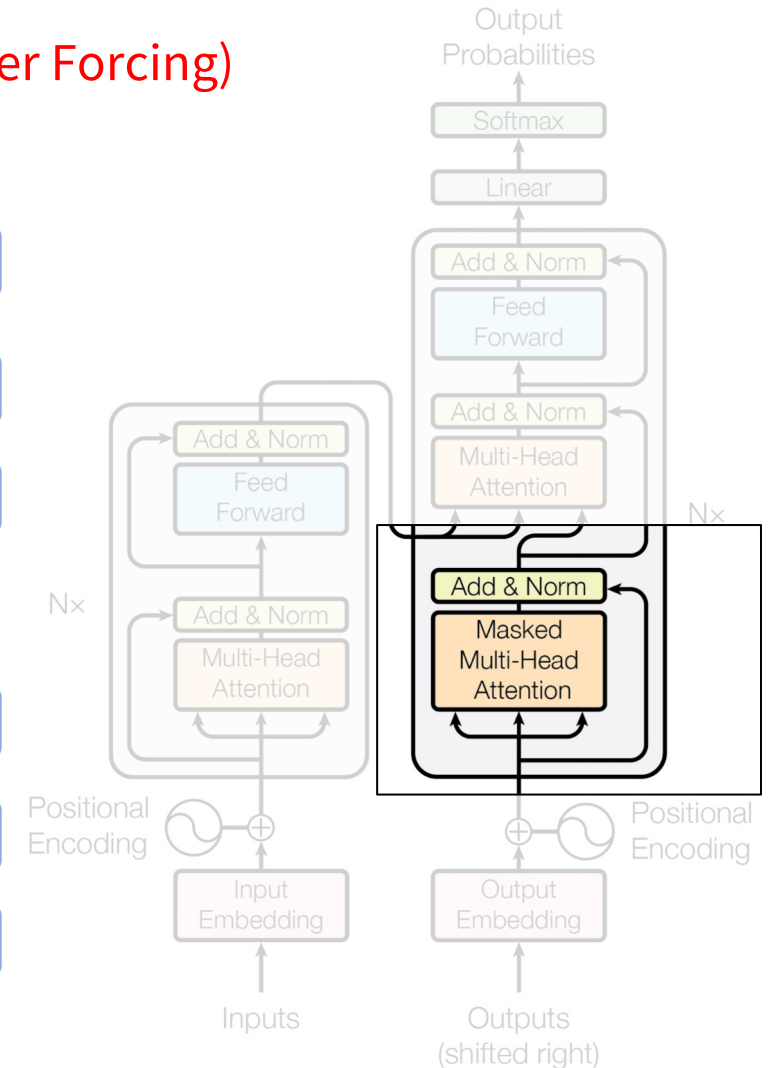
Outputs at time T should only pay attention to outputs until time $T-1$



Masked Multi Head Attention

Decoding step by step (using Teacher Forcing)

1	<sos>	Ich	have	einen	apfel	gegessen	<eos>
2	<sos>	Ich	have	einen	apfel	gegessen	<eos>
3	<sos>	Ich	have	einen	apfel	gegessen	<eos>
4	<sos>	Ich	have	einen	apfel	gegessen	<eos>
5	<sos>	Ich	have	einen	apfel	gegessen	<eos>
6	<sos>	Ich	have	einen	apfel	gegessen	<eos>
7	<sos>	Ich	have	einen	apfel	gegessen	<eos>

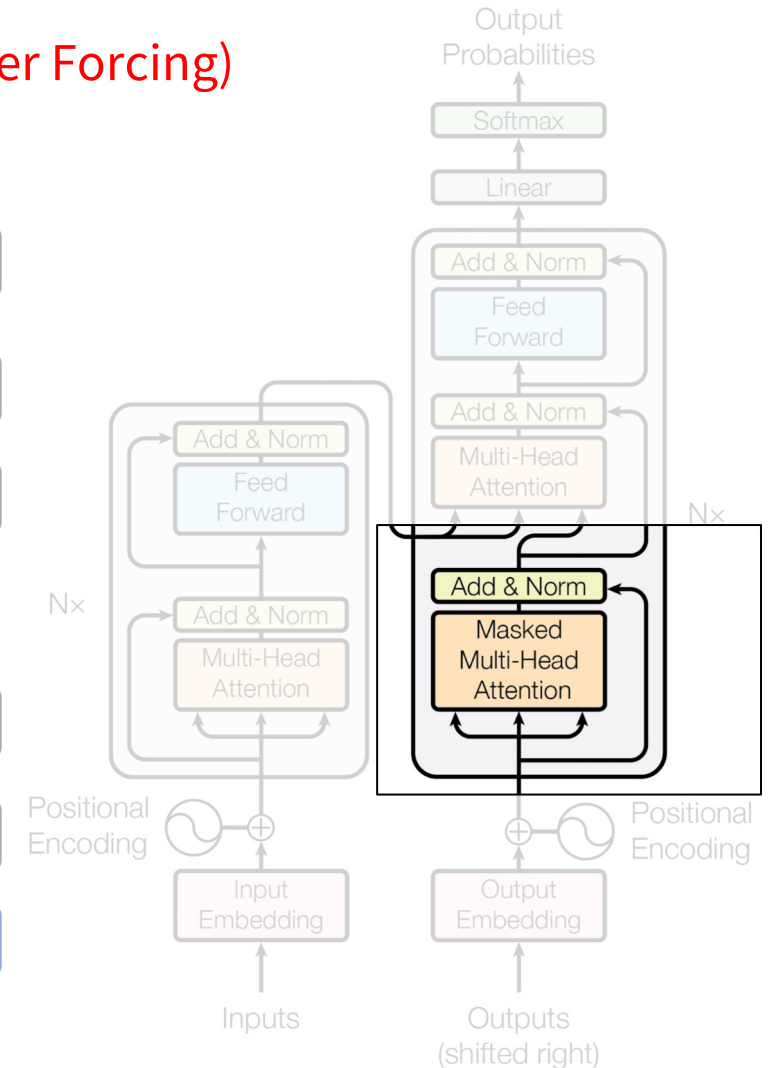


Masked Multi Head Attention

Decoding step by step (using Teacher Forcing)

1	<sos>	Ich	have	einen	apfel	gegessen	<eos>
2	<sos>	Ich	have	einen	apfel	gegessen	<eos>
3	<sos>	Ich	have	einen	apfel	gegessen	<eos>
4	<sos>	Ich	have	einen	apfel	gegessen	<eos>
5	<sos>	Ich	have	einen	apfel	gegessen	<eos>
6	<sos>	Ich	have	einen	apfel	gegessen	<eos>
7	<sos>	Ich	have	einen	apfel	gegessen	<eos>

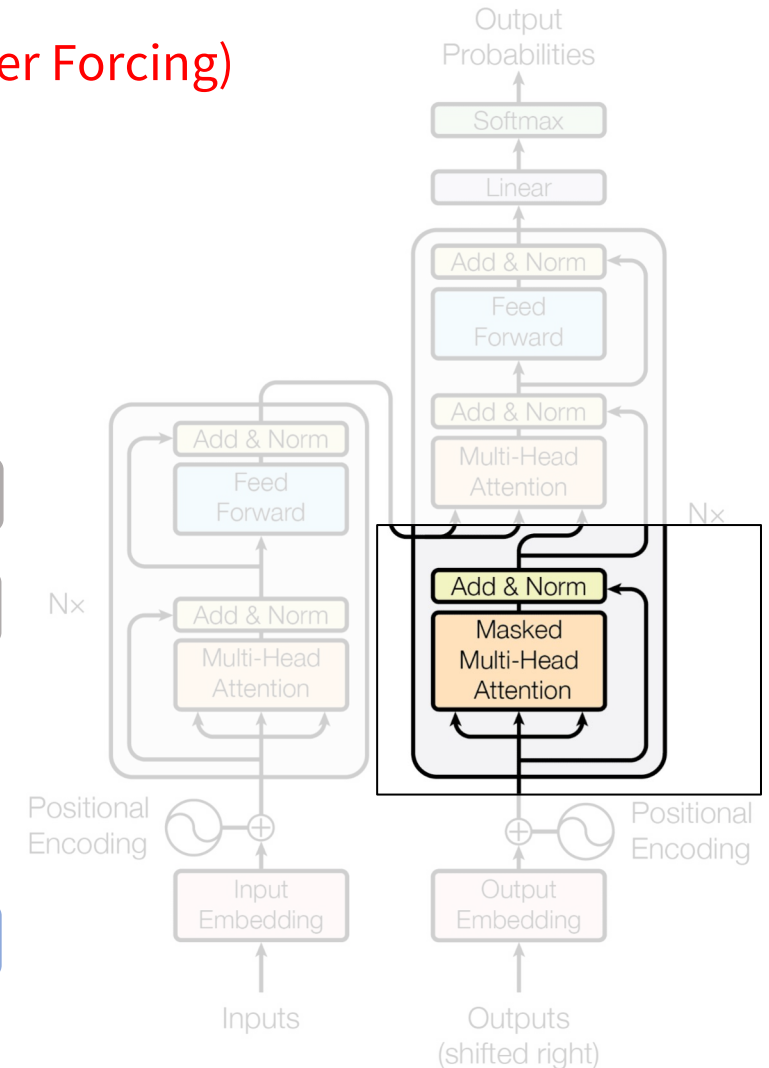
Mask the available attention values ?



Masked Multi Head Attention

Decoding step by step (using Teacher Forcing)

1	<sos>	- ∞	- ∞	- ∞	- ∞	- ∞
2	<sos>	Ich	- ∞	- ∞	- ∞	- ∞
3	<sos>	Ich	have	- ∞	- ∞	- ∞
4	<sos>	Ich	have	einen	- ∞	- ∞
5	<sos>	Ich	have	einen	apfel	- ∞
6	<sos>	Ich	have	einen	apfel	gegessen
7	<sos>	Ich	have	einen	apfel	gegessen

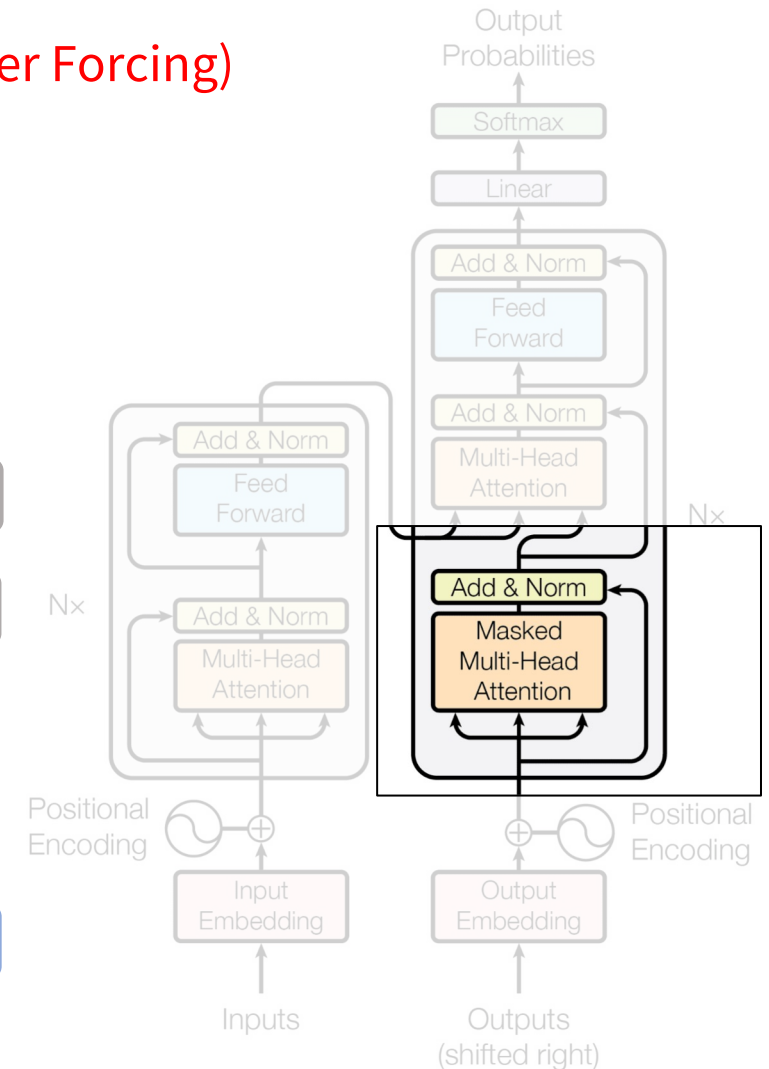


Masked Multi Head Attention

Decoding step by step (using Teacher Forcing)

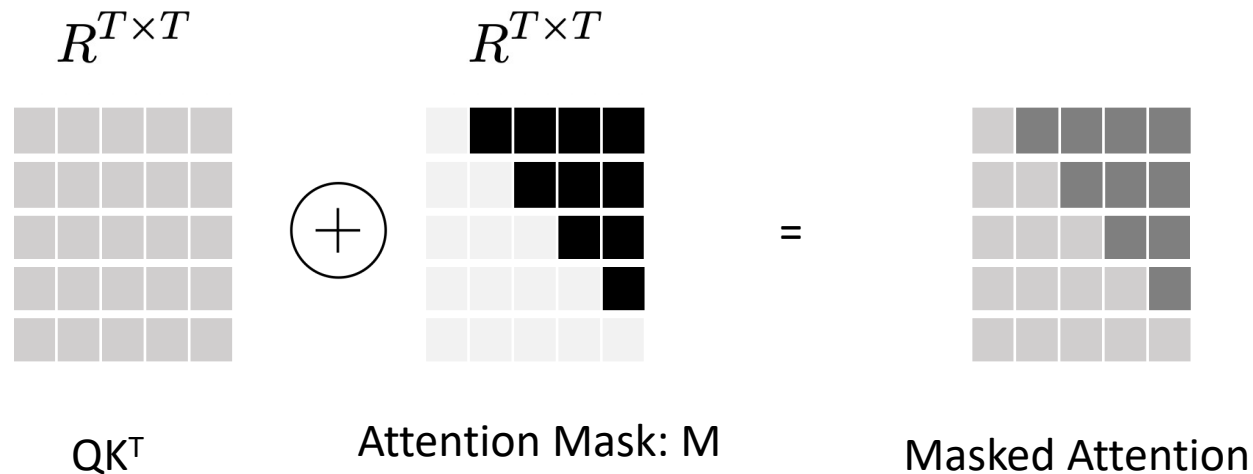
1	<sos>	- ∞	- ∞	- ∞	- ∞	- ∞	- ∞
2	<sos>	Ich	- ∞	- ∞	- ∞	- ∞	- ∞
3	<sos>	Ich	have	- ∞	- ∞	- ∞	- ∞
4	<sos>	Ich	have	einen	- ∞	- ∞	- ∞
5	<sos>	Ich	have	einen	apfel	- ∞	- ∞
6	<sos>	Ich	have	einen	apfel	gegessen	- ∞
7	<sos>	Ich	have	einen	apfel	gegessen	<eos>

Softmax -> - ∞ -> 0

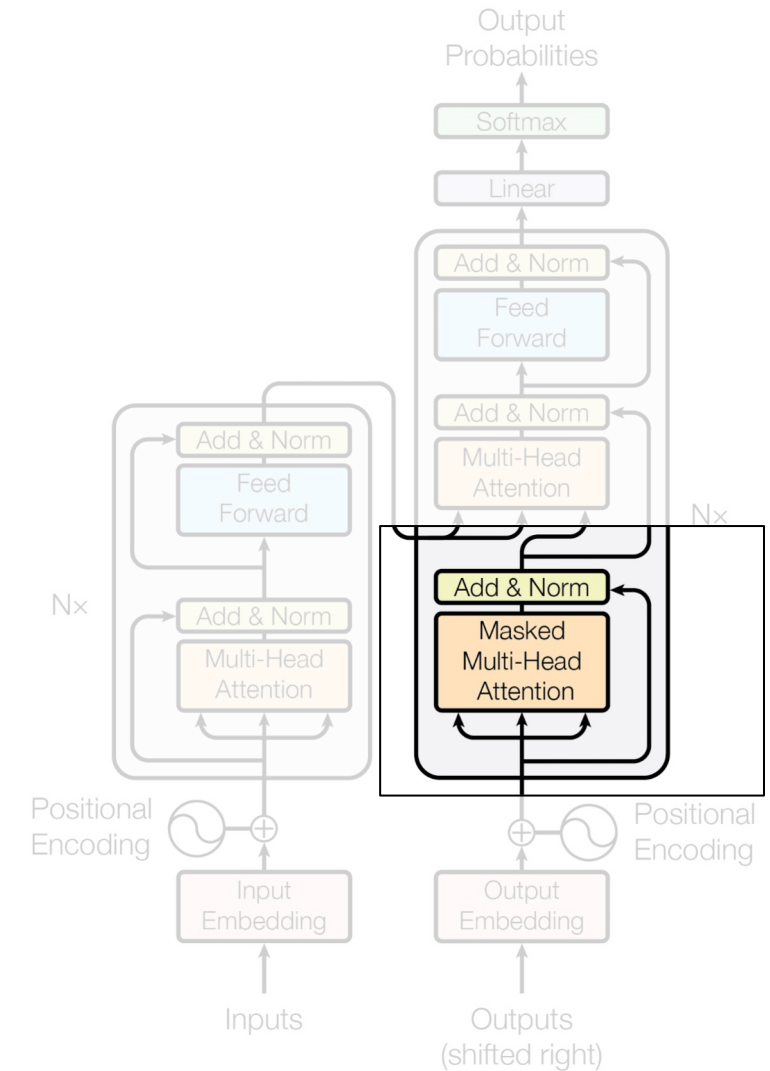


Masked Multi Head Attention

Masked Multi Head Attention

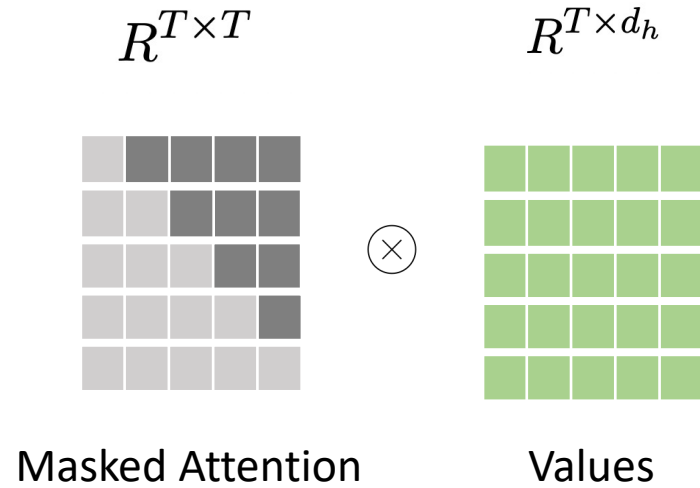


Masked Multi Head Attention : Z'

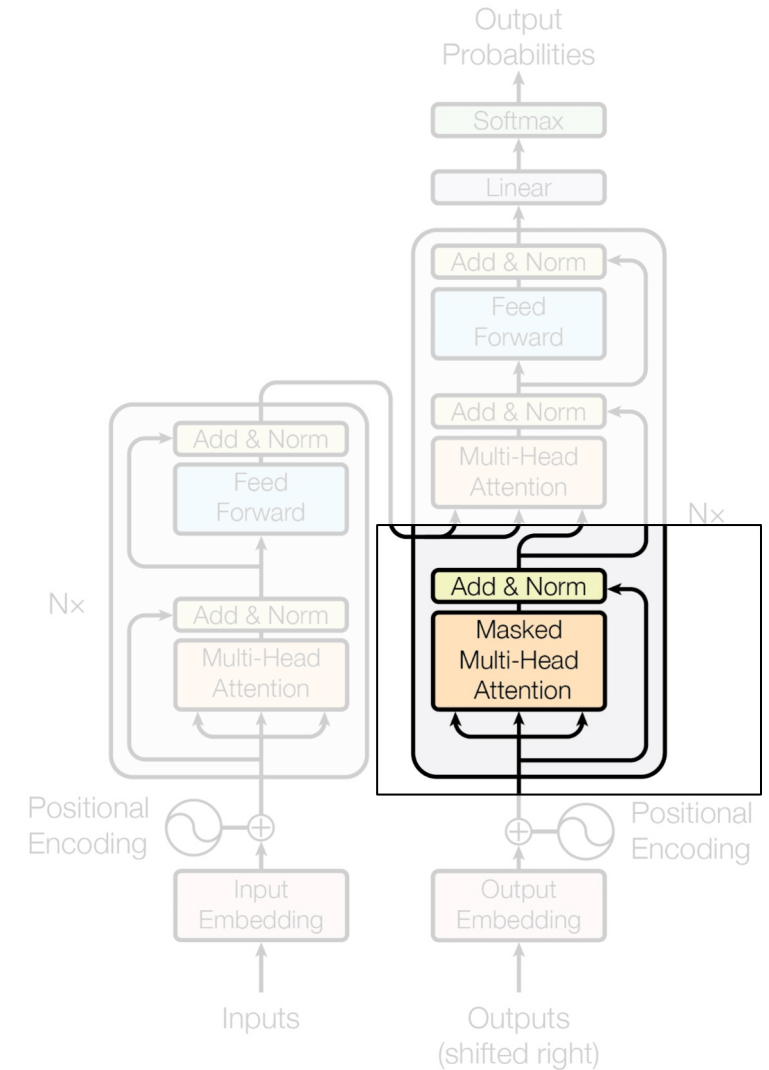


Masked Multi Head Attention

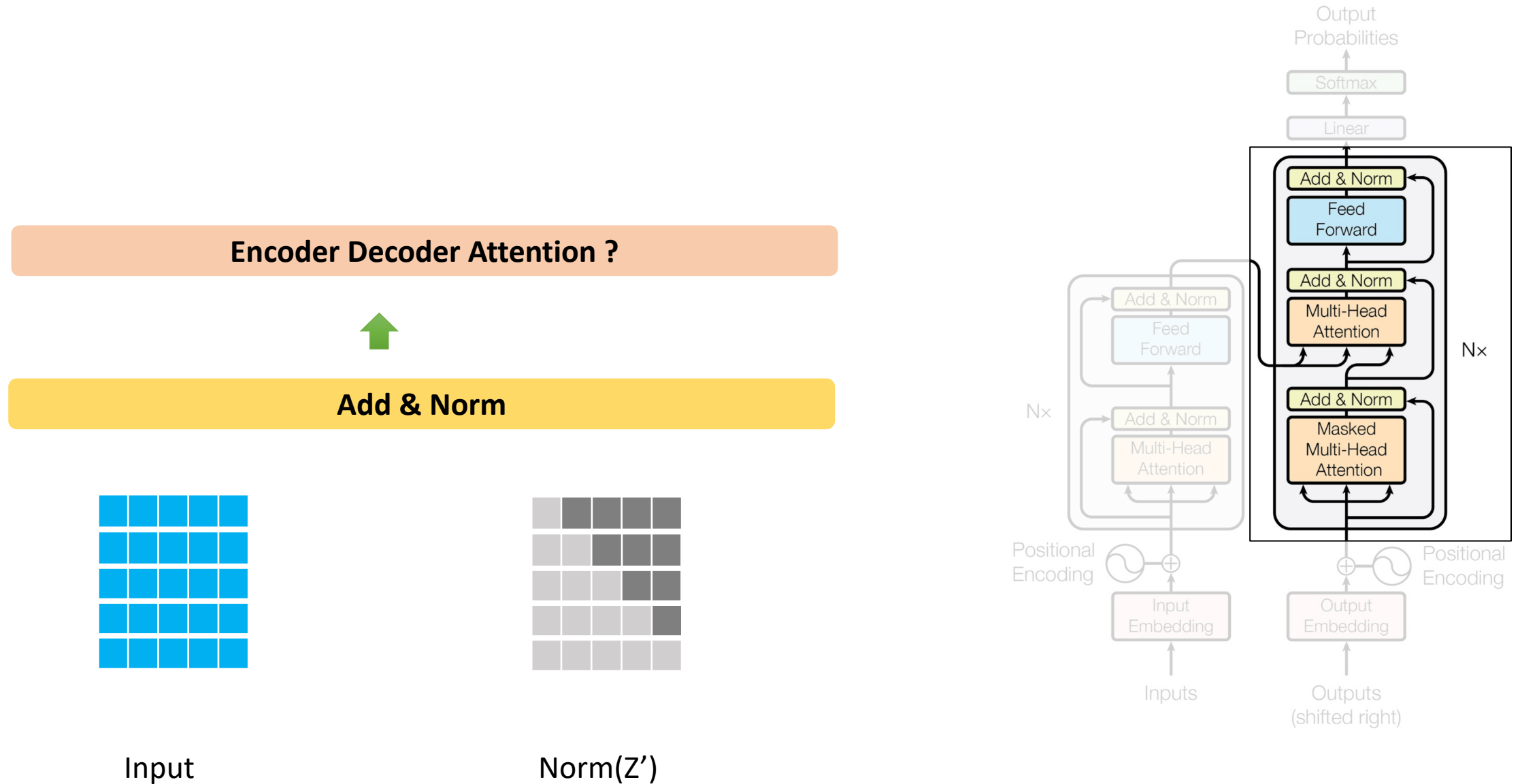
Masked Multi Head Attention



Masked Multi Head Attention : Z'

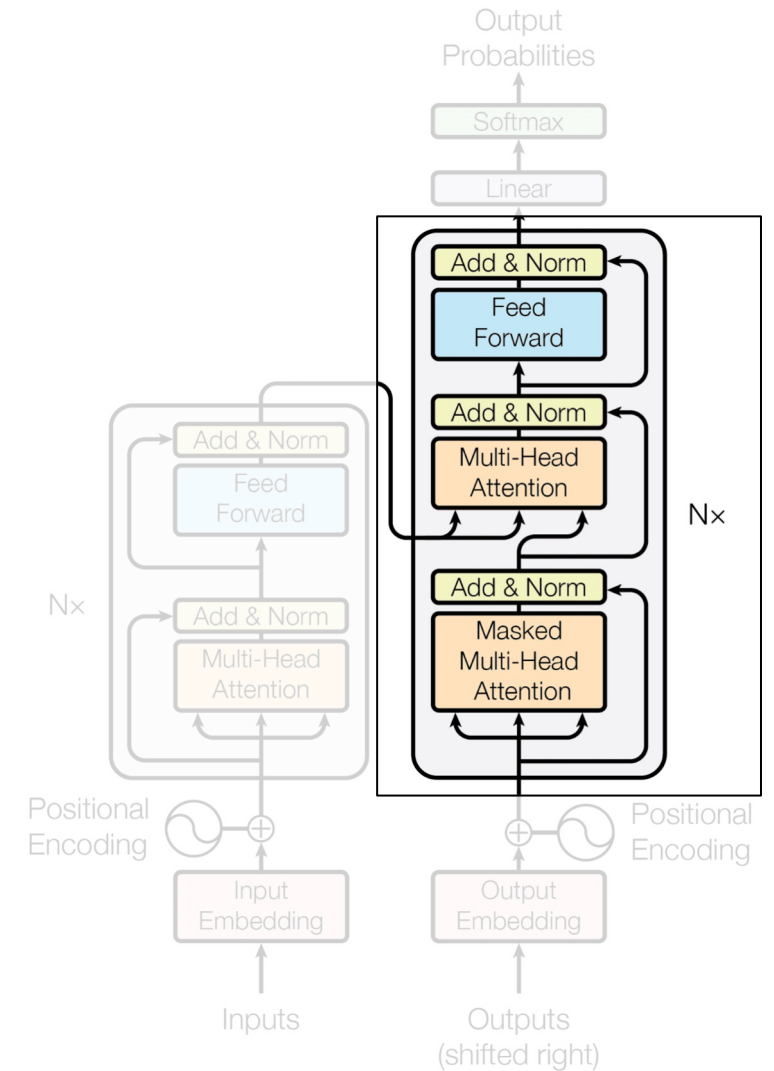


Encoder Decoder Attention



Encoder Decoder Attention

Encoder Decoder Attention ?



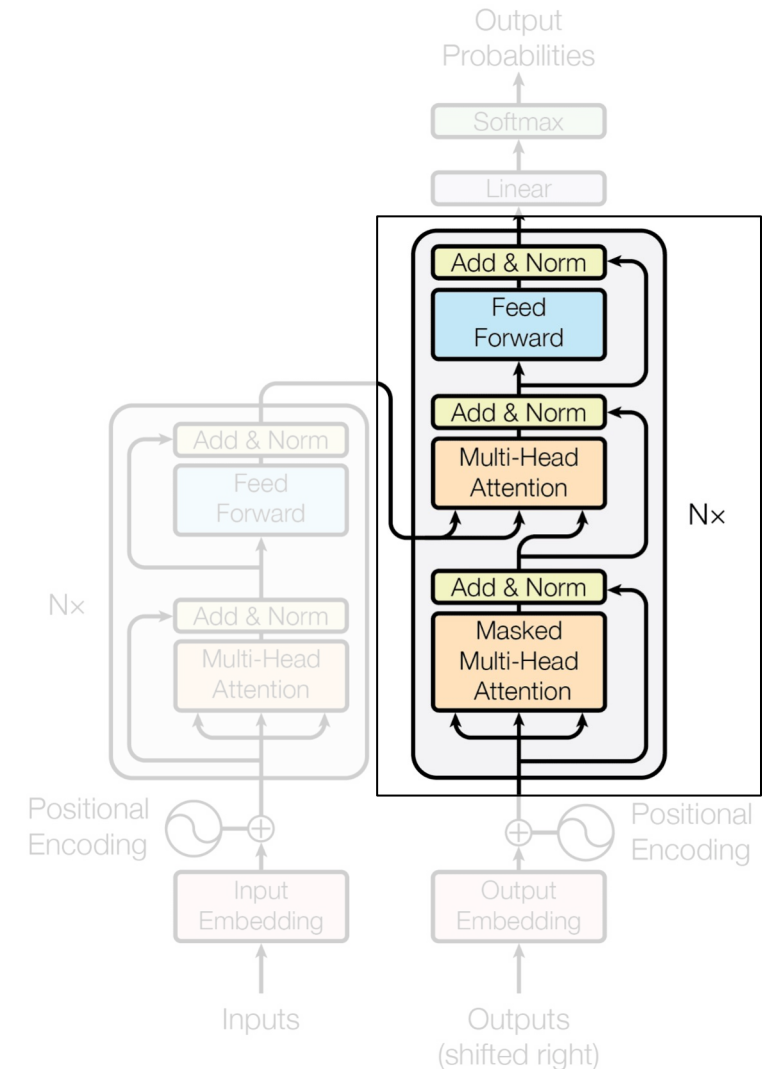
Encoder Decoder Attention

Encoder **Self** Attention

1. Queries from Encoder Inputs
2. Keys from Encoder Inputs
3. Values from Encoder Inputs

Decoder **Masked Self** Attention

1. Queries from Decoder Inputs
2. Keys from Decoder Inputs
3. Values from Decoder Inputs



Attention

{Key, Value store}

{Query: "Order details of order_104"}

{Query: "Order details of order_106"}

```
{ "order_100": { "items": "a1", "delivery_date": "a2", ... },  
  "order_101": { "items": "b1", "delivery_date": "b2", ... },  
  "order_102": { "items": "c1", "delivery_date": "c2", ... },  
  "order_103": { "items": "d1", "delivery_date": "d2", ... },  
  "order_104": { "items": "e1", "delivery_date": "e2", ... },  
  "order_105": { "items": "f1", "delivery_date": "f2", ... },  
  "order_106": { "items": "g1", "delivery_date": "g2", ... },  
  "order_107": { "items": "h1", "delivery_date": "h2", ... },  
  "order_108": { "items": "i1", "delivery_date": "i2", ... },  
  "order_109": { "items": "j1", "delivery_date": "j2", ... },  
  "order_110": { "items": "k1", "delivery_date": "k2", ... }
```

Encoder Decoder Attention

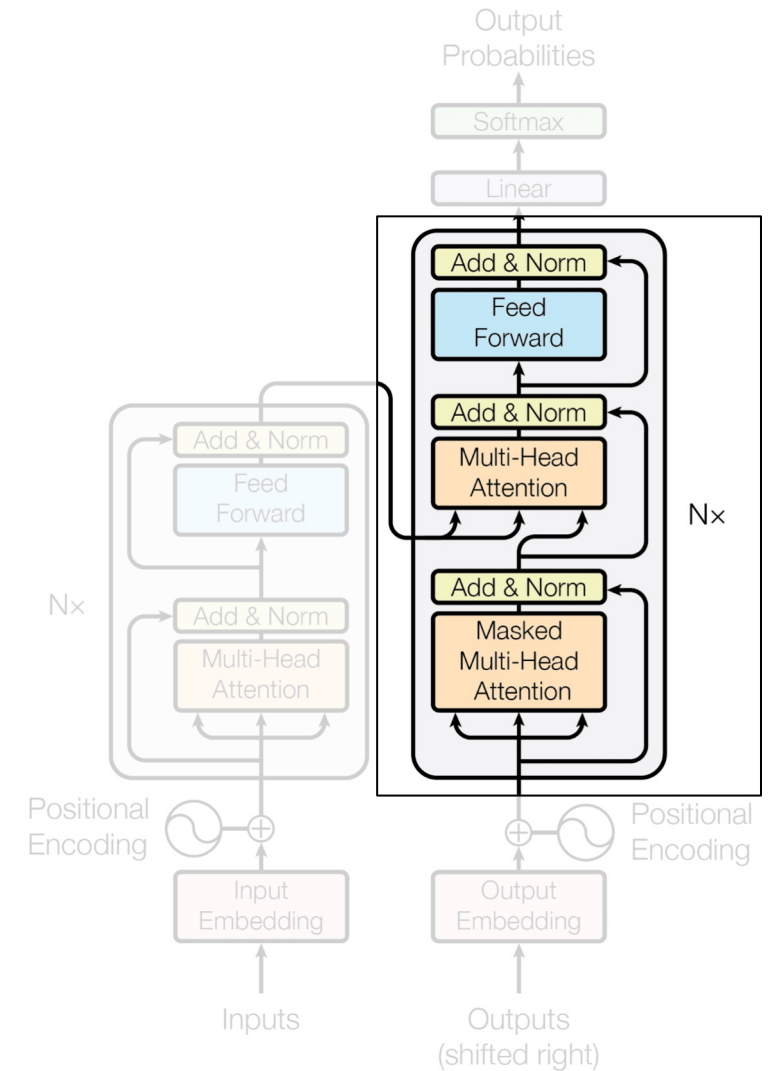
Encoder

Keys from **Encoder Outputs**
Values from **Encoder Outputs**

Decoder

Queries from **Decoder Inputs**

NOTE: Every decoder block receives the same FINAL encoder output



Encoder Decoder Attention



$$R^{T_d \times d_{model}}$$

Z''

$$R^{T_d \times T_e}$$

$$\text{softmax}\left(\frac{Q_d K_e^T}{\sqrt{d_{model}}}\right).$$

$$V_e R^{T_e \times d_{model}}$$

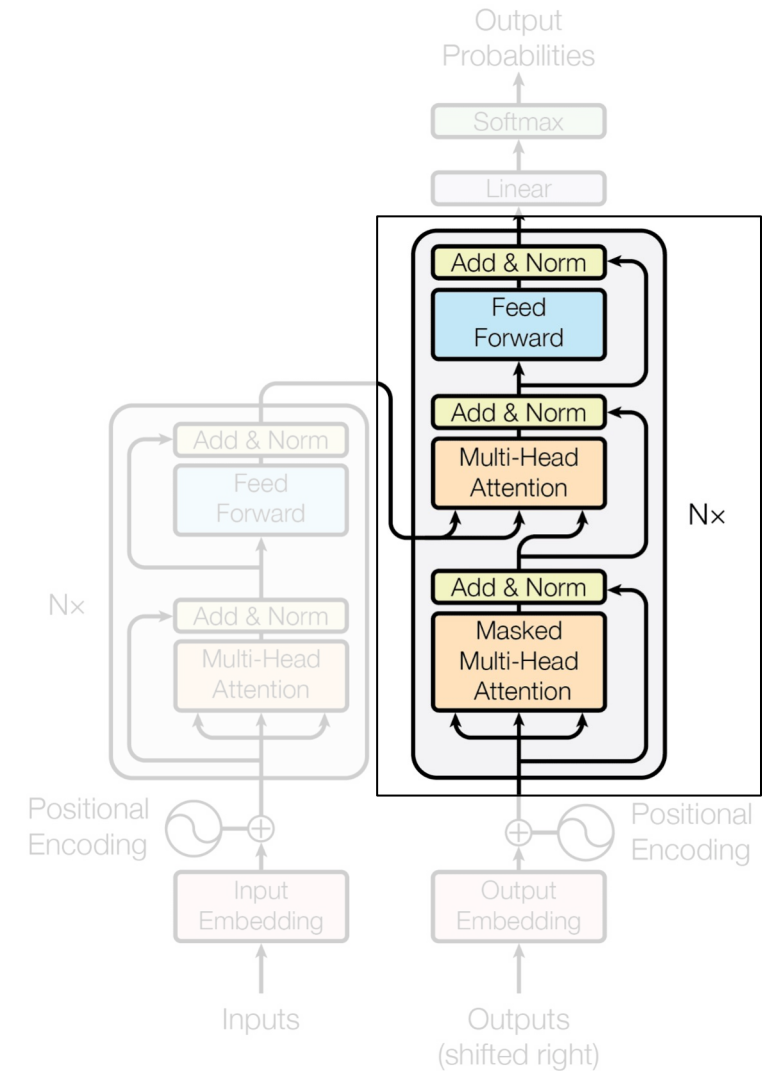
$$R^{T_d \times T_e}$$

$$\text{softmax}\left(\frac{Q_d K_e^T}{\sqrt{d_{model}}}\right)$$

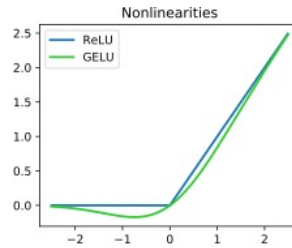
$$R^{T_d \times d_{model}}$$

$Q_d \quad K_e$

$$R^{T_e \times d_{model}}$$

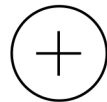
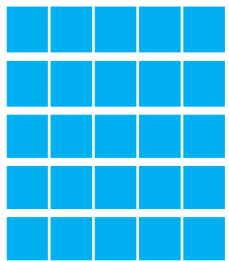


Encoder Decoder Attention

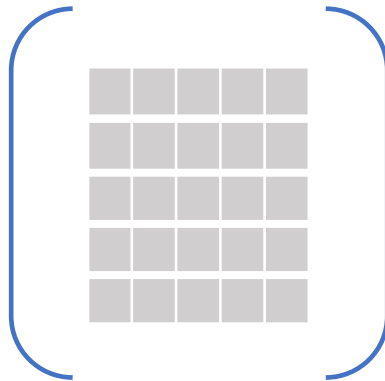


- Non Linearity
- Complex Relationships
- Learn from each other

Feed Forward

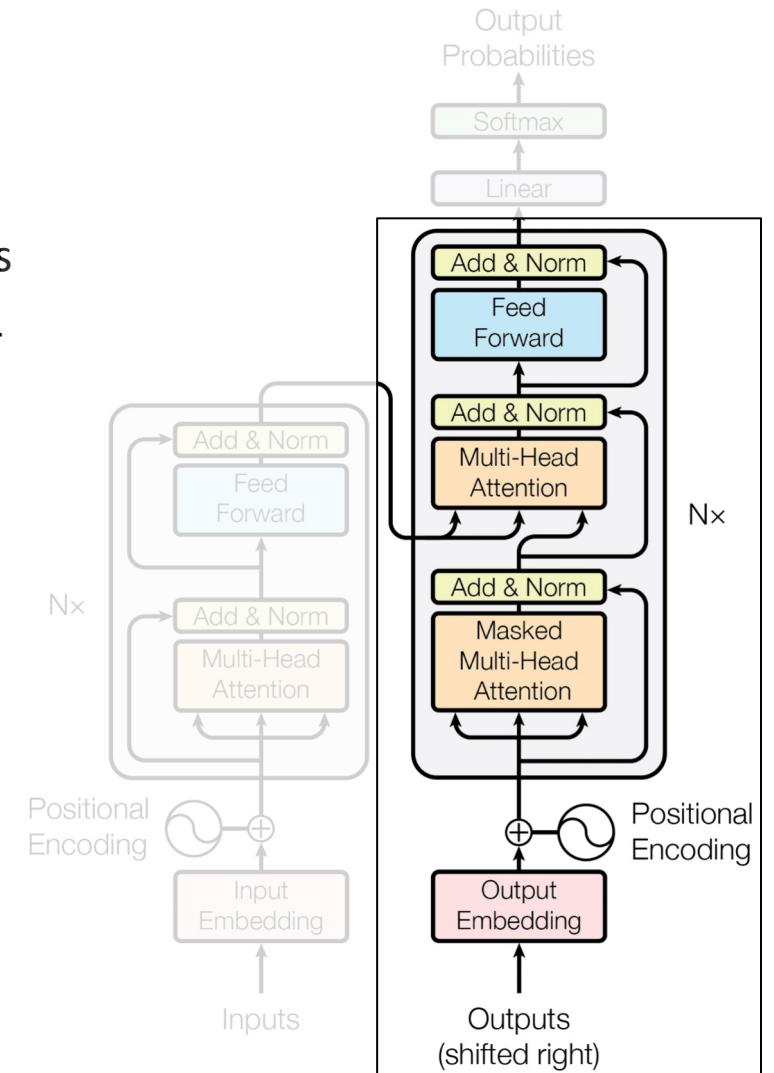


Residuals



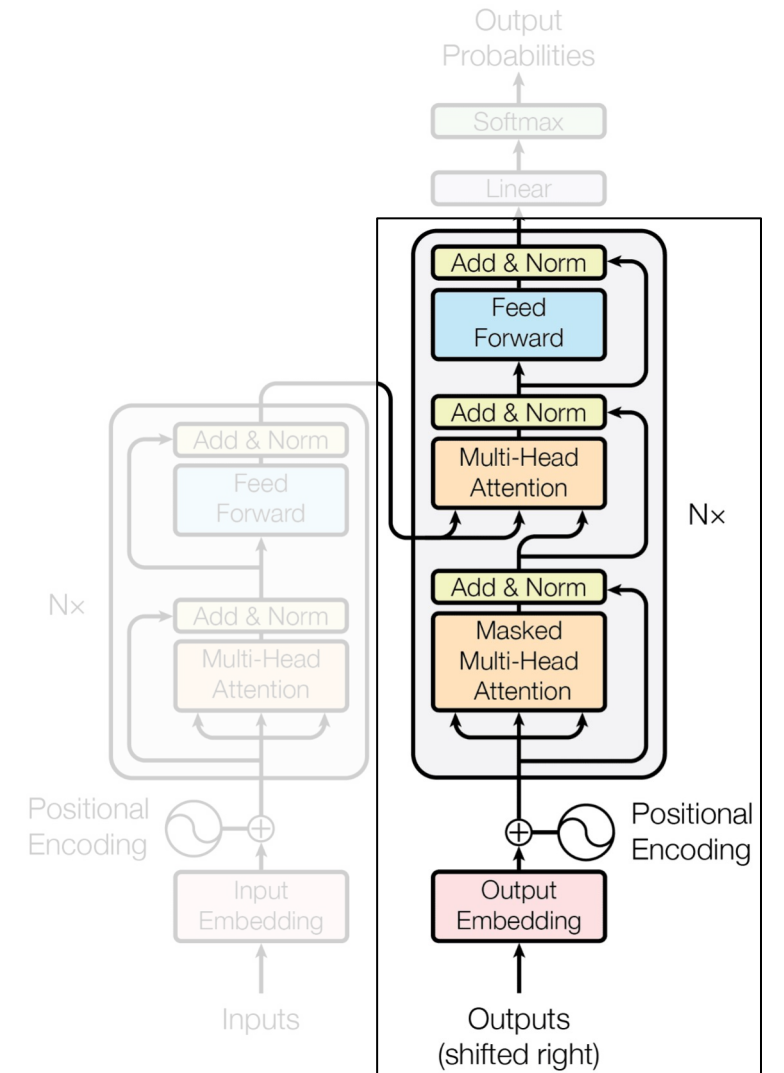
Add n Norm Decoder Self Attn

Norm(Z'')

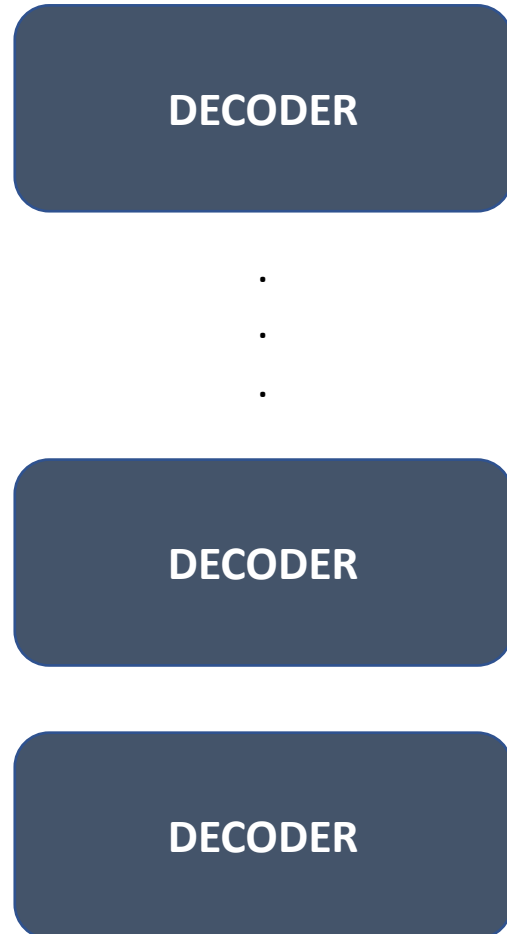


Decoder

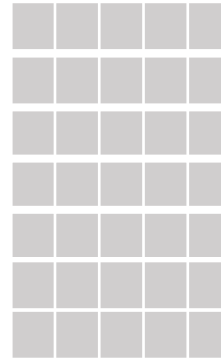
DECODER



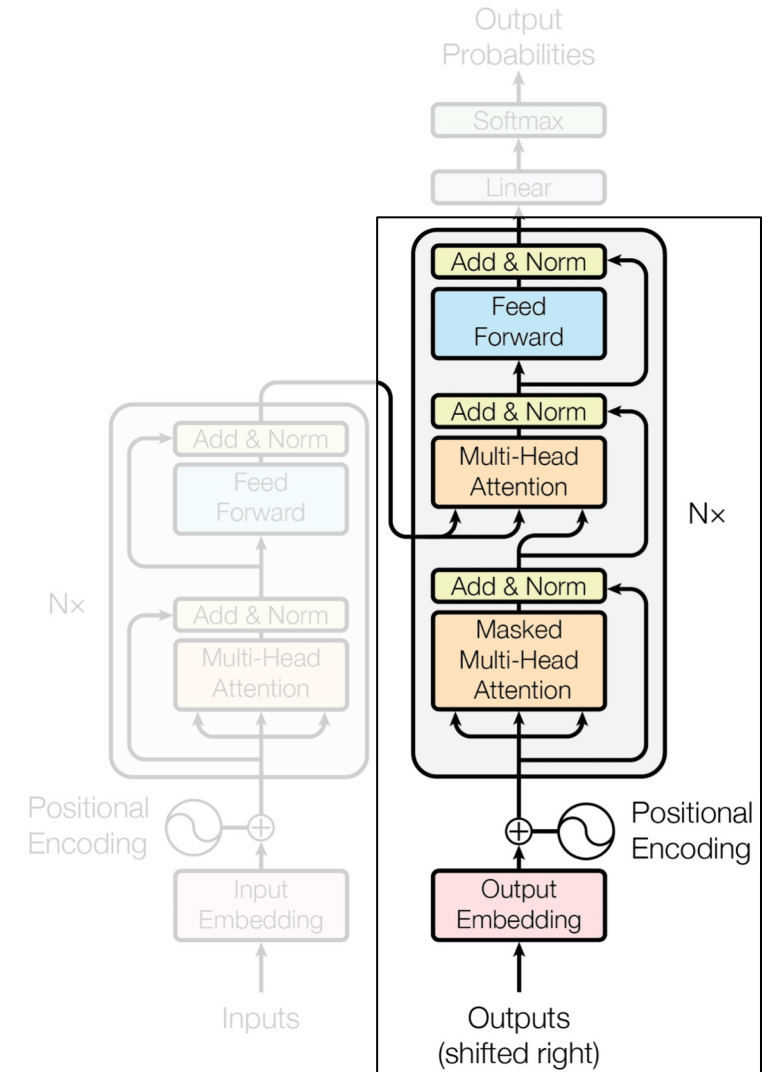
Decoder



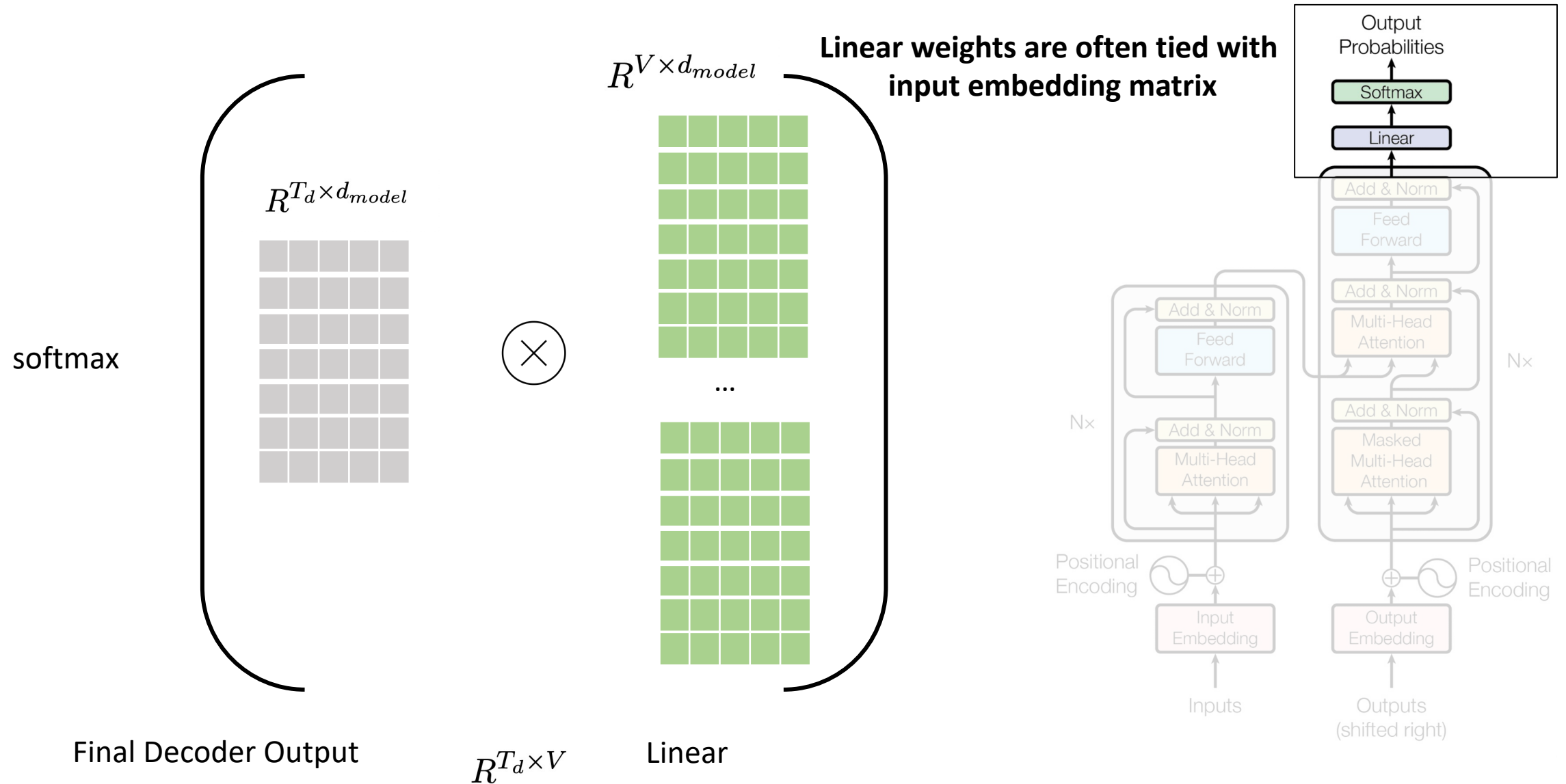
$$R^{T_d \times d_{model}}$$



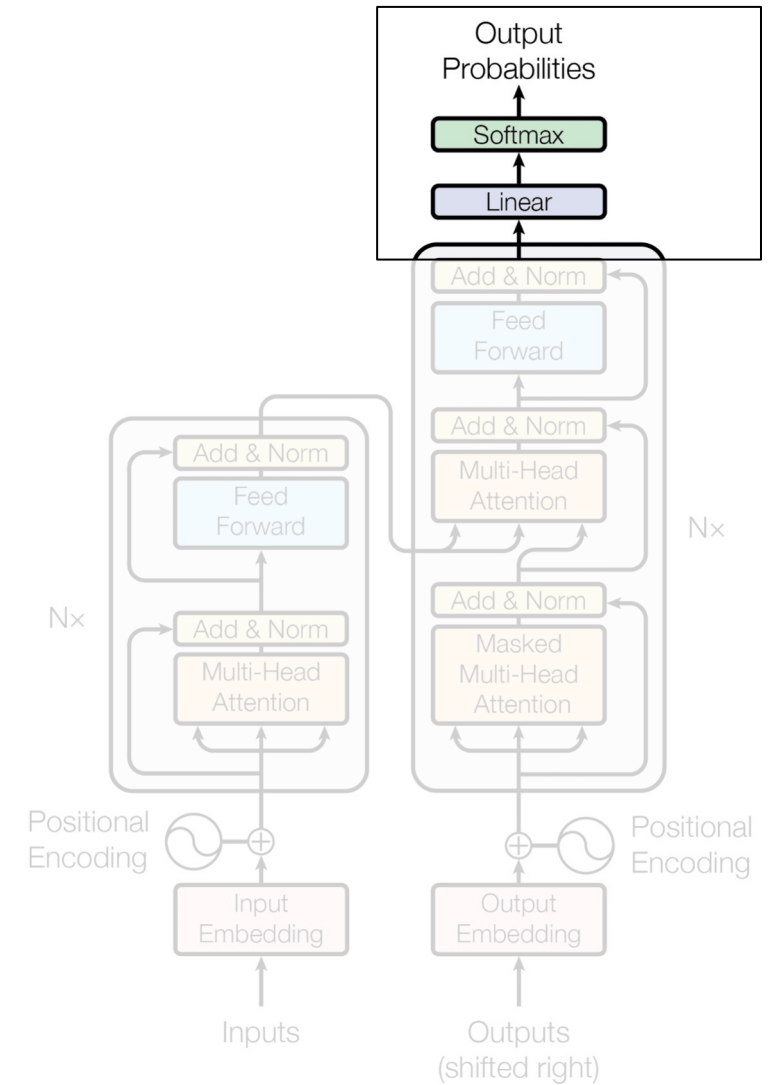
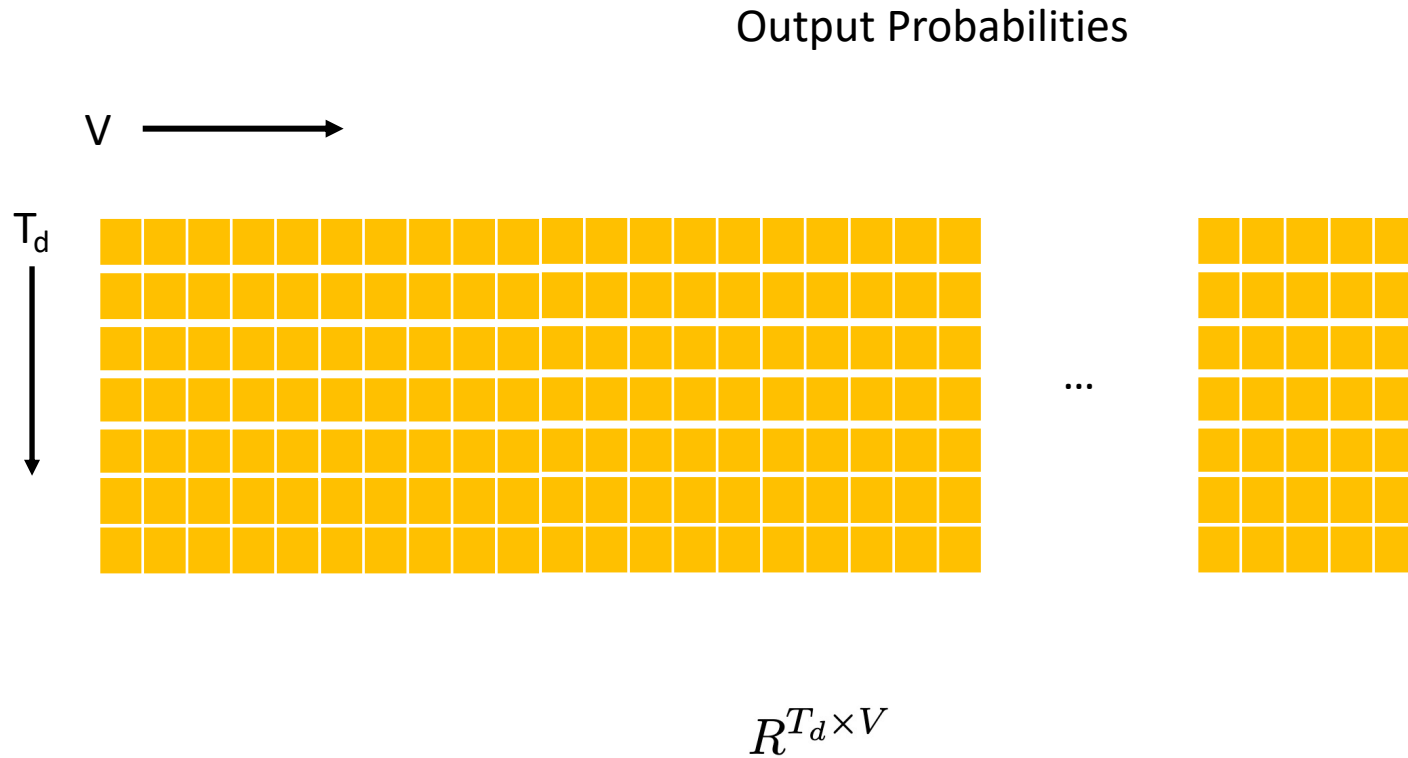
Decoder output



Linear



Softmax



Poll 2 (@1297)

Which of the following are true about transformers?

- a. Transformers can always be run in parallel
- b. Transformer decoders can only be parallelized during training
- c. Positional encodings help parallelize the transformer encoder
- d. Queries, keys, and values are obtained by splitting the input into 3 equal segments
- e. Multiheaded attention helps transformers find different kinds of relations between the tokens
- f. During decoding, decoder outputs function as queries and keys while the values come from the encoder

Poll 2 (@1126)

Which of the following are true about transformers?

- a. Transformers can always be run in parallel
- b. **Transformer decoders can only be parallelized during training**
- c. Positional encodings help parallelize the transformer encoder
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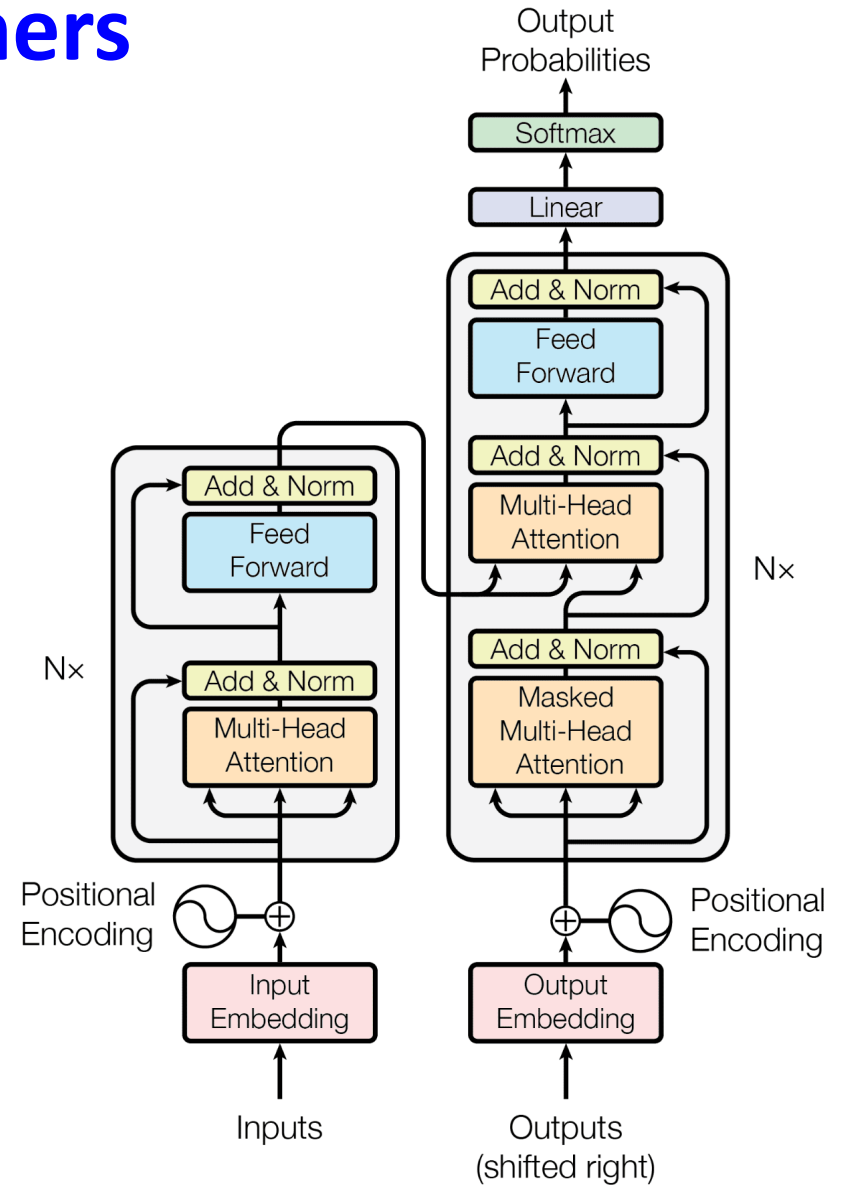
Transformers

Targets
Ich have einen apfel gegessen



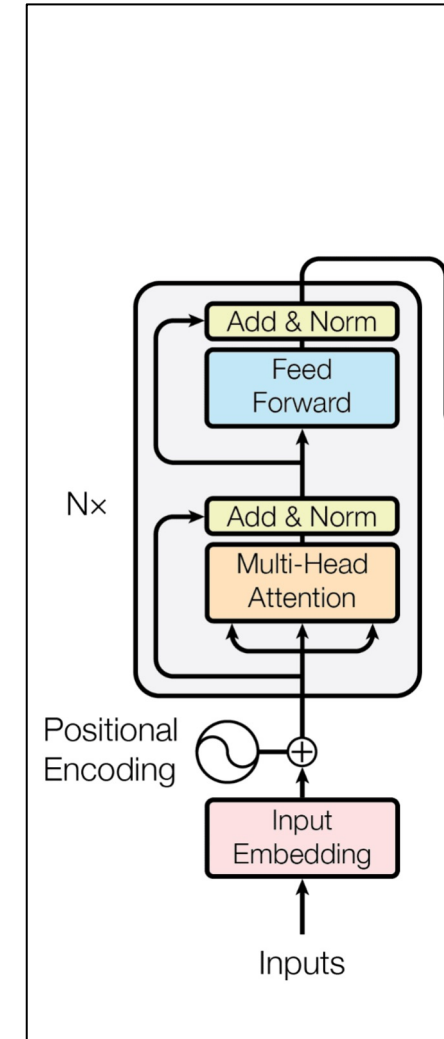
Inputs
I ate an apple

Machine Translation



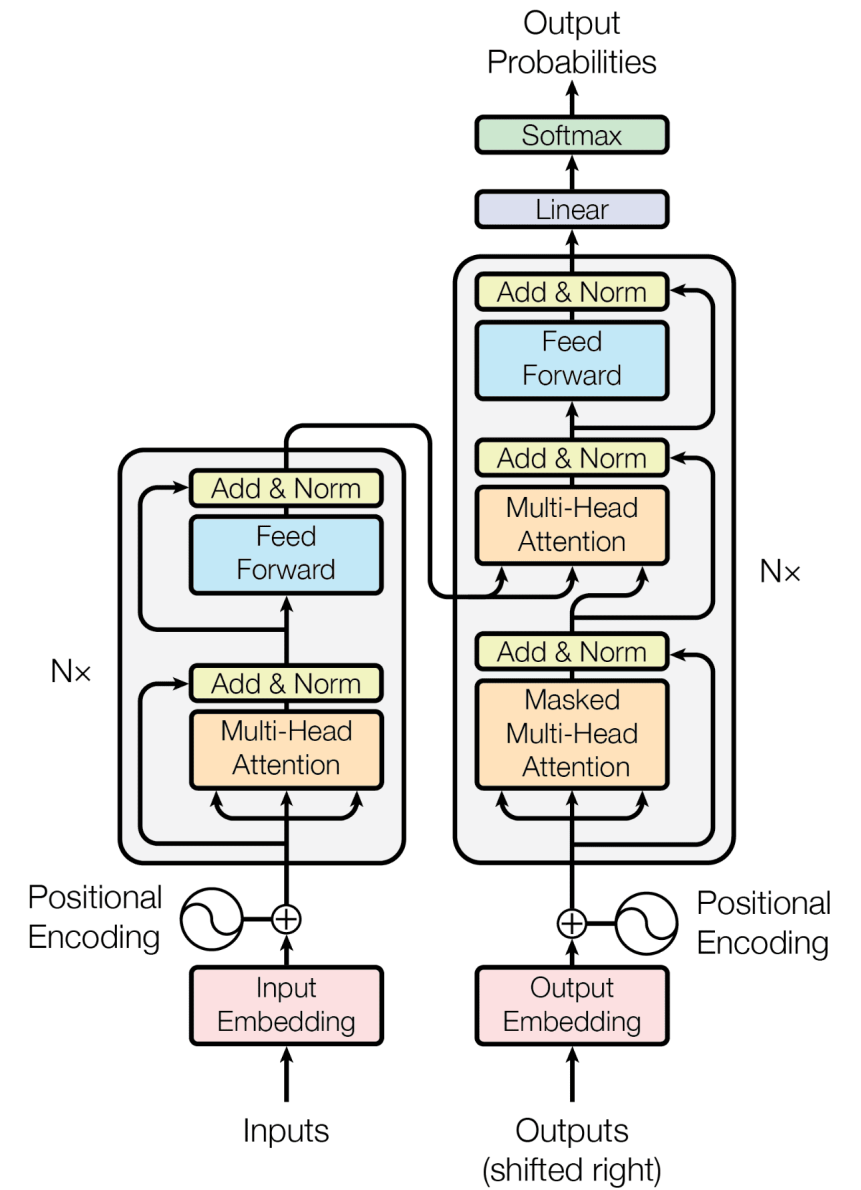
Transformers

- ✓ Tokenization
- ✓ Input Embeddings
- ✓ Position Encodings
- ✓ Residuals
- ✓ Query
- ✓ Key
- ✓ Value
- ✓ Add & Norm
- ✓ Encoder
- ✓ Decoder
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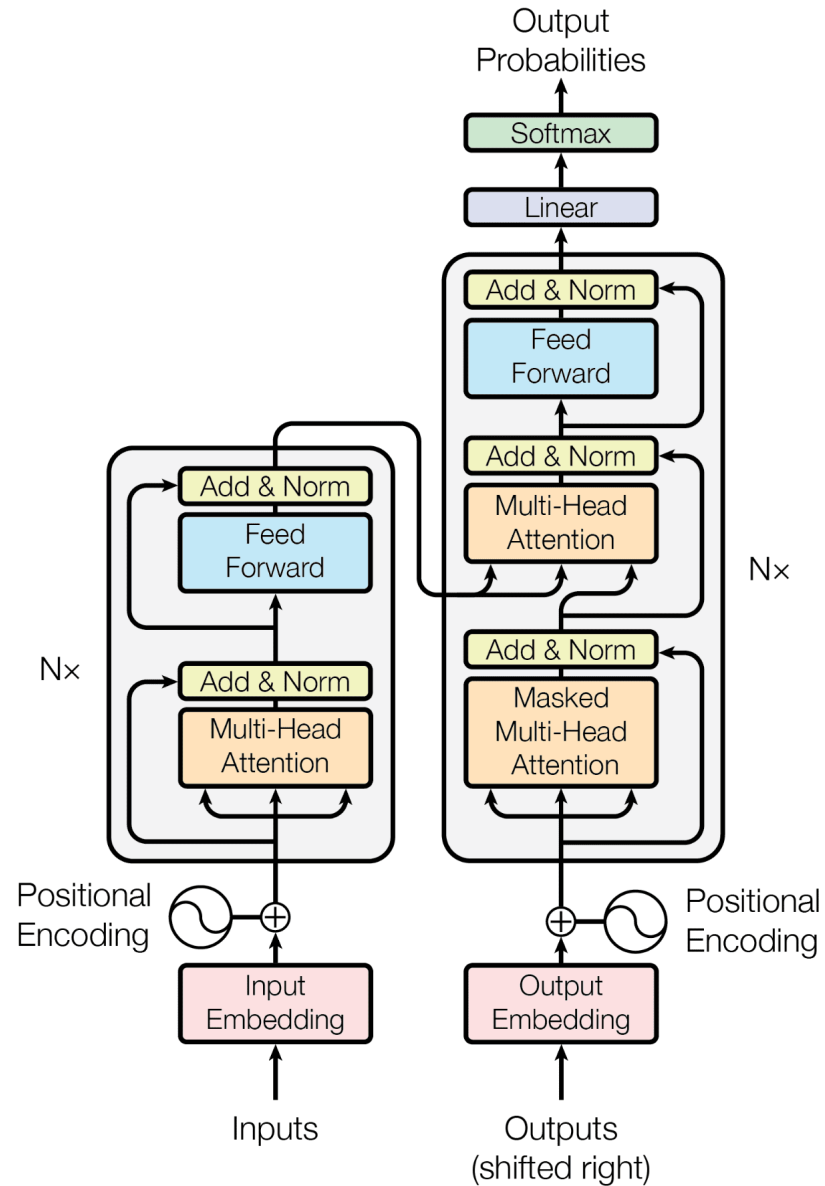


Part 2

LLM

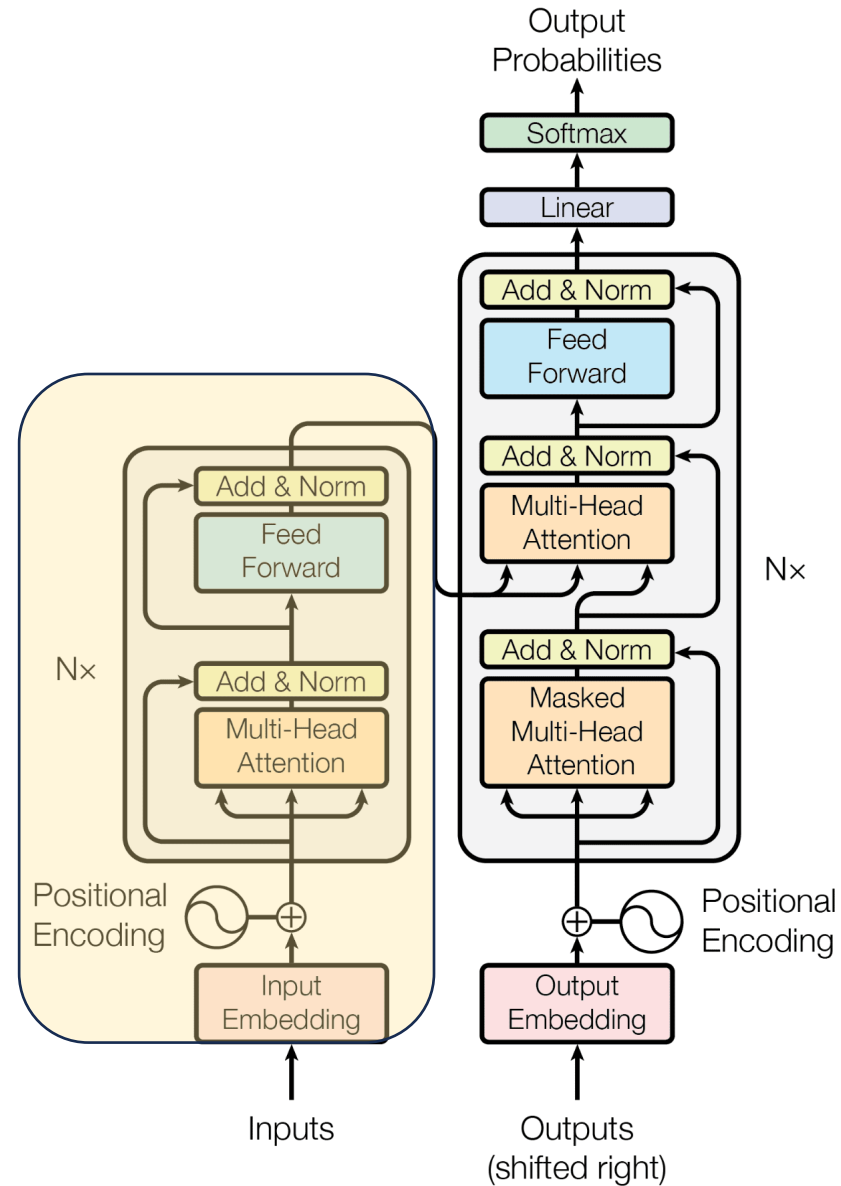


Transformers, mid-2017



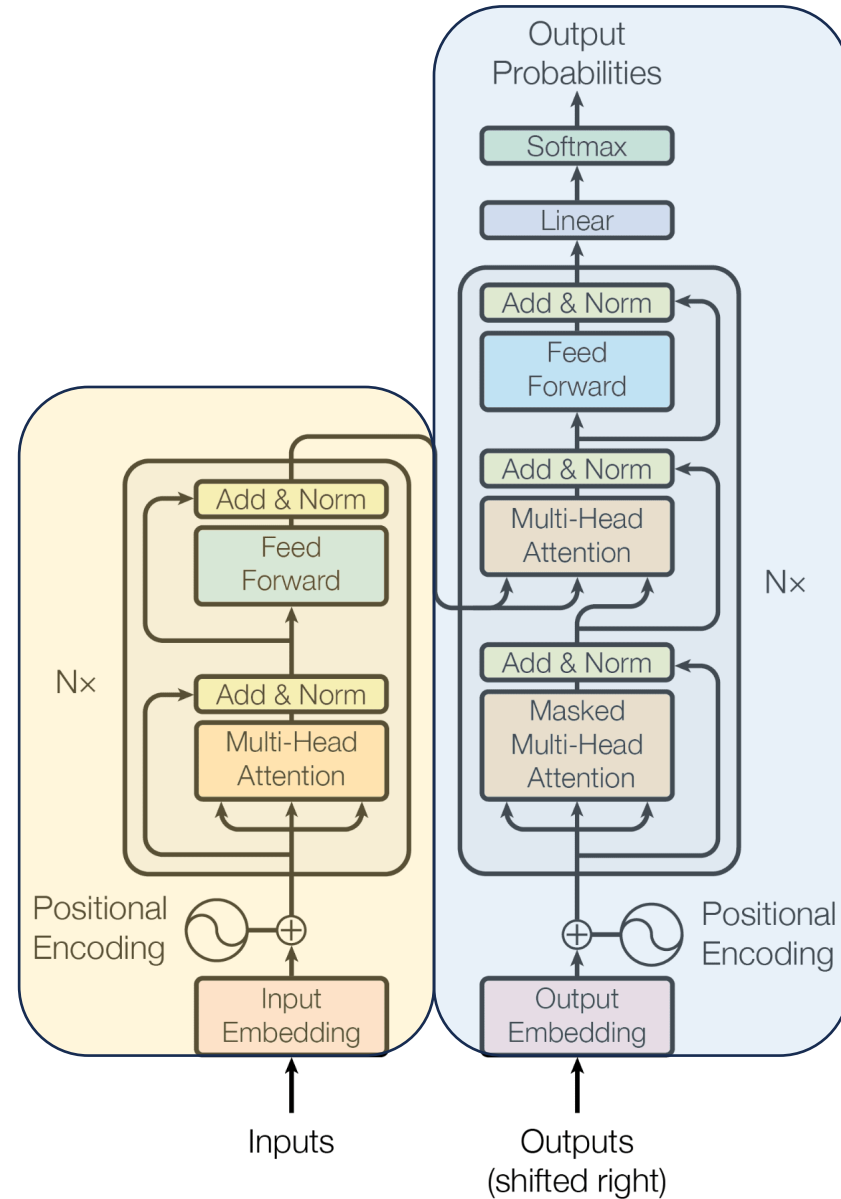
Transformers, mid-2017

Representation



Transformers, mid-2017

Representation

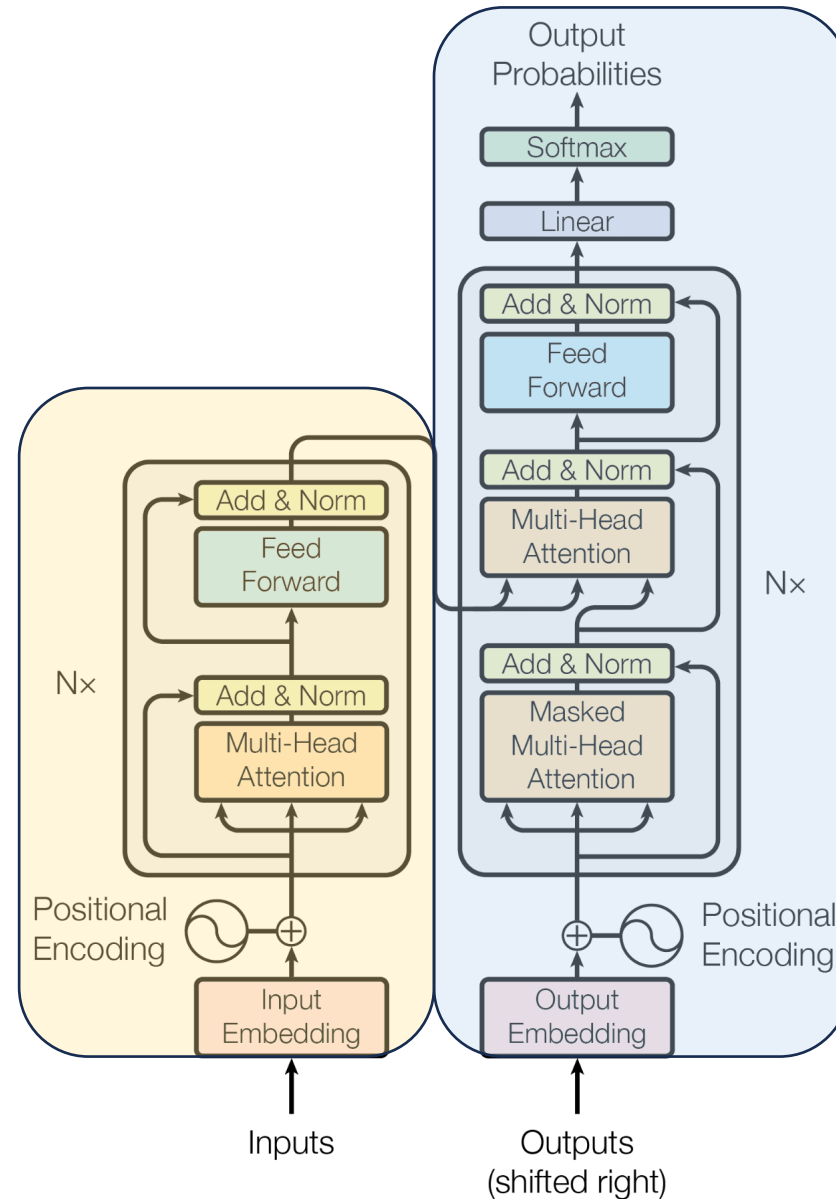


Generation

Transformers, mid-2017

Input – input tokens
Output – hidden states

Representation



Input – output tokens and hidden states*
Output – output tokens

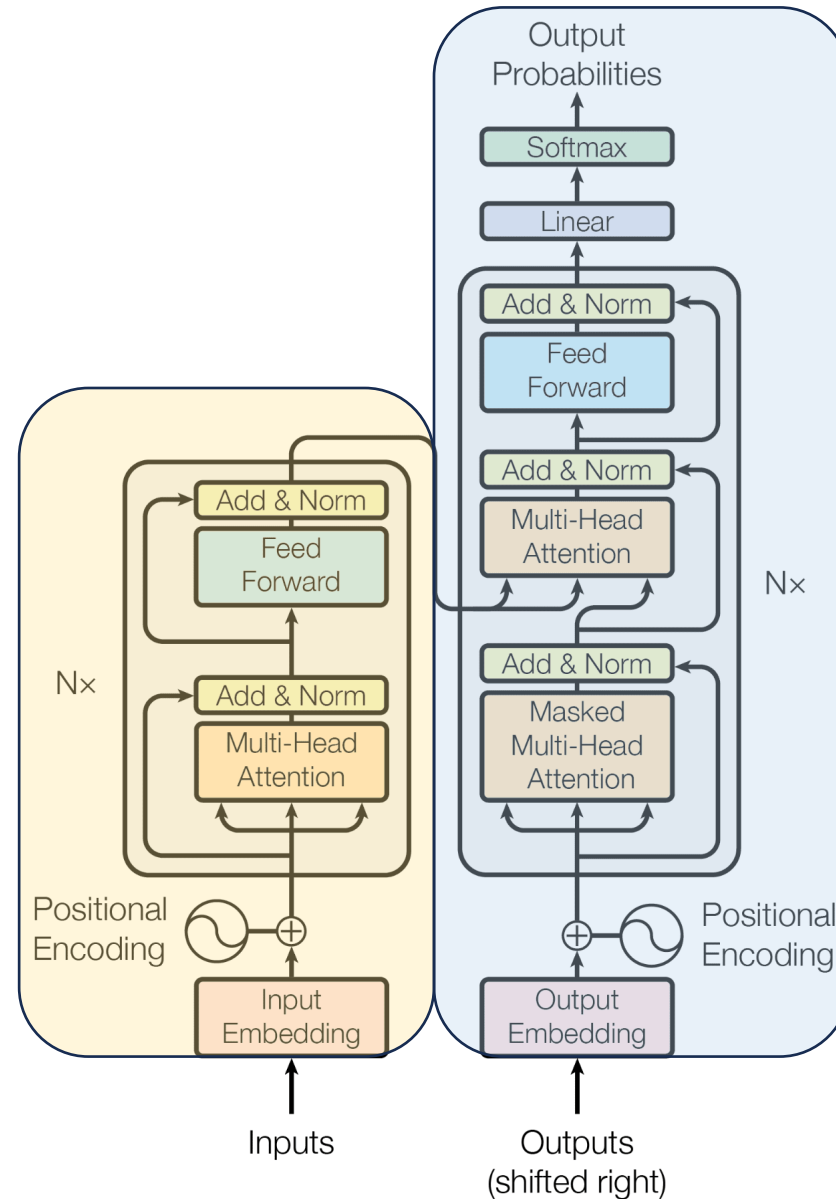
Generation

Transformers, mid-2017

Input – input tokens
Output – hidden states

Model can see all timesteps

Representation



Input – output tokens and hidden states*
Output – output tokens

Model can only see previous timesteps

Generation

Transformers, mid-2017

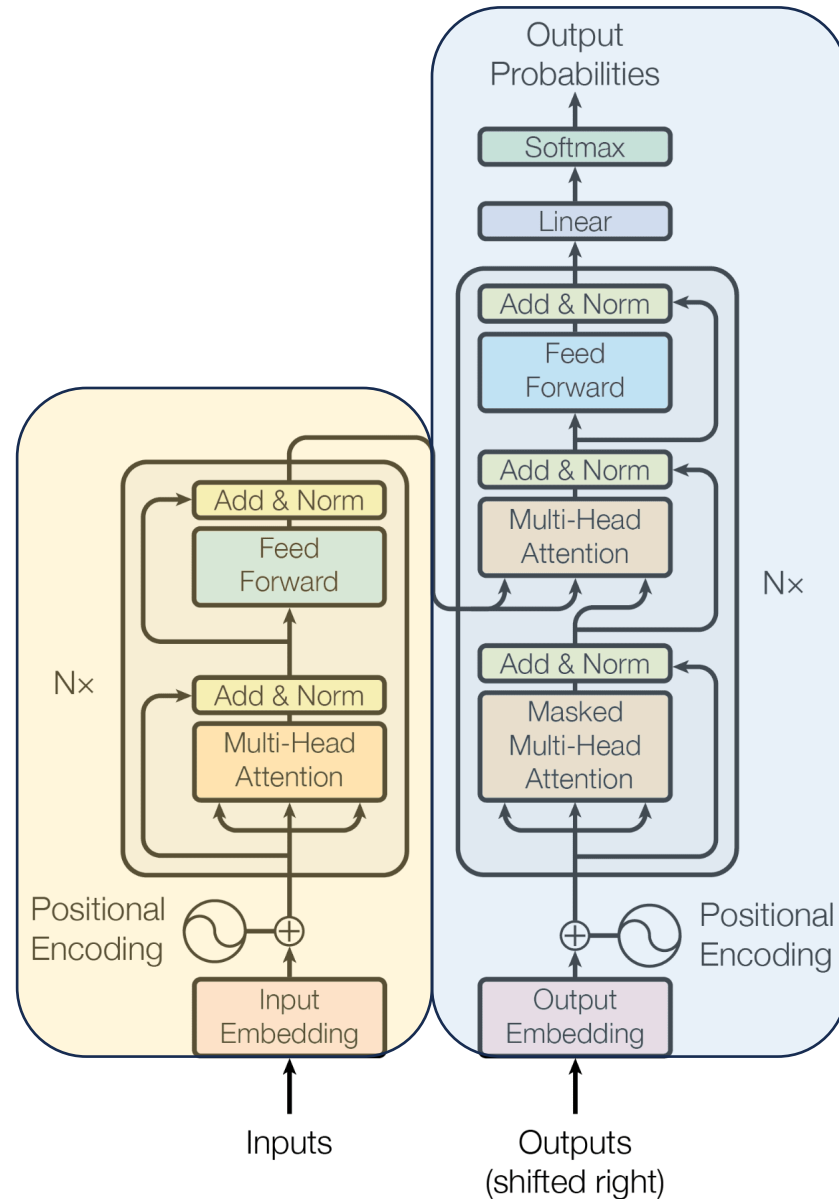
Input – input tokens

Output – hidden states

Model can see all timesteps

Does not usually output tokens, so no inherent auto-regressivity

Representation



Input – output tokens and hidden states*

Output – output tokens

Model can only see previous timesteps

Model is auto-regressive with previous timesteps' outputs

Generation

Transformers, mid-2017

Input – input tokens

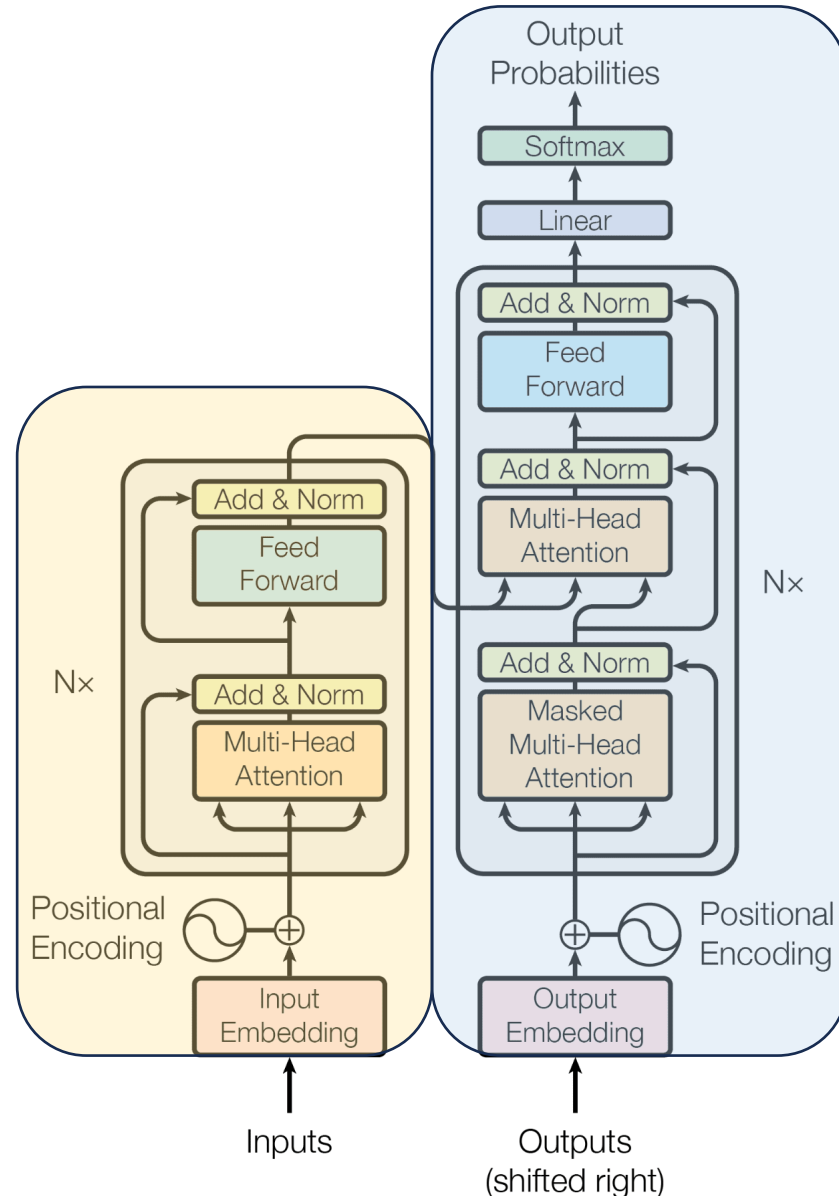
Output – hidden states

Model can see all timesteps

Does not usually output tokens, so no inherent auto-regressivity

Can also be adapted to generate tokens by appending a module that maps hidden state dimensionality to vocab size

Representation



Input – output tokens and hidden states*

Output – output tokens

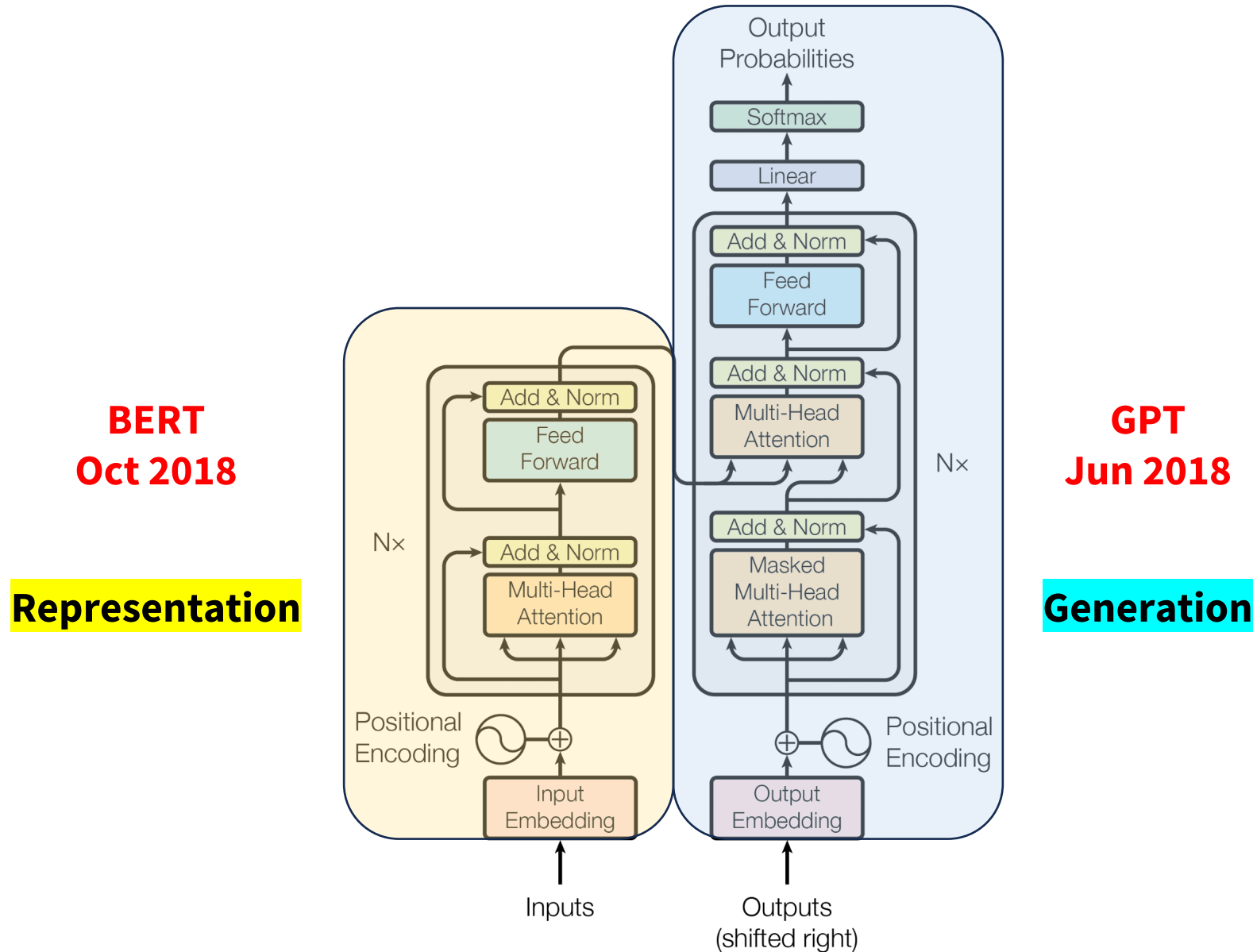
Model can only see previous timesteps

Model is auto-regressive with previous timesteps' outputs

Can also be adapted to generate hidden states by looking before token outputs

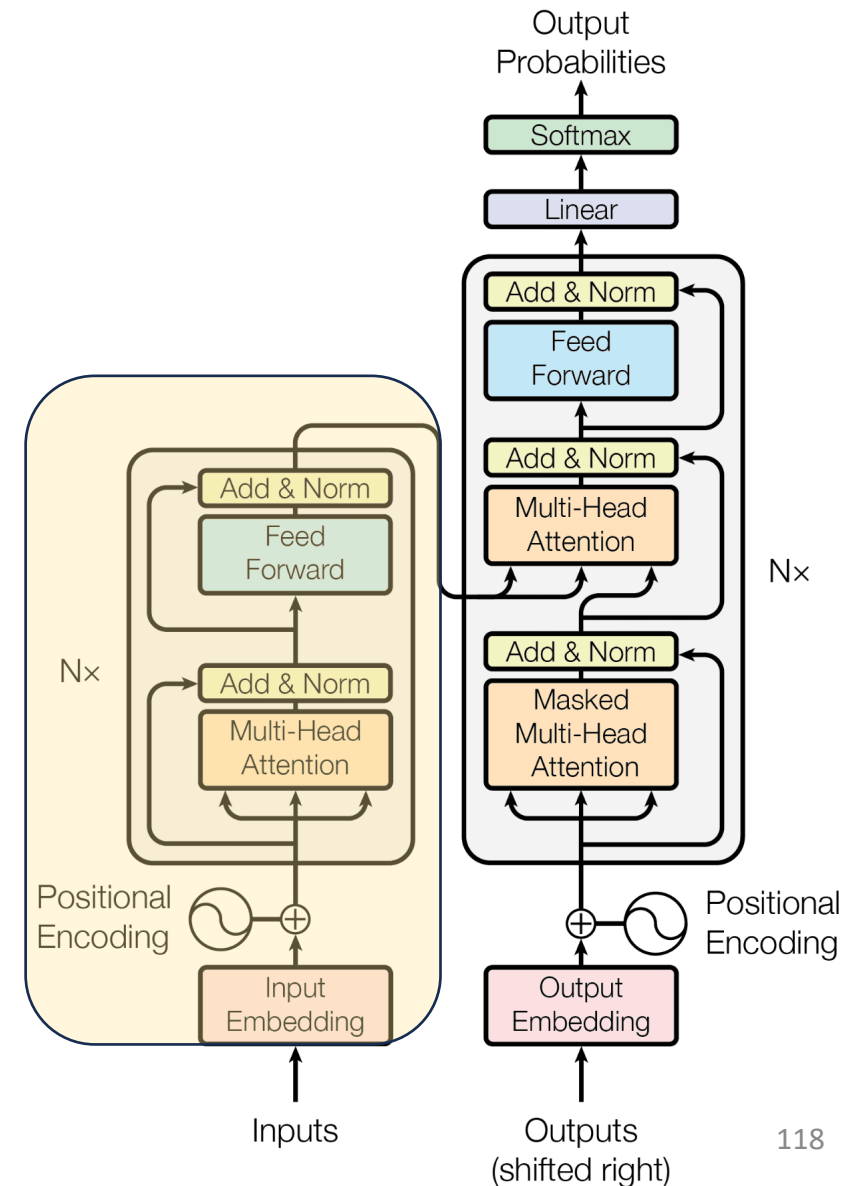
Generation

2018 – The Inception of the LLM Era



BERT - Bidirectional Encoder Representations

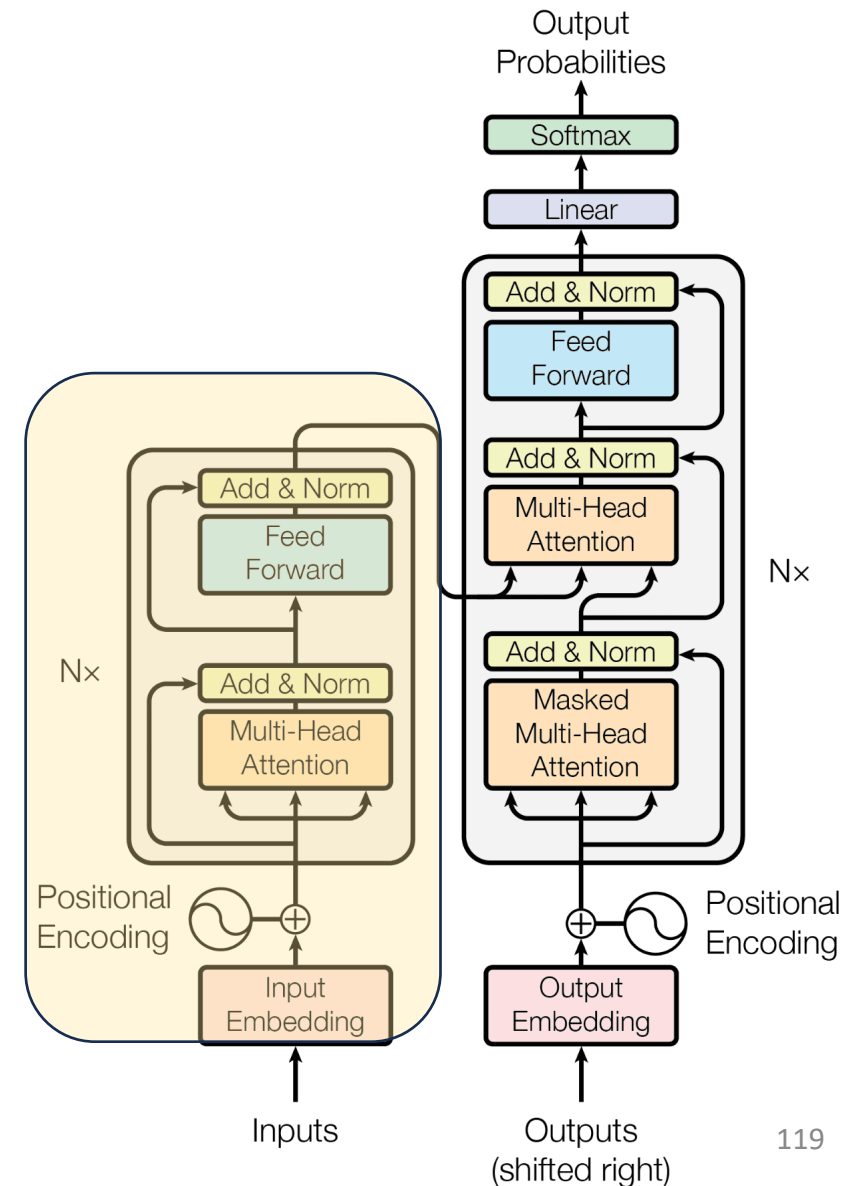
- One of the biggest challenges in LM-building used to be the lack of task-specific training data.
- What if we learn an effective representation that can be applied to a variety of downstream tasks?
 - Word2vec (2013)
 - GloVe (2014)



BERT - Bidirectional Encoder Representations

BERT Pre-Training Corpus:

- English Wikipedia - 2,500 million words
- Book Corpus - 800 million words



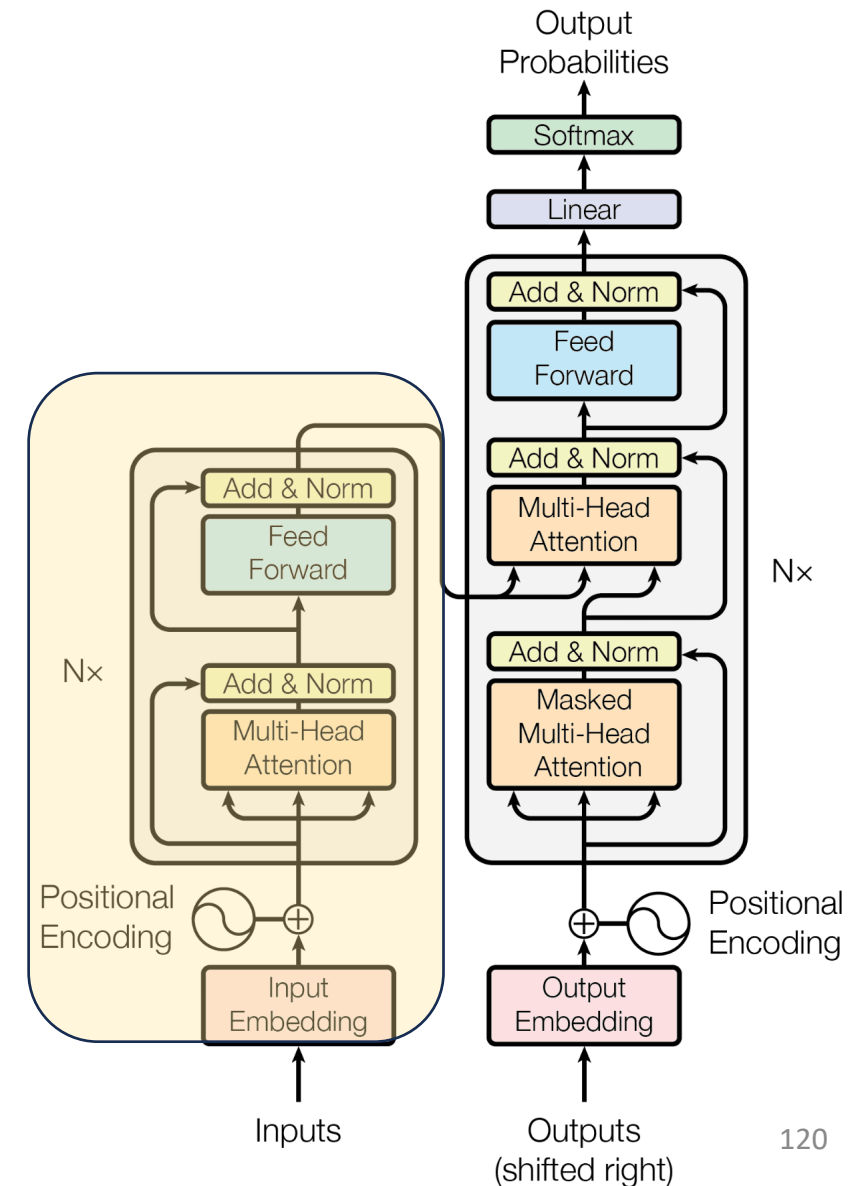
BERT - Bidirectional Encoder Representations

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BERT Pre-Training Tasks:

- MLM (Masked Language Modeling)
- NSP (Next Sentence Prediction)



BERT - Bidirectional Encoder Representations

BERT Pre-Training Corpus:

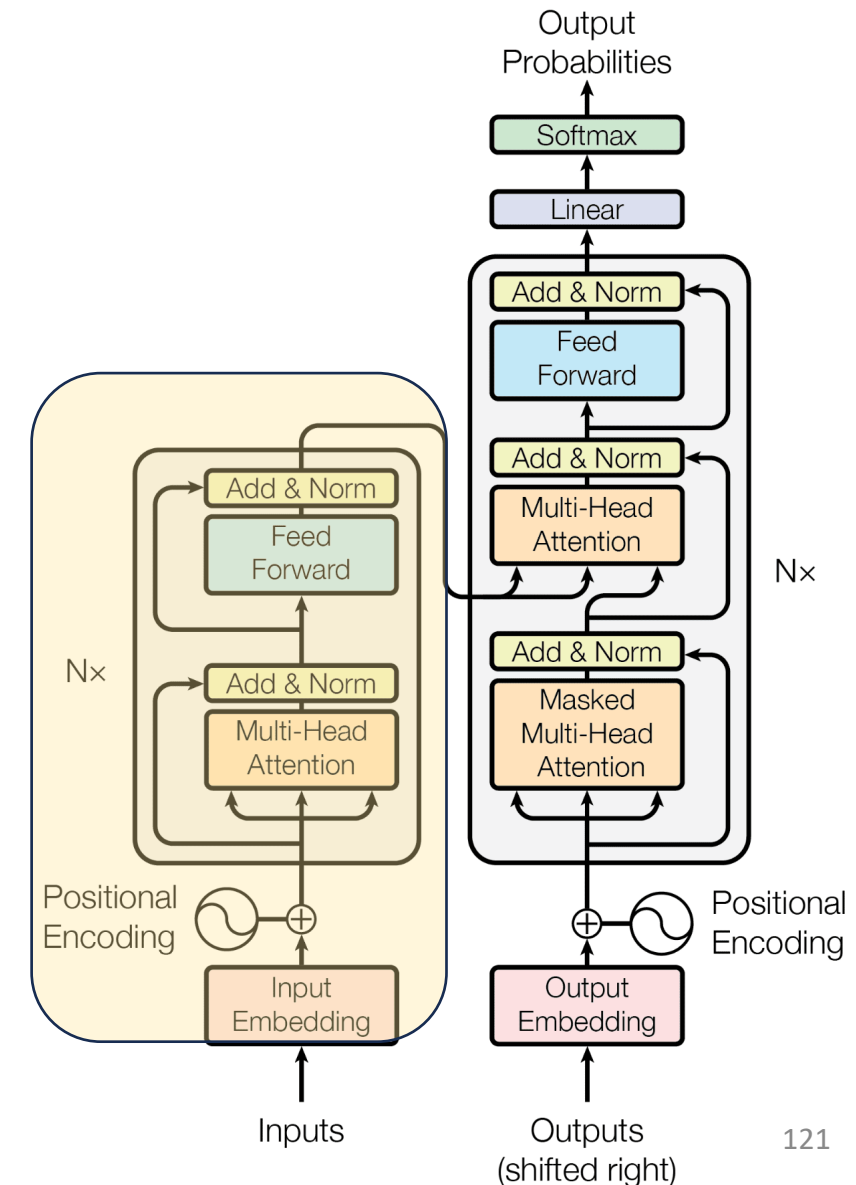
- English Wikipedia - 2,500 million words
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BERT Pre-Training Tasks:

- MLM (Masked Language Modeling)
- NSP (Next Sentence Prediction)

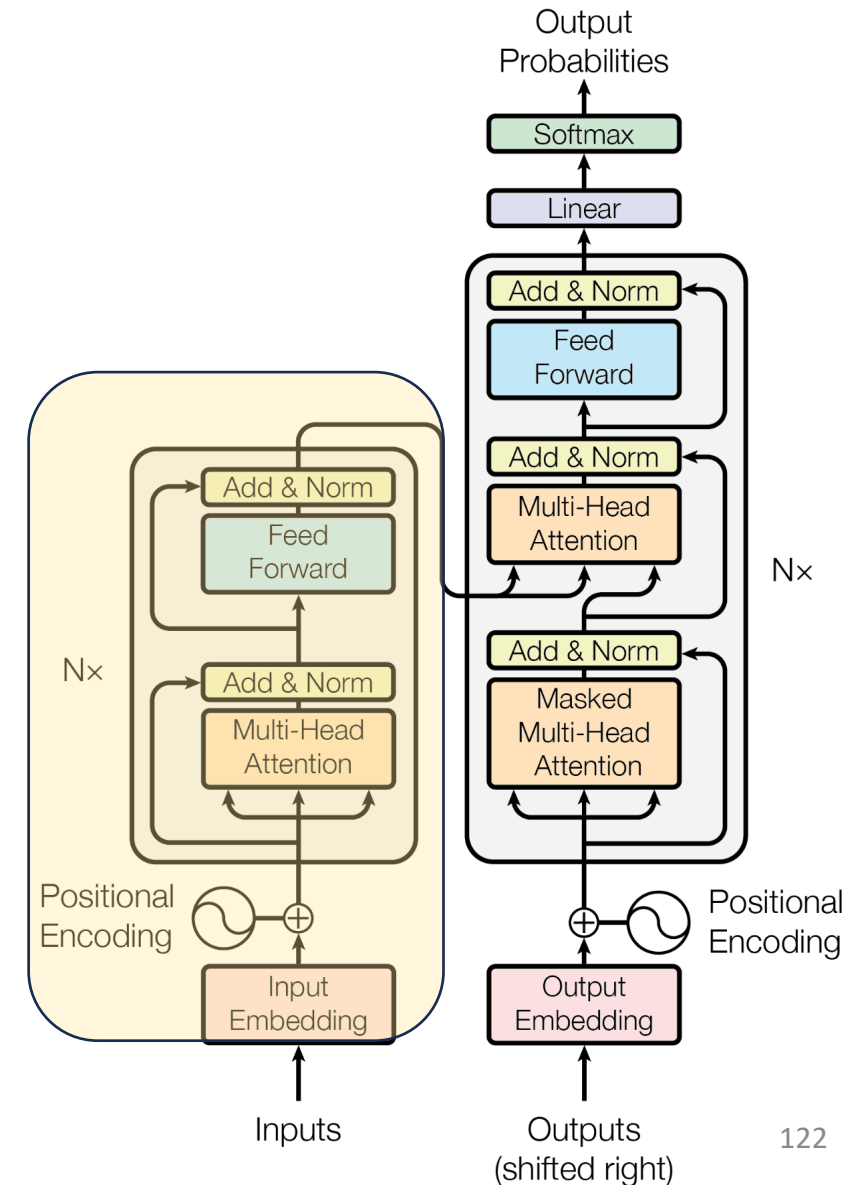
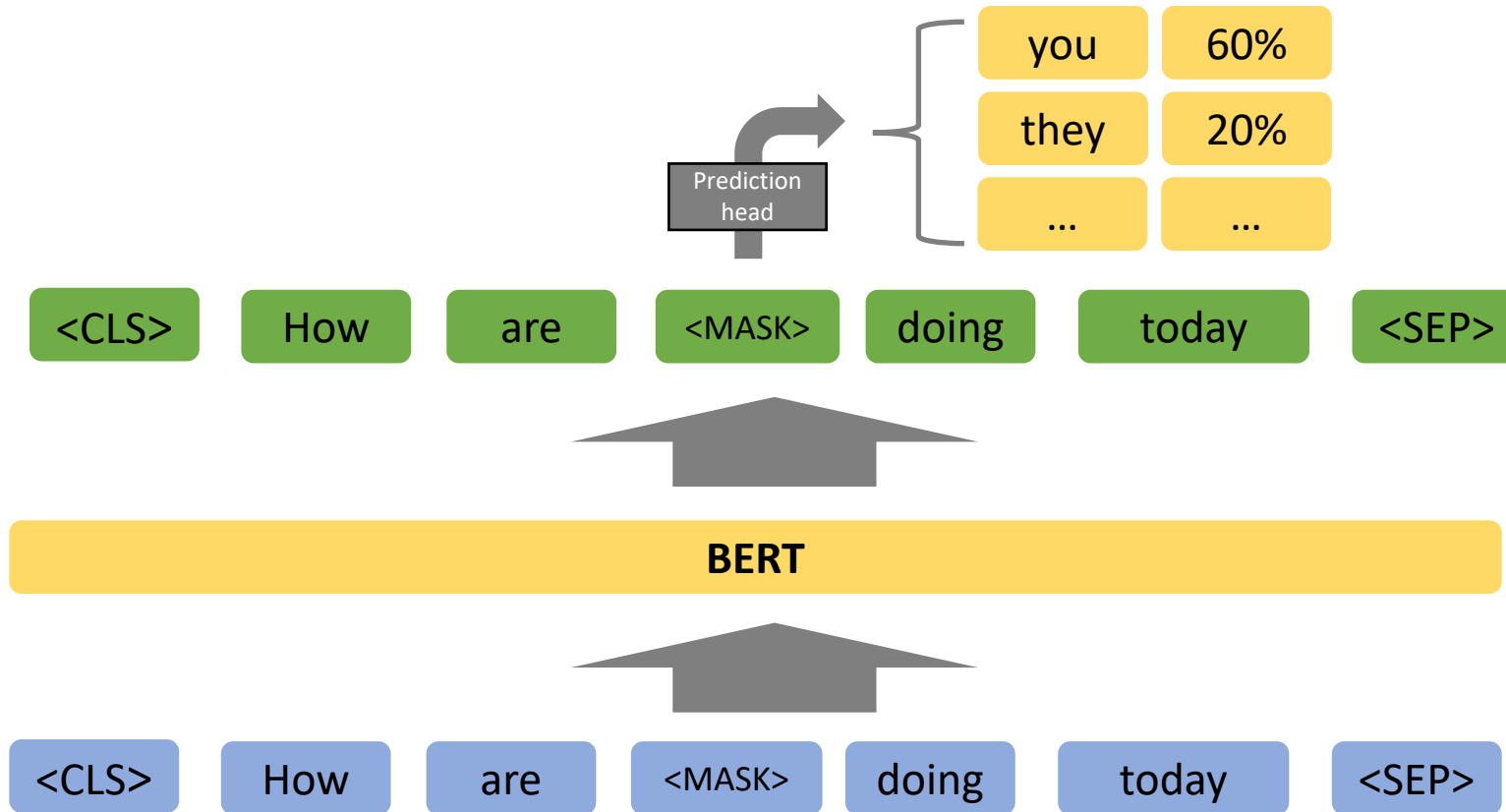
BERT Pre-Training Results:

- BERT-Base – 110M Params
- BERT-Large – 340M Params



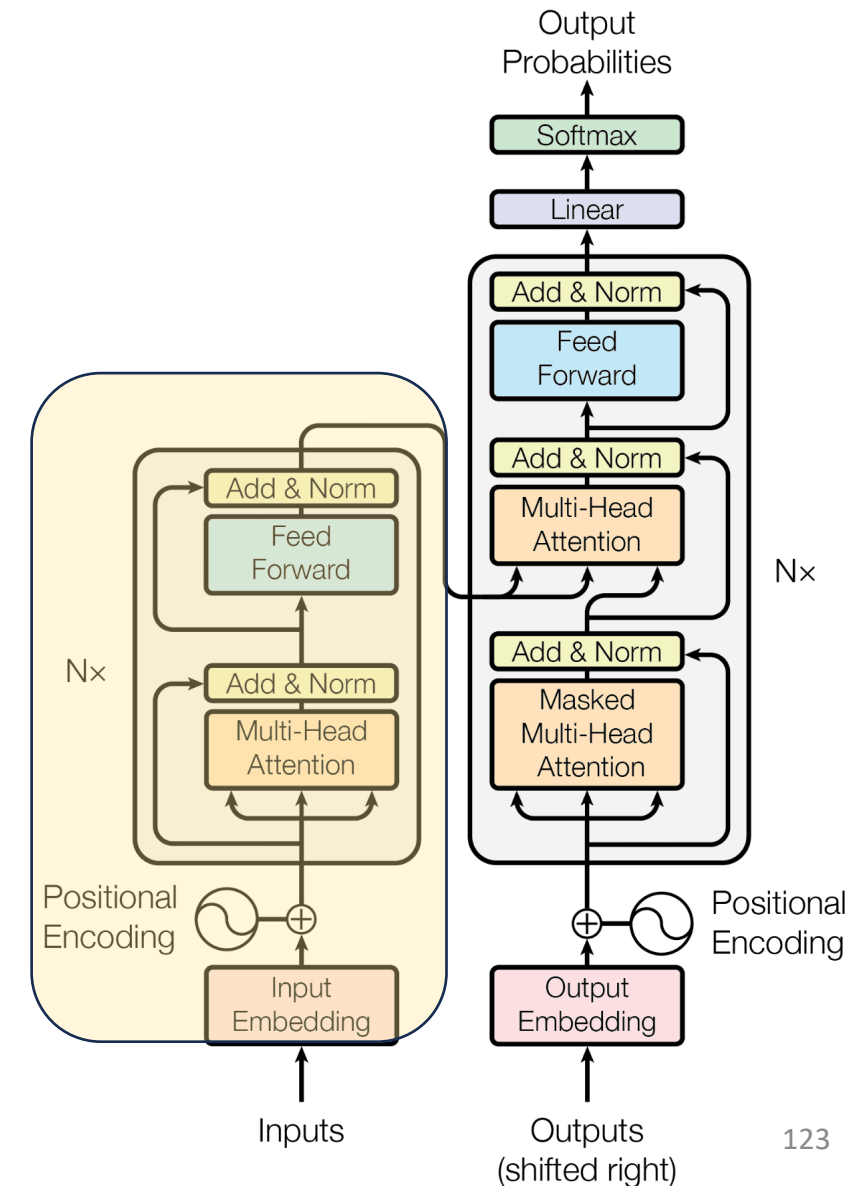
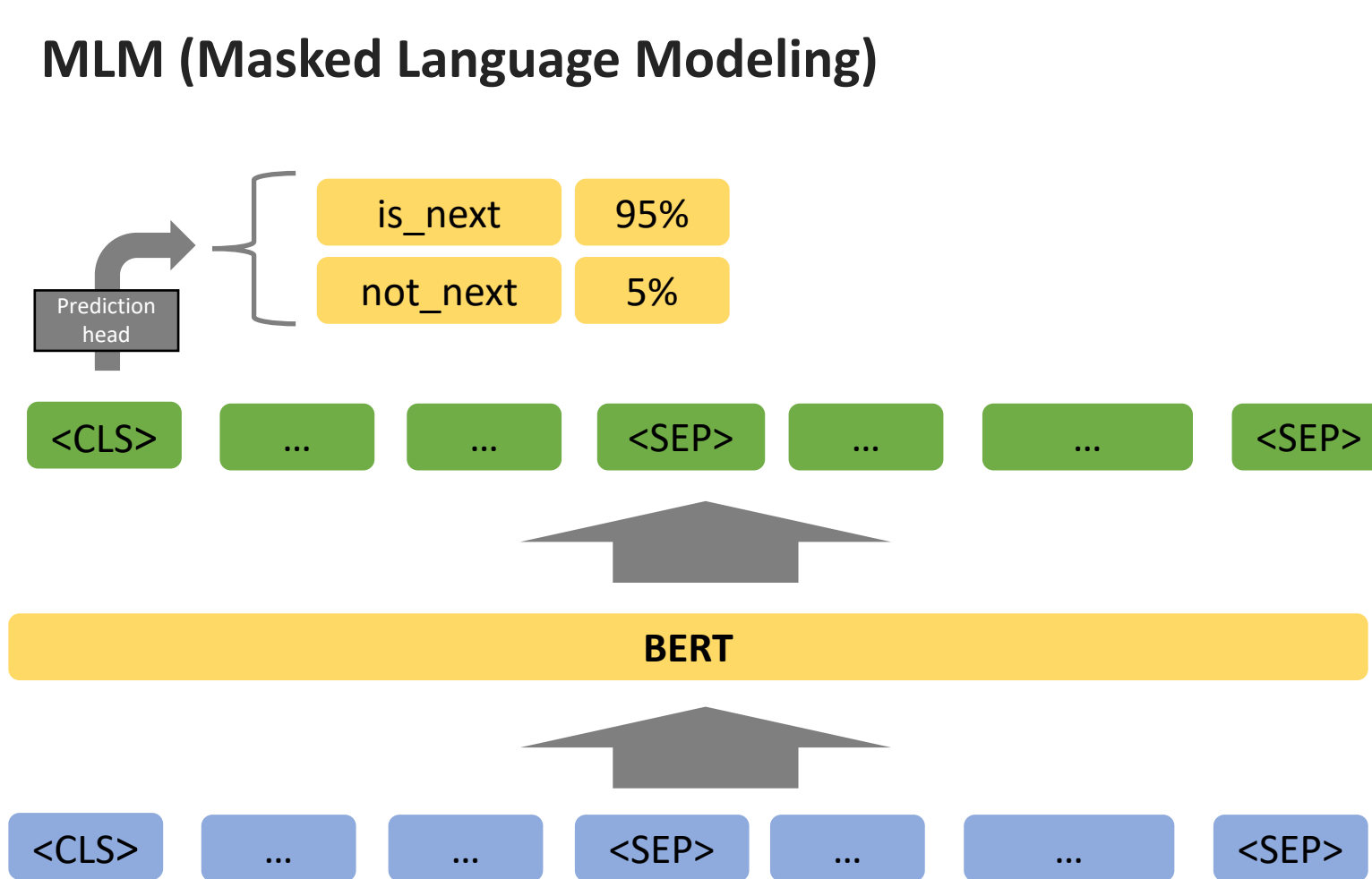
BERT - Bidirectional Encoder Representations

MLM (Masked Language Modeling)



BERT - Bidirectional Encoder Representations

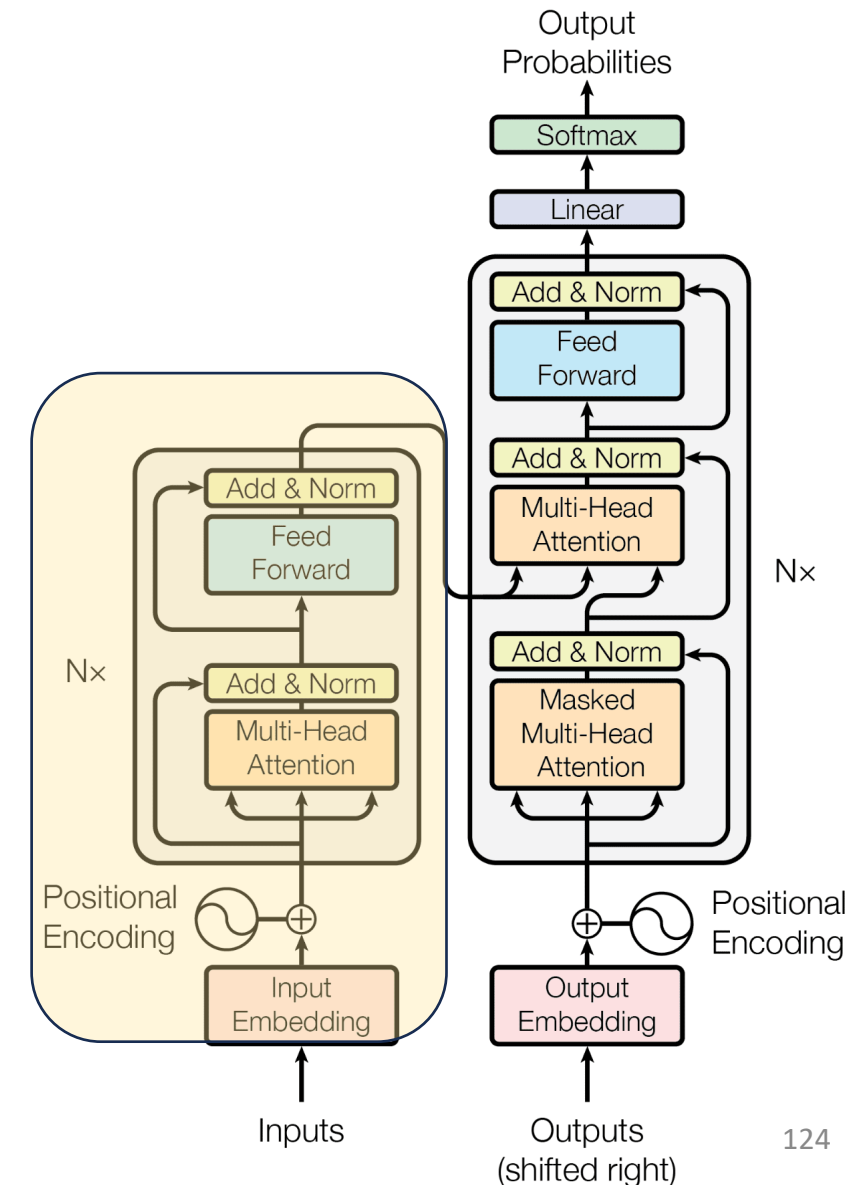
MLM (Masked Language Modeling)



BERT - Bidirectional Encoder Representations

BERT Fine-Tuning:

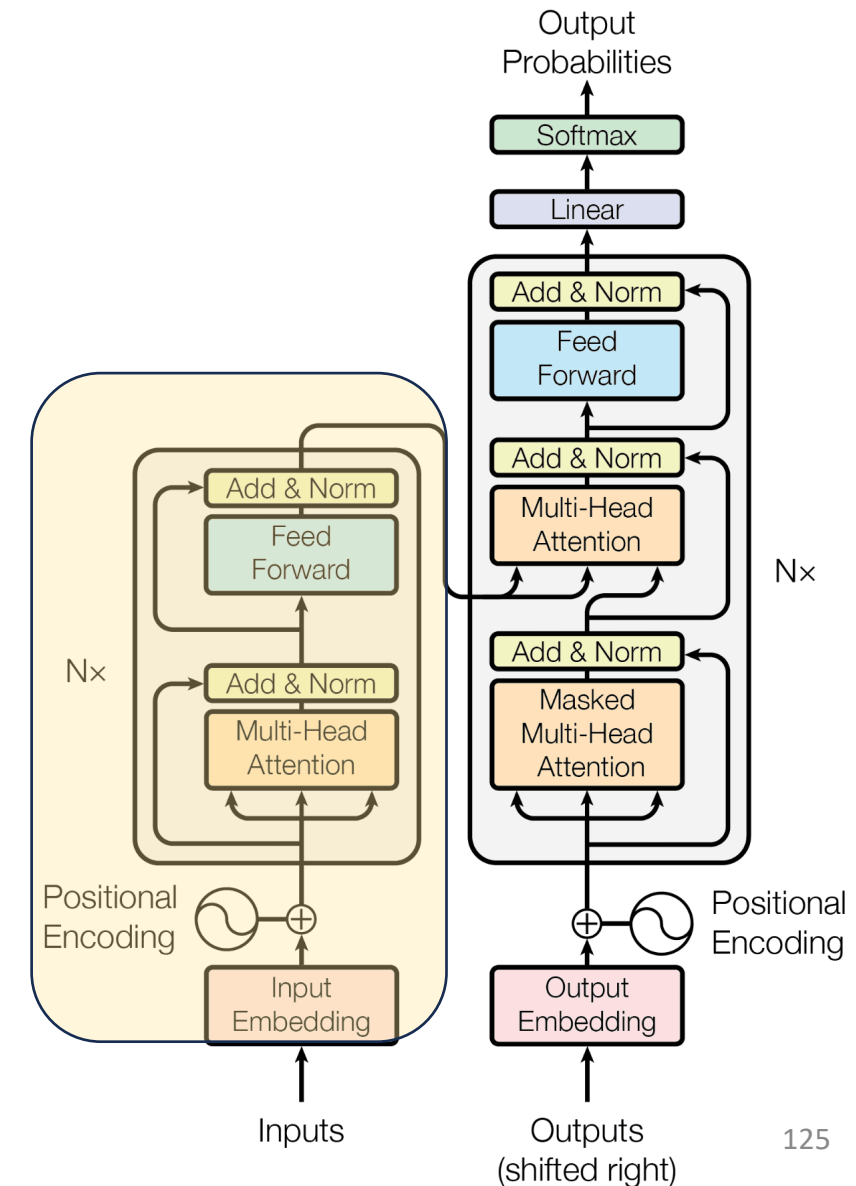
- Simply add a task-specific module after the last encoder layer to map it to the desired dimension.
 - Classification Tasks:
 - Add a feed-forward layer on top of the encoder output for the [CLS] token
 - Question Answering Tasks:
 - Train two extra vectors to mark the beginning and end of answer from paragraph
 - ...



BERT - Bidirectional Encoder Representations

BERT Evaluation:

- General Language Understanding Evaluation (GLUE)
 - Sentence pair tasks
 - Single sentence classification
- Stanford Question Answering Dataset (SQuAD)



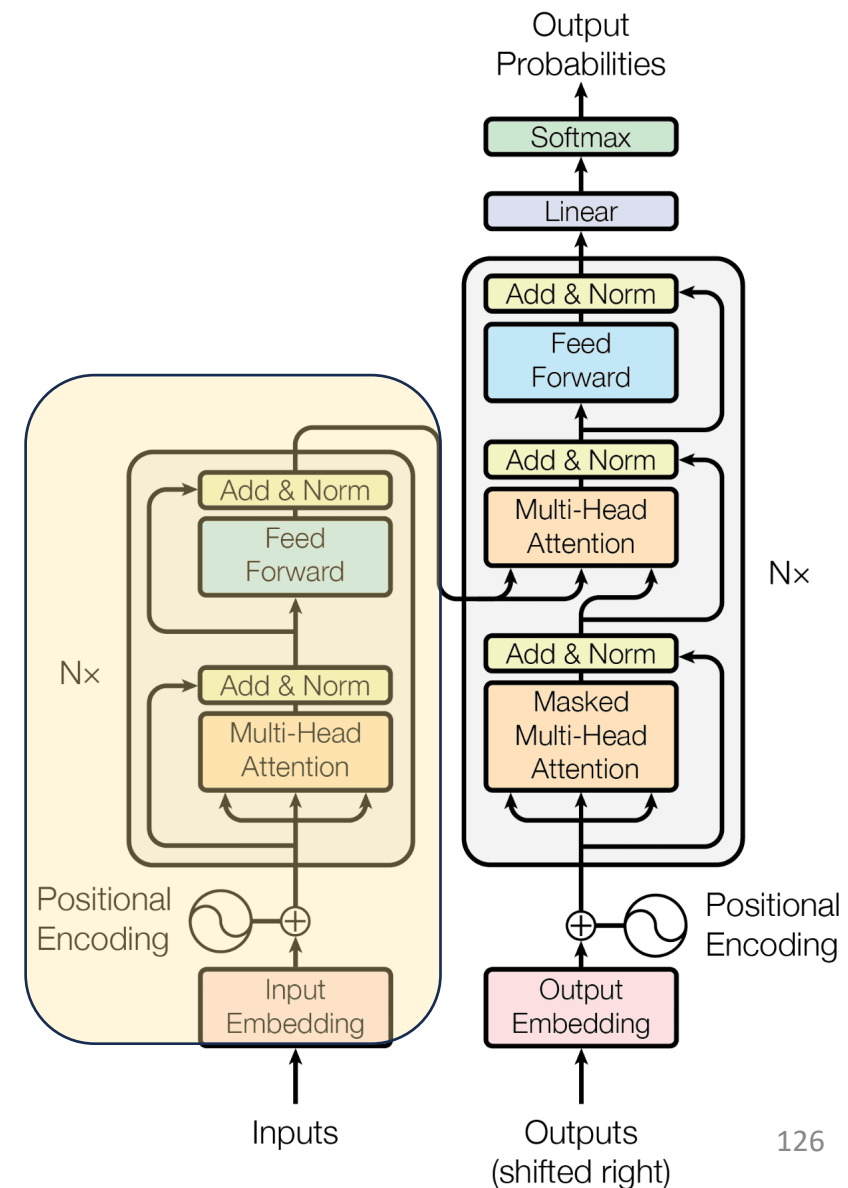
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BERT Evaluation:

System	MNLI-(m/mm) 392k	QQP 363k	QNLI 108k	SST-2 67k	CoLA 8.5k	STS-B 5.7k	MRPC 3.5k	RTE 2.5k	Average -
Pre-OpenAI SOTA	80.6/80.1	66.1	82.3	93.2	35.0	81.0	86.0	61.7	74.0
BiLSTM+ELMo+Attn	76.4/76.1	64.8	79.8	90.4	36.0	73.3	84.9	56.8	71.0
OpenAI GPT	82.1/81.4	70.3	87.4	91.3	45.4	80.0	82.3	56.0	75.1
BERT _{BASE}	84.6/83.4	71.2	90.5	93.5	52.1	85.8	88.9	66.4	79.6
BERT _{LARGE}	86.7/85.9	72.1	92.7	94.9	60.5	86.5	89.3	70.1	82.1

System	Dev		Test	
	EM	F1	EM	F1
Leaderboard (Oct 8th, 2018)				
Human	-	-	82.3	91.2
#1 Ensemble - nlnet	-	-	86.0	91.7
#2 Ensemble - QANet	-	-	84.5	90.5
#1 Single - nlnet	-	-	83.5	90.1
#2 Single - QANet	-	-	82.5	89.3
Published				
BiDAF+ELMo (Single)	-	85.8	-	-
R.M. Reader (Single)	78.9	86.3	79.5	86.6
R.M. Reader (Ensemble)	81.2	87.9	82.3	88.5
Ours				
BERT _{BASE} (Single)	80.8	88.5	-	-
BERT _{LARGE} (Single)	84.1	90.9	-	-
BERT _{LARGE} (Ensemble)	85.8	91.8	-	-
BERT _{LARGE} (Sgl.+TriviaQA)	84.2	91.1	85.1	91.8
BERT _{LARGE} (Ens.+TriviaQA)	86.2	92.2	87.4	93.2

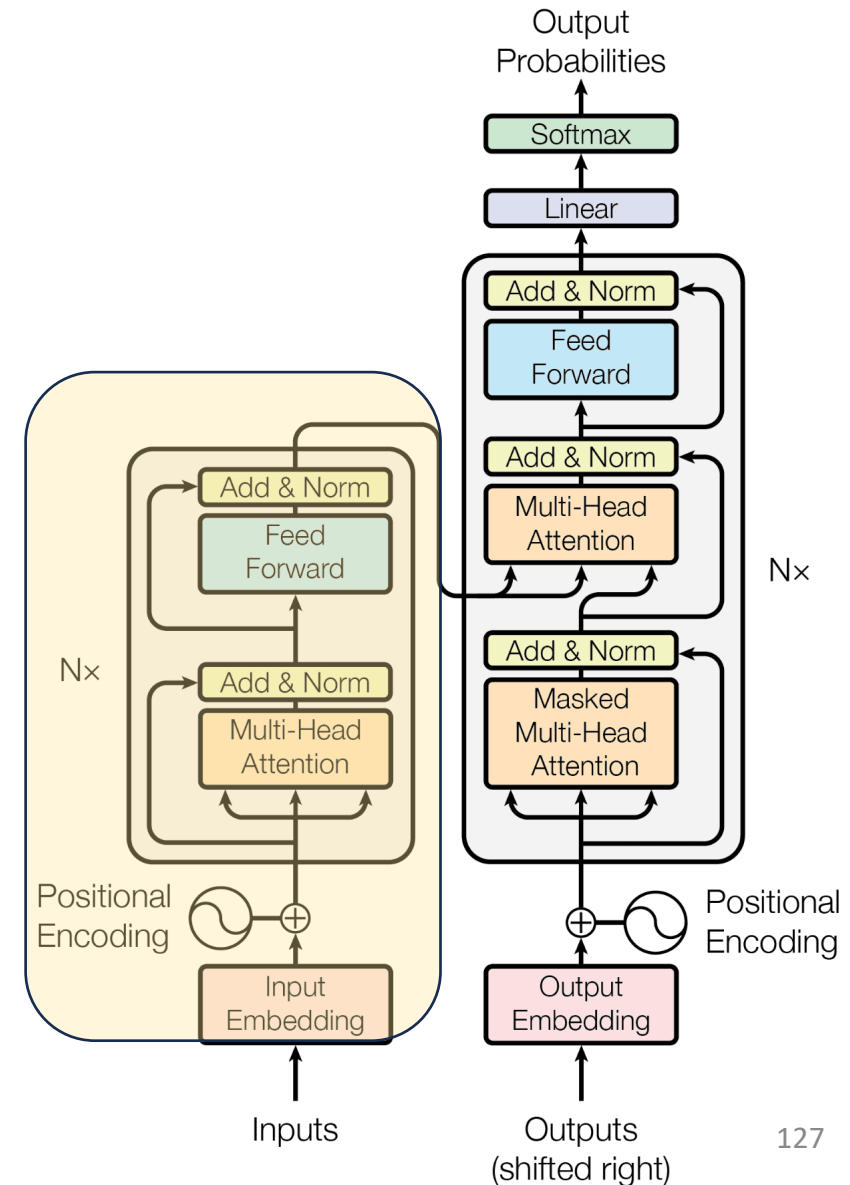
Table 2: SQuAD results. The BERT ensemble is 7x systems which use different pre-training checkpoints and fine-tuning seeds.



BERT - Bidirectional Encoder Representations

What is our takeaway from BERT?

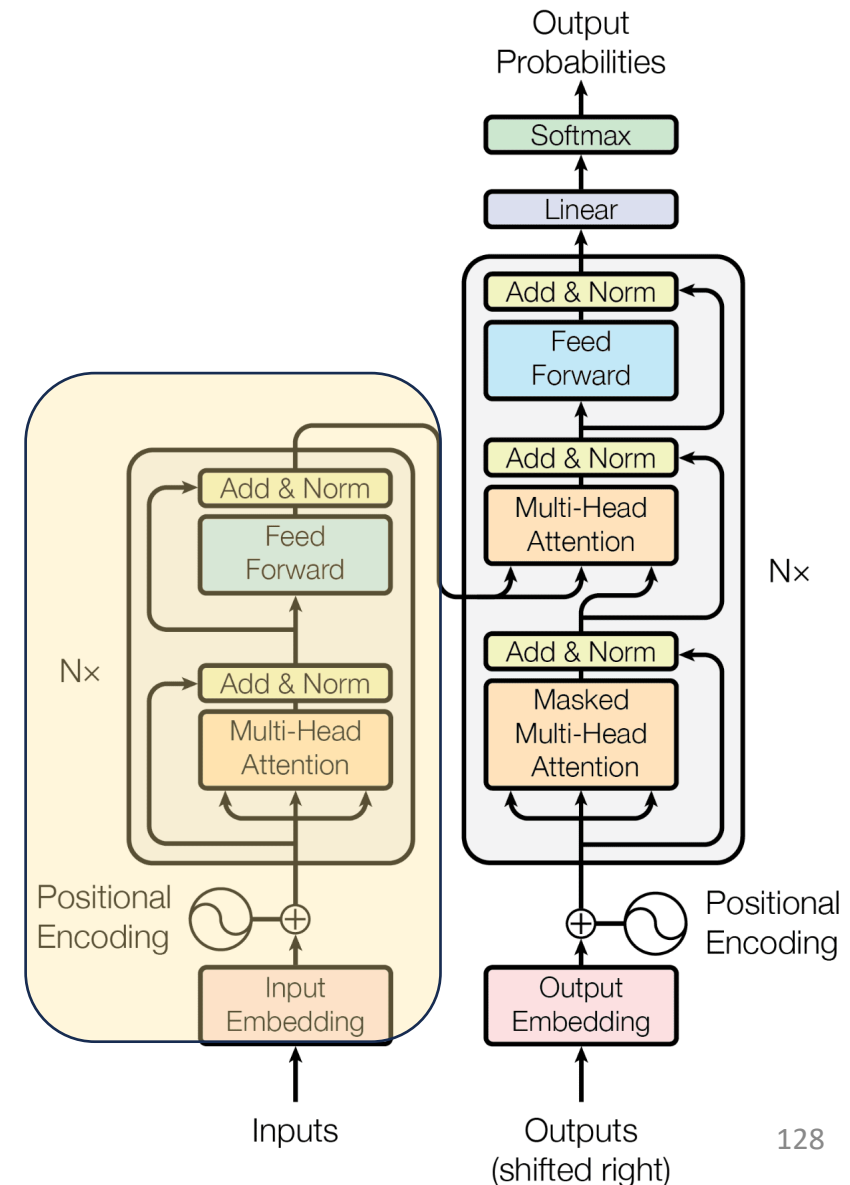
- **Pre-training tasks can be invented flexibly...**
 - Effective representations can be derived from a flexible regime of pre-training tasks.



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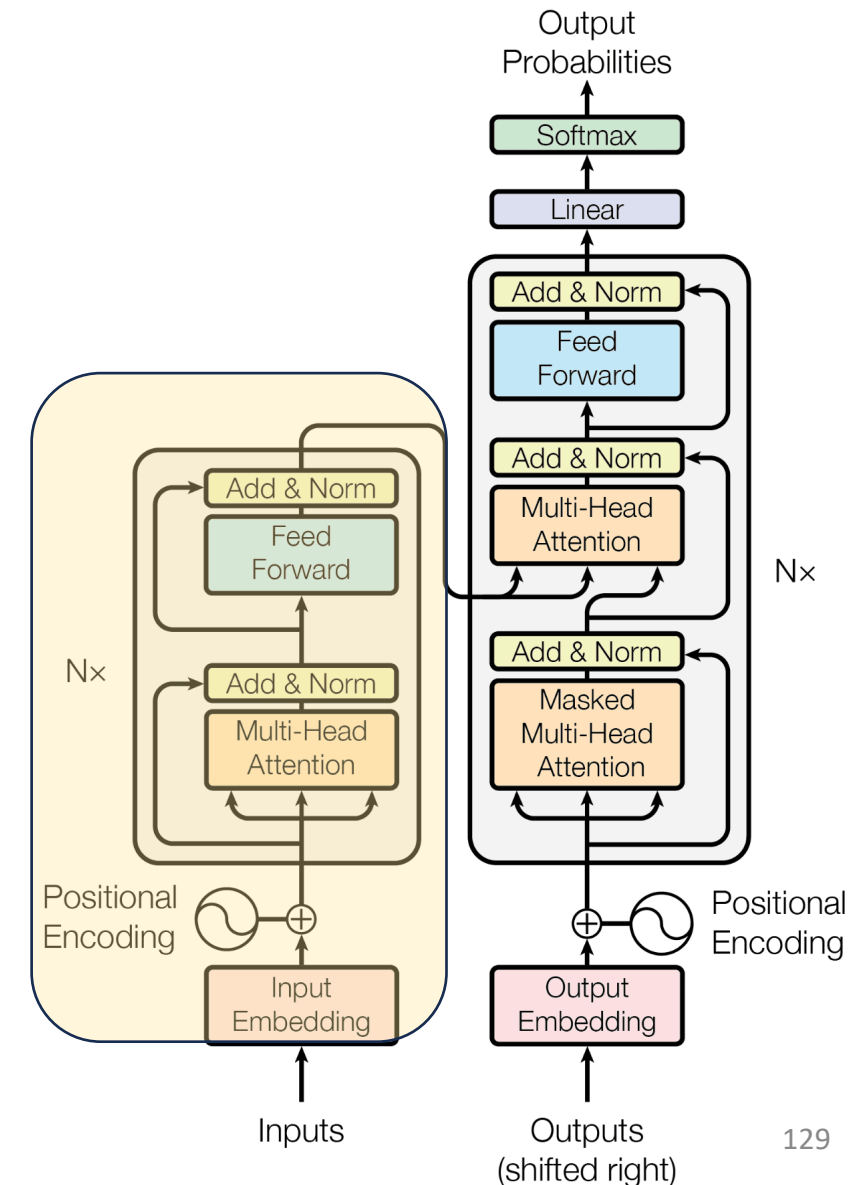
- **Pre-training tasks can be invented flexibly...**
 - Effective representations can be derived from a flexible regime of pre-training tasks.
- **Different NLP tasks seem to be highly transferable with each other...**
 - As long as we have effective representations, that seems to form a general model which can serve as the backbone for many specialized models.



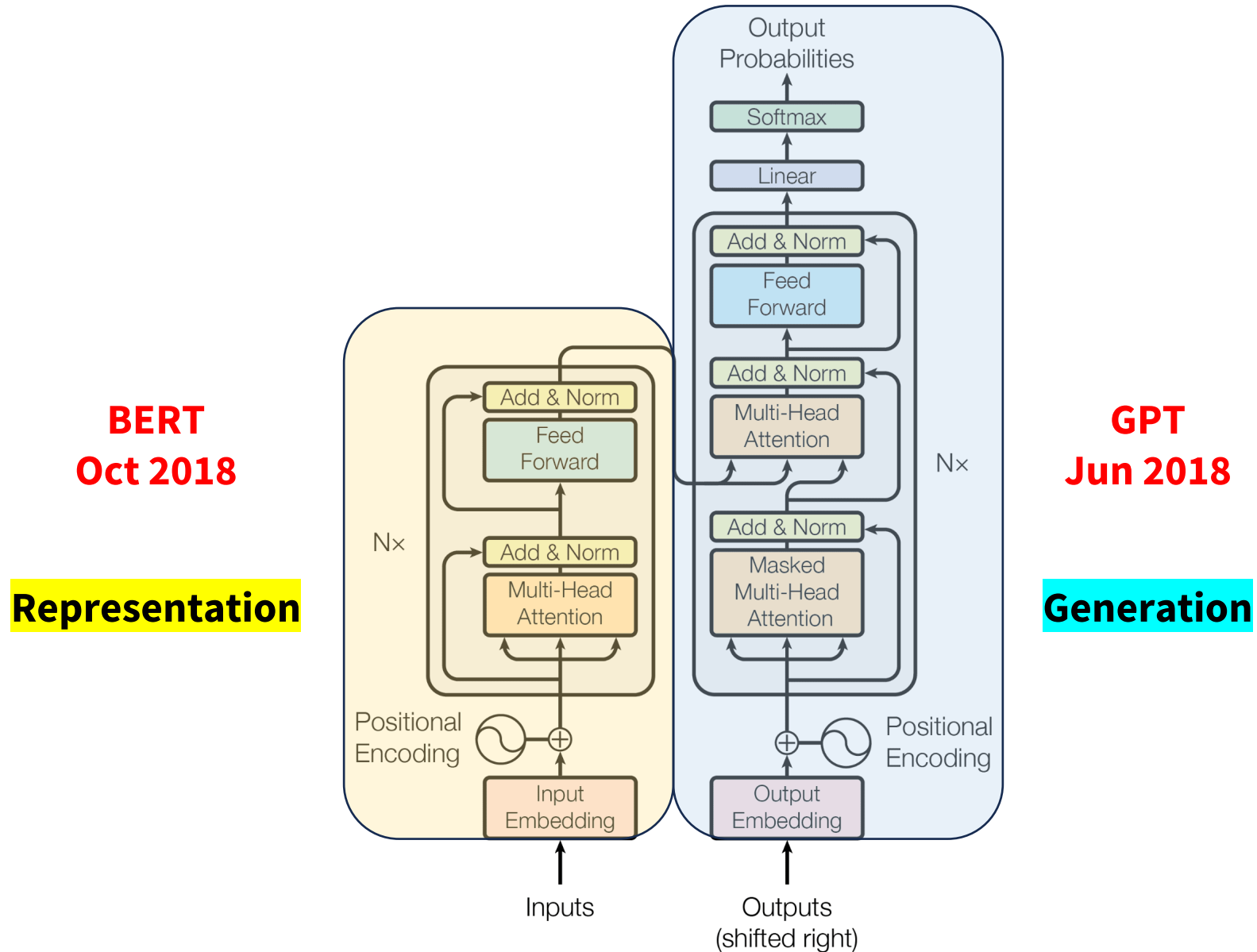
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- **And scaling works!!!**
 - 340M is considered large in 2018

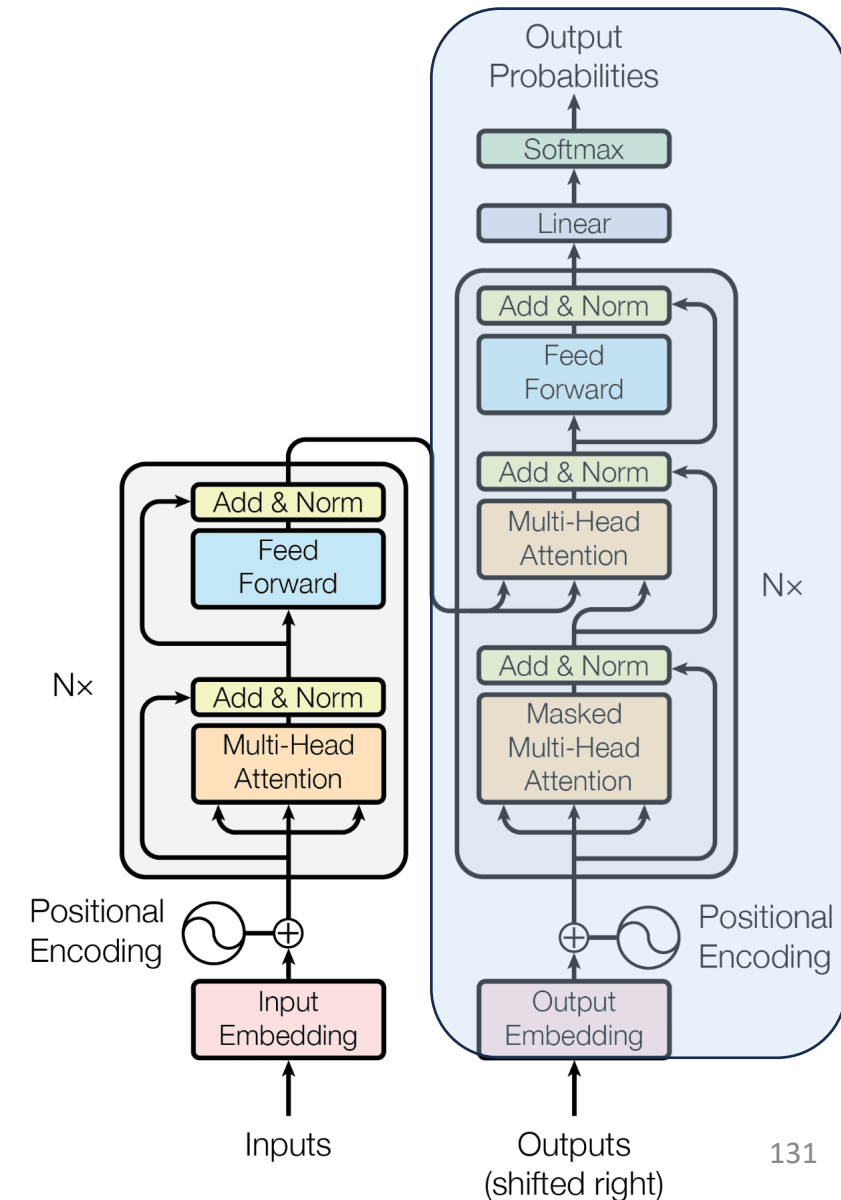


2018 – The Inception of the LLM Era



GPT – **Generative** Pretrained Transformer

- Similarly motivated as BERT, though differently designed
 - Can we leverage large amounts of unlabeled data to pretrain an LM that understands general patterns?



GPT – **Generative** Pretrained Transformer

GPT Pre-Training Corpus:

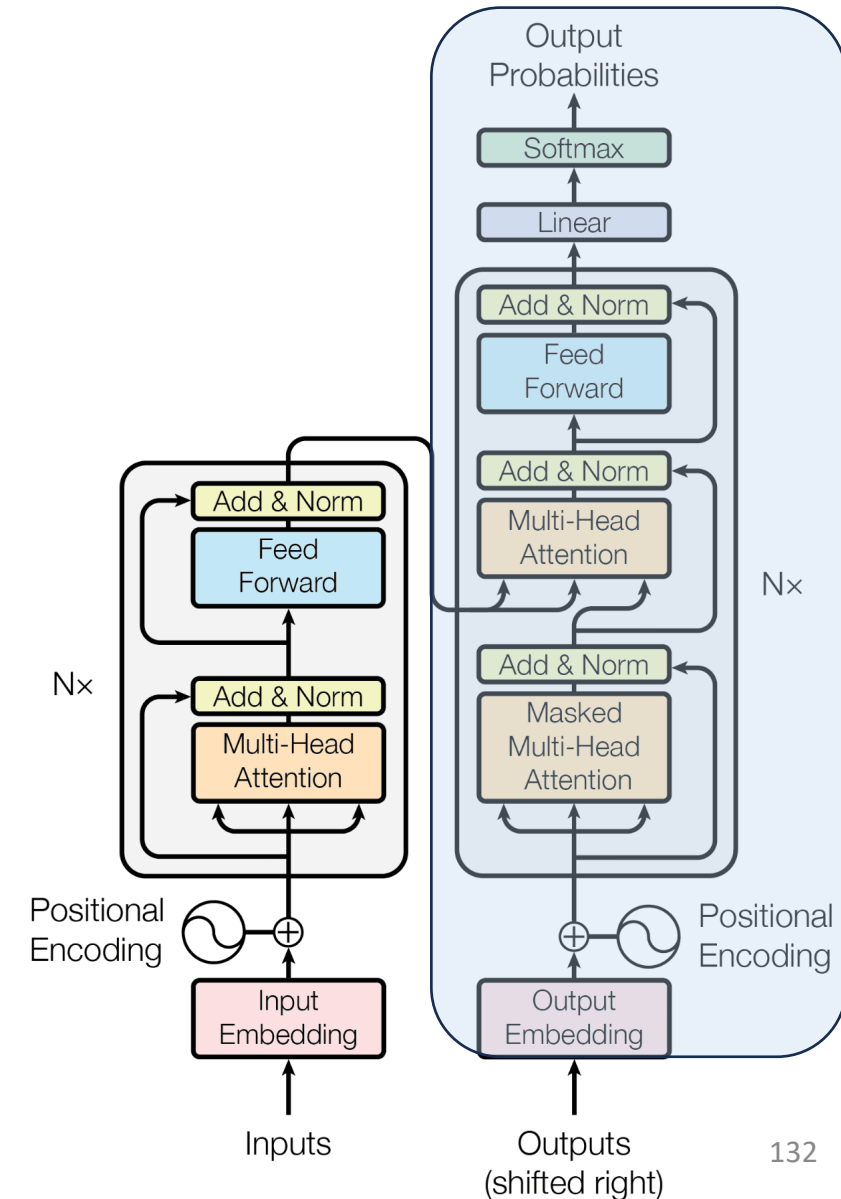
- Similarly, BooksCorpus and English Wikipedia

GPT Pre-Training Tasks:

- Predict the next token, given the previous tokens
 - More learning signals than MLM

GPT Pre-Training Results:

- GPT – 117M Params
 - Similarly competitive on GLUE and SQuAD



GPT – **Generative** Pretrained Transformer

GPT Fine-Tuning:

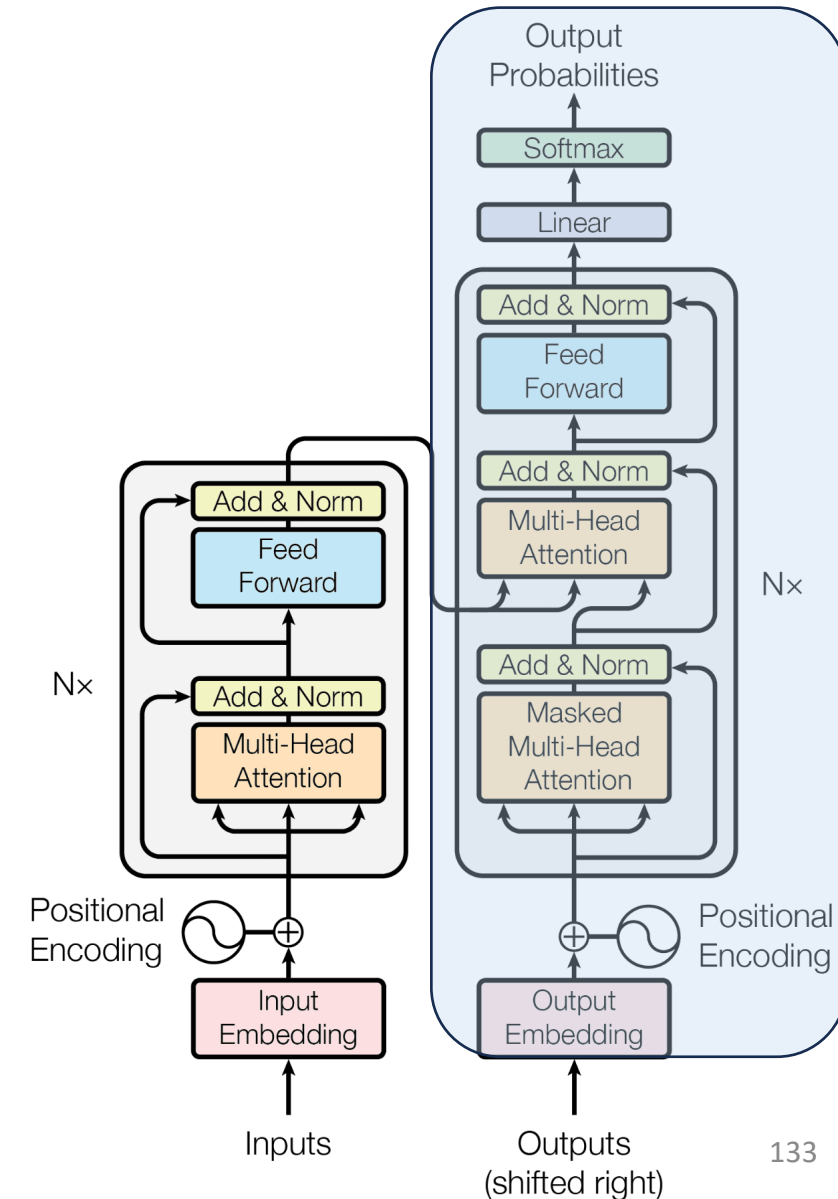
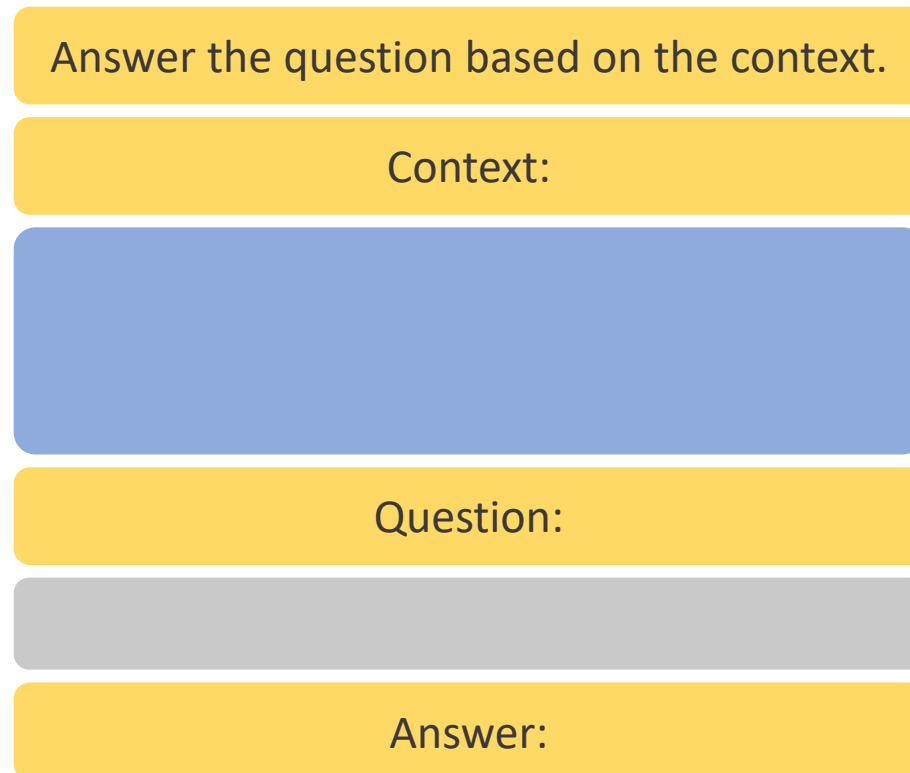
- Prompt-format task-specific text as a continuous stream for the model to fit

QA

Summarization



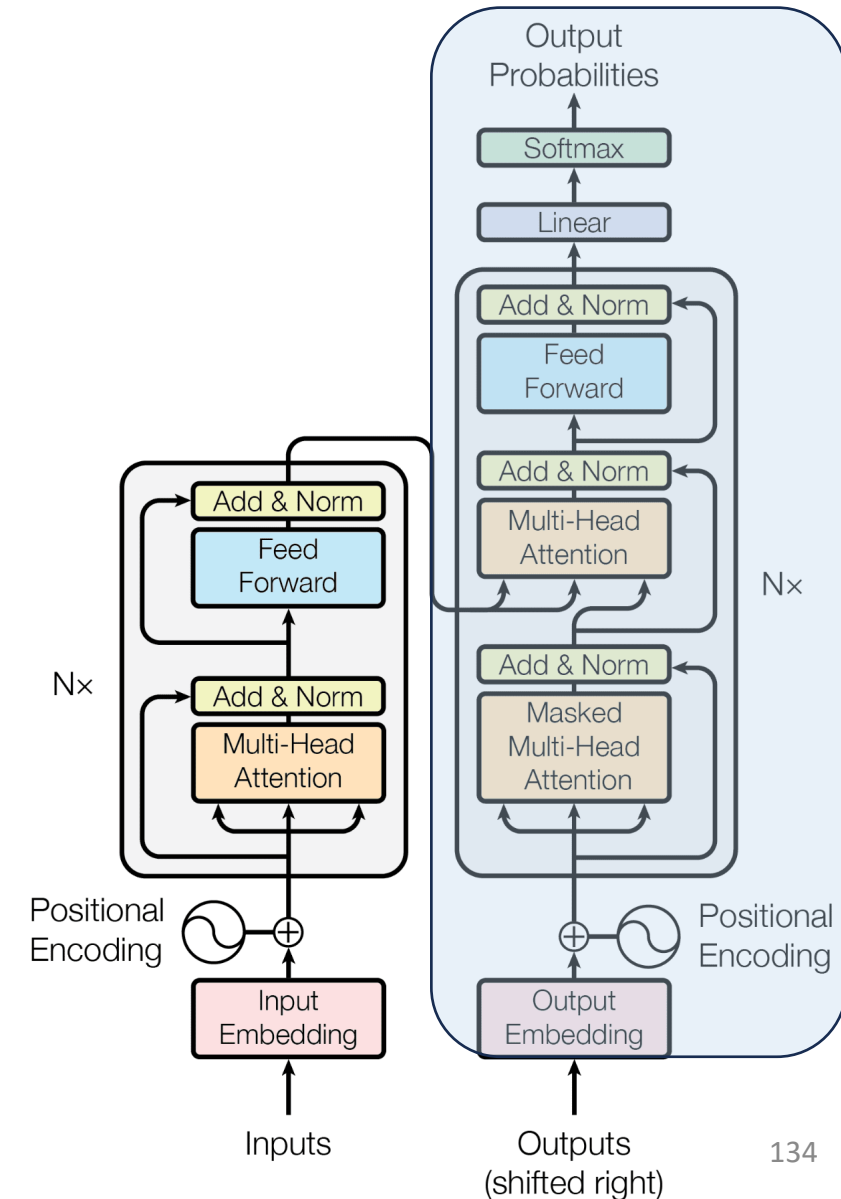
Answer the question based on the context.



GPT – **Generative** Pretrained Transformer

What is our takeaway from GPT?

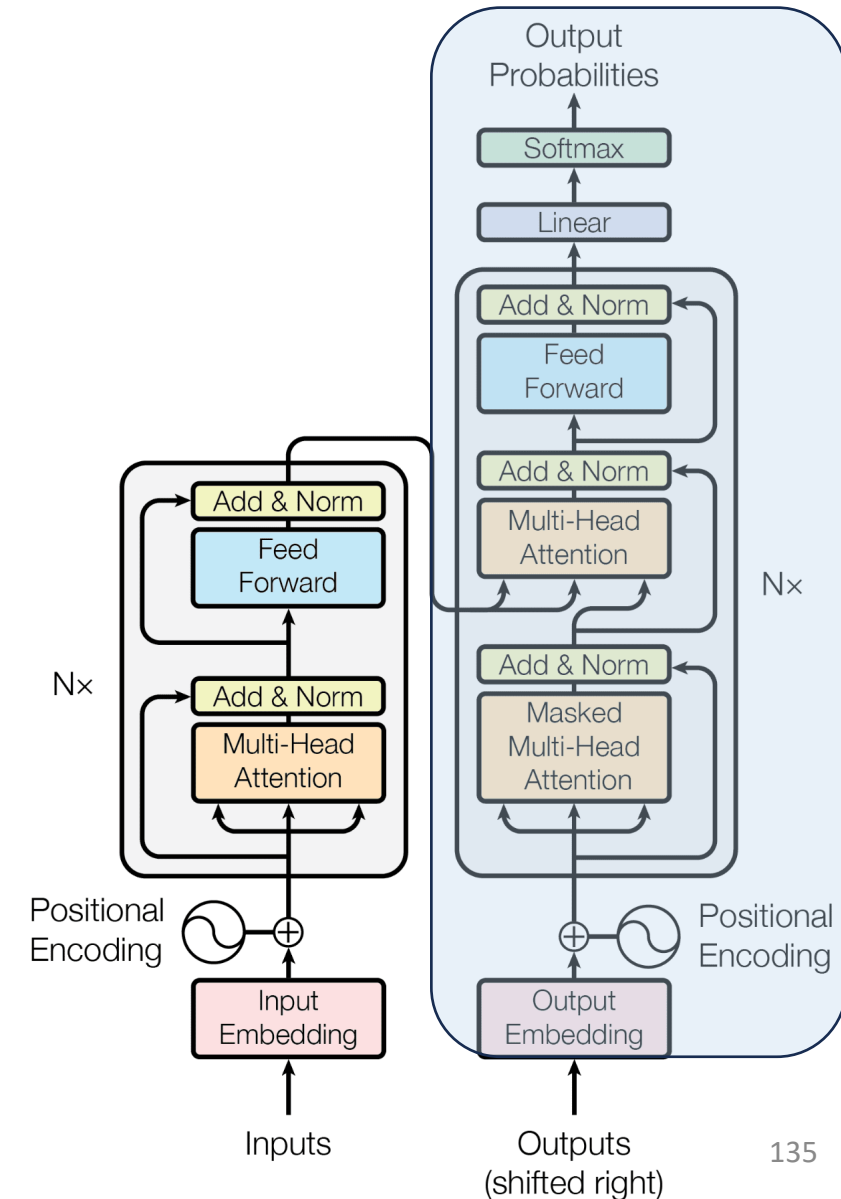
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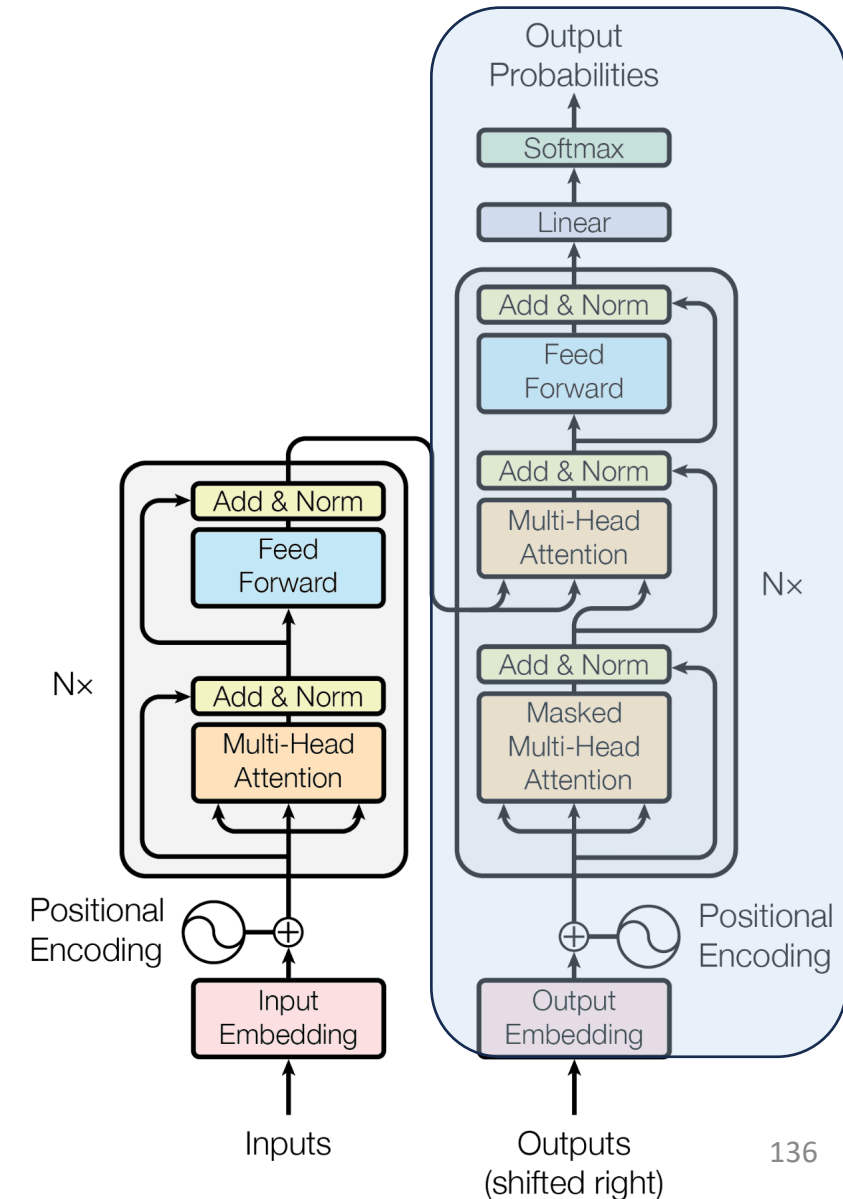
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 - Specifically, a generatively pretrained model seems to have a decent zero-shot performance on a range of NLP tasks.



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Poll

Piazza @1291

The original GPT's parameter count is closest to...

- A. 117
- B. 117K
- C. 117M
- D. 117B

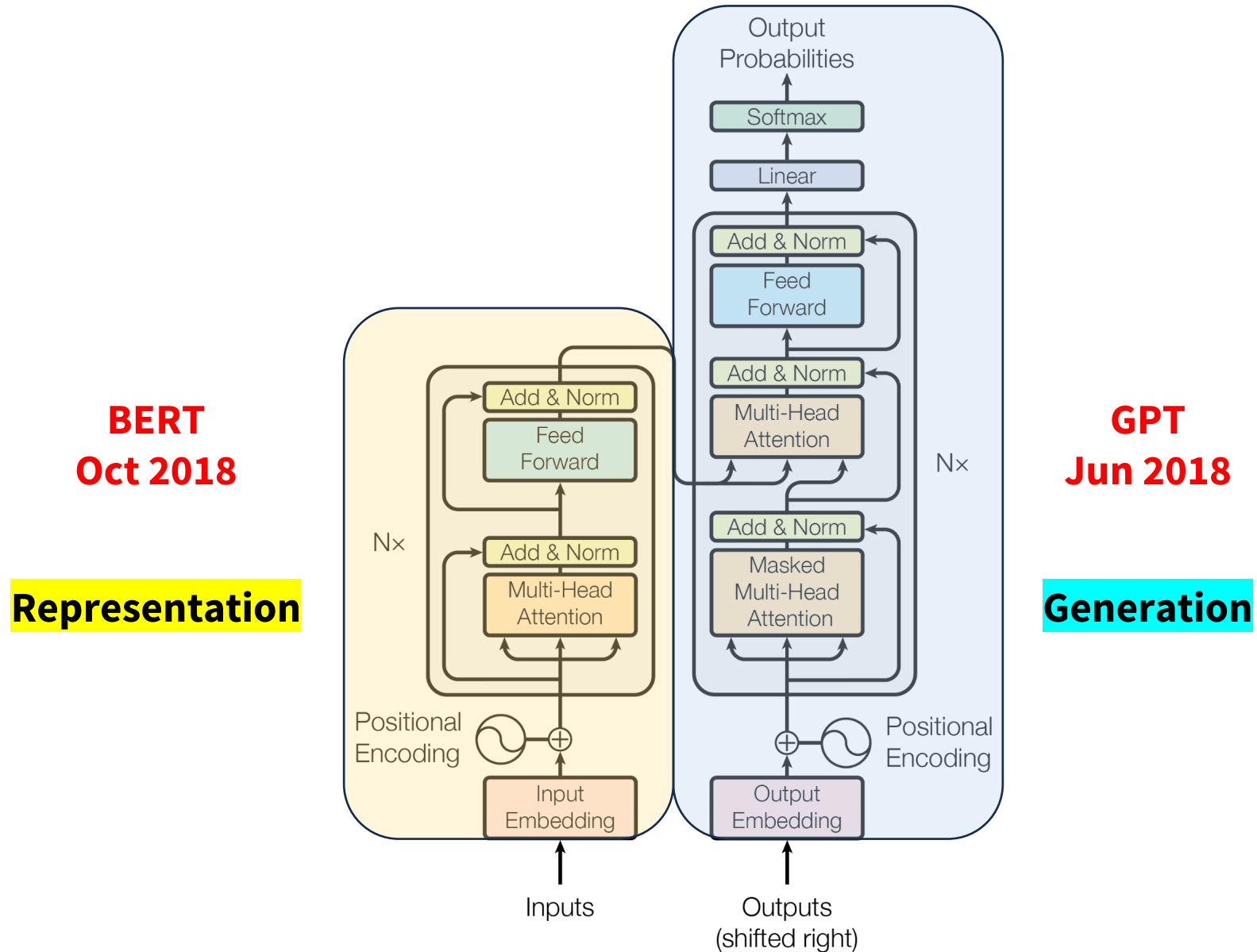
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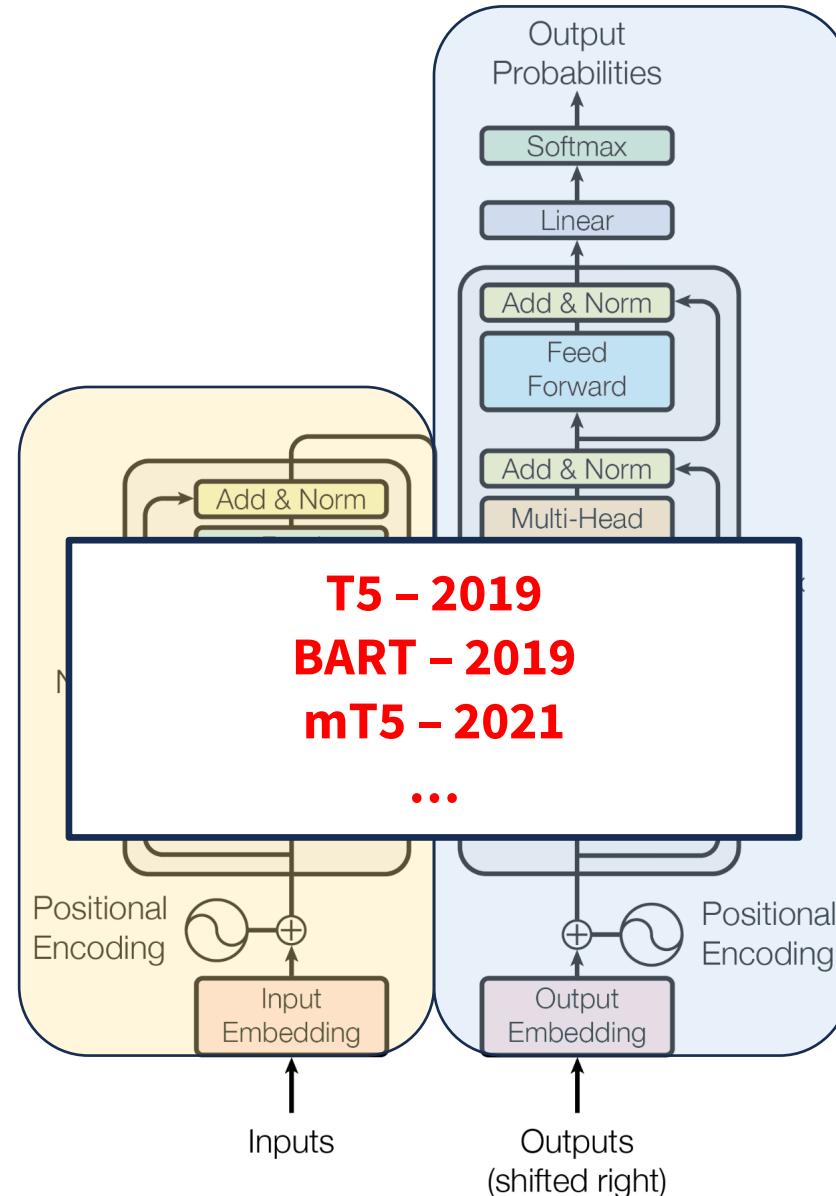
The LLM Era – Paradigm Shift in Machine Learning



The LLM Era – Paradigm Shift in Machine Learning

BERT – 2018
DistilBERT – 2019
RoBERTa – 2019
ALBERT – 2019
ELECTRA – 2020
DeBERTa – 2020
...

Representation



GPT – 2018
GPT-2 – 2019
GPT-3 – 2020
GPT-Neo – 2021
GPT-3.5 (ChatGPT) – 2022
LLaMA – 2023
GPT-4 – 2023
...

Generation

The LLM Era – Paradigm Shift in Machine Learning

From both BERT and GPT, we learn that...

- Transformers seem to provide a new class of generalist models that are capable of capturing knowledge which is more fundamental than task-specific abilities.

Before LLMs

Since LLMs

- **Feature Engineering**
 - How do we design or select the best features for a task?

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- **Interpretability and Explainability**
 - How can we understand the inner workings of our own models?

The LLM Era – Paradigm Shift in Machine Learning

- What has caused this paradigm shift?

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 - Problem in recurrent networks
 - Information is effectively lost during encoding of long sequences
 - Sequential nature disables parallel training and favors late timestep inputs

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- What has caused this paradigm shift?
 - **Problem in recurrent networks**
 - Information is effectively lost during encoding of long sequences
 - Sequential nature disables parallel training and favors late timestep inputs
 - **Solution: Attention mechanism**
 - Handling long-range dependencies
 - Parallel training
 - Dynamic attention weights based on inputs

The LLM Era – Paradigm Shift in Machine Learning

- Attention and Transformer – is this the end?

The LLM Era – Paradigm Shift in Machine Learning

- **Attention and Transformer – is this the end?**
 - **Problem in current Transformer-based LLMs??**

Poll

Piazza @1292

What might be a flaw of our current Transformer-based LLMs?

Freeform response

The LLM Era – Paradigm Shift in Machine Learning

- **Attention and Transformer – is this the end?**
 - **Problem in current Transformer-based LLMs??**
 - True understanding the material vs. memorization and pattern-matching
 - Cannot reliably follow rules – factual hallucination e.g. inability in arithmetic

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 - **Solution: ???**